



Multiscale coupling and multiphysics approaches in earth sciences: Applications

Klaus Regenauer-Lieb^{1,2,*}, Manolis Veveakis², Thomas Poulet², Florian Wellmann², Ali Karrech³, Jie Liu¹, Juerg Hauser², Christoph Schrank^{1,4}, Oliver Gaede^{1,4}, Florian Fousseis⁵, and Mike Trefry^{1,6}

¹Faculty of Science, Laboratory for Multiscale Earth System Dynamics and Geothermal Research, School of Earth and Environment, The University of Western Australia, M004, 35 Stirling Hwy, 6009 Crawley, WA, Australia

²CSIRO Earth science and Resource Engineering, 26 Dick Perry Ave. 6151 Kensington, WA, Australia

³Faculty of Engineering, School of Civil and Resource Engineering, The University of Western Australia, Computation and Mathematics, M051, 35 Stirling Hwy, 6009 Crawley, WA, Australia

⁴Science and Engineering Faculty, Queensland University of Technology, GPO Box 2434, Brisbane, QLD, 4001, Australia

⁵School of Geosciences Grant Institute, The University of Edinburgh, West Mains Road Edinburgh EH9 3JW, Great Britain

⁶CSIRO Land and Water, Underwood Ave, 6014 Floreat, WA, Australia

(Received: 22 September 2013. Accepted: 30 September 2013)

ABSTRACT

Geoscientists are confronted with the challenge of assessing nonlinear phenomena that result from multiphysics coupling across multiple scales from the quantum level to the scale of the earth and from femtosecond to the 4.5 Ga of history of our planet. We neglect in this review electromagnetic modelling of the processes in the Earth's core, and focus on four types of couplings that underpin fundamental instabilities in the Earth. These are thermal (T), hydraulic (H), mechanical (M) and chemical (C) processes which are driven and controlled by the transfer of heat to the Earth's surface. Instabilities appear as faults, folds, compaction bands, shear/fault zones, plate boundaries and convective patterns. Convective patterns emerge from buoyancy overcoming viscous drag at a critical Rayleigh number. All other processes emerge from non-conservative thermodynamic forces with a critical critical dissipative source term, which can be characterised by the modified Gruntfest number Gr . These dissipative processes reach a quasi-steady state when, at maximum dissipation, THMC diffusion (Fourier, Darcy, Biot, Fick) balance the source term. The emerging steady state dissipative patterns are defined by the respective diffusion length scales. These length scales provide a fundamental thermodynamic yardstick for measuring instabilities in the Earth. The implementation of a fully coupled THMC multiscale theoretical framework into an applied workflow is still in its early stages. This is largely owing to the four fundamentally different lengths of the THMC diffusion yardsticks spanning micro-metre to tens of kilometres compounded by the additional necessity to consider microstructure information in the formulation of enriched continua for THMC feedback simulations (i.e., micro-structure enriched continuum formulation). Another challenge is to consider the important factor time which implies that the geomaterial often is very far away from initial yield and flowing on a time scale that cannot be accessed in the laboratory. This leads to the requirement of adopting a thermodynamic framework in conjunction with flow theories of plasticity. This framework allows, unlike consistency plasticity, the description of both solid mechanical and fluid dynamic instabilities. In the applications we show the similarity of THMC feedback patterns across scales such as brittle and ductile folds and faults. A particular interesting

* Author to whom correspondence should be addressed.
Email: klaus.regenauer-lieb@uwa.edu.au

case is discussed in detail, where out of the fluid dynamic solution, ductile compaction bands appear which are akin and can be confused with their brittle siblings. The main difference is that they require the factor time and also a much lower driving forces to emerge. These low stress solutions cannot be obtained on short laboratory time scales and they are therefore much more likely to appear in nature than in the laboratory. We finish with a multiscale description of a seminal structure in the Swiss Alps, the Glarus thrust, which puzzled geologists for more than 100 years. Along the Glarus thrust, a km-scale package of rocks (nappe) has been pushed 40 km over its footwall as a solid rock body. The thrust itself is a m-wide ductile shear zone, while in turn the centre of the thrust shows a mm-cm wide central slip zone experiencing periodic extreme deformation akin to a stick-slip event. The m-wide creeping zone is consistent with the THM feedback length scale of solid mechanics, while the ultralocalised central slip zones is most likely a fluid dynamic instability.

Keywords: Multiscaling, Multiphysics, Microstructure, Homogenisation, Complex Systems, Fluid Dynamics, Solid Mechanics, Geomechanics.

Section: Life, Climate & Environmental Sciences

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1. INTRODUCTION

Earth processes occur across 15 orders of magnitude in spatial scales (from nanometers to thousands of kilometres), and across at least 30 orders of magnitude in time scale (from femtoseconds to hundreds of millions of years). Quantifying physical processes across the length and timescales is beyond the reach of traditional theories. However, in recent decades, the advent of data intensive computing has revolutionised sciences.⁽¹⁾ Multi-scale, multi-physics computer simulations have become available to explore domains that are inaccessible to both theory and experiment. In this review we present a first overview of the current approaches to coupling the scales using methods borrowed from other disciplines. We then follow with a discussion of applications that grew out

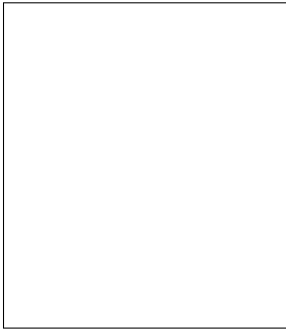
of these first implementations and have been developed under a geoscience-specific workflow laid out in the theory review.⁽²⁾

2. MICROSTRUCTURE MODELLING OF EARTH MATERIALS

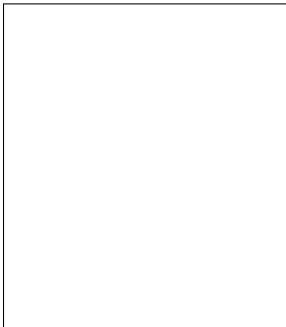
Quantum mechanical models allow probing of the physics of geomaterials at extreme geological conditions such as the conditions of temperatures well in excess of 3000 °C and pressures in excess of 130 GPa pressure found deep in the earth.⁽³⁾ The technique is ideally suited for super-computer implementations as molecular dynamics has a long track record of at least 50 years in the area of condensed matter physics. In earth sciences the current status of these calculations is mainly limited to applications in mineralogy and nanochemistry, and the approach has not yet bridged the scales from thousands of atoms to describing the behaviour of larger rock specimens consisting of mineral aggregates.

A notable exception is recent progress reported for the rheology of MgO^(4,5) allowing a first assessment of deformation mechanism deep in the Earth's mantle using a statistical mechanics point of view and upscaling based on discrete interactions over multiple scales. In these multi-scale simulations the electronic structure is explicitly taken into account followed by a mesoscopic scale simulation of dislocation dynamics benchmarked in the laboratory.

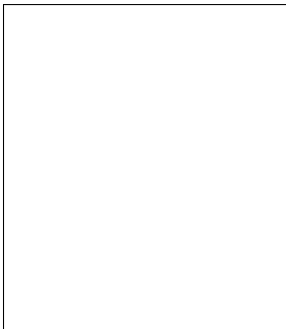
A complementary thermodynamic technique has recently been proposed⁽⁶⁾ for derivation of flow laws of mantle materials. It starts with a description on the opposite scale using a thermodynamic homogenisation technique which will be discussed in details in this review using examples of modelling problem at larger scale than the scale of grain aggregates. For the grain scale averaging model Ref. [6] uses the working hypothesis that a polycrystalline aggregate can be represented by a distribution function characterising the state of individual grains by three thermodynamic state variables: elastic strain, dislocation density and grain size. Through the



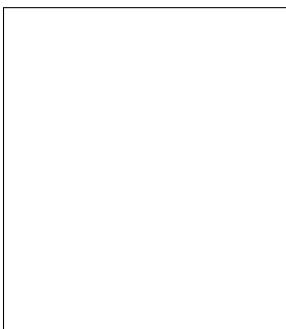
Klaus Regenauer-Lieb



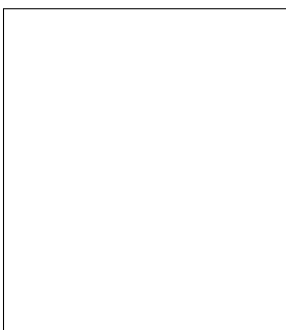
Manolis Veveakis



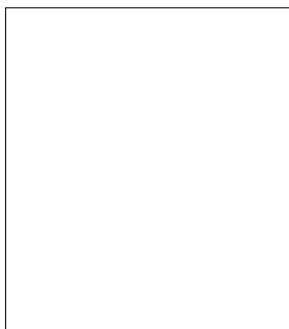
Thomas Poulet



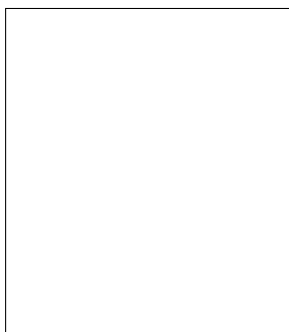
Florian Wellmann



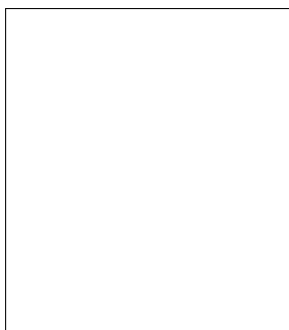
Ali Karrech



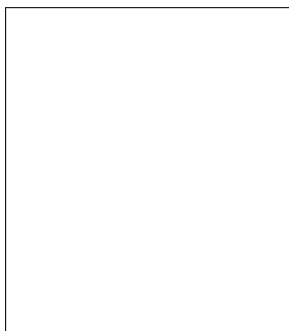
Jie Liu



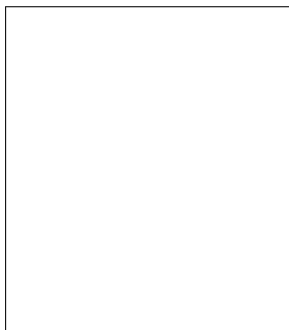
Juerg Hauser



Christoph Schrank

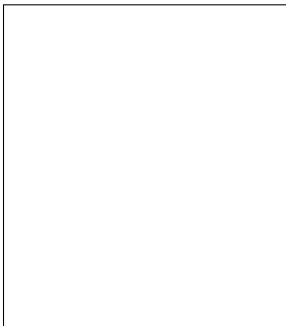


Oliver Gaede



Florian Füsseis

Mike Trefry



assumption of maximum entropy production (minimum Helmholtz free energy) the rheology of Olivine aggregates can be derived.

Statistical mechanics based approaches for derivation of polycrystalline flow laws have also been reported for multiscale formulation of deformation of ice where well established material science concepts have been used to describe its deformation behaviour.⁽⁷⁾ Ice is one of the few earth materials that creeps in nature at relatively high rates up 10^{-6} s^{-1} – 10^{-12} s^{-1} so that laboratory experiments can be compared to numerical experiments. The complex behaviour of polycrystalline ice is controlled by random assemblages of grains with preferred size-, shape- and lattice orientations creating strong anisotropy at grain scale. A number of different techniques are used for upscaling the material behaviour into an effective medium and calibration by laboratory experiments is feasible. The largest scale of computations has been achieved by an explicit coupling of a viscoplastic solver based on Fast Fourier Transform (FFT) with an operator splitting algorithm of the individual micro-processes involved.⁽⁷⁾ The example of the simulation is shown in Figure 1.

The numerical simulation is clearly reproducing a number of features observed in the laboratory experiment such as the appearance of kink bands in both numerical and

experiments (here shown by intracrystalline variations in orientations), serrations and bulging on the grain boundaries. However, the model also shows some discrepancies such as the difference in grain boundary migration from the initial configuration in laboratory and simulation results. At present these numerical tools can be used to test the importance of the assumed micro-mechanism and the degree of internal couplings through comparison of simulation and experiment. The authors⁽⁷⁾ conclude that although the hexagonal symmetry of ice crystals makes this material one of the simplest earth materials to consider, we are at present limited by computational power to realistically use this explicit microstructural technique to estimate the mechanical response for an entire ice-sheet flow model.

The ice example illustrates the challenge posed by modelling multiscale behaviour of geomaterials. Ice only requires the incorporation of 1 unit vector for the lattice owing to its simple symmetry. Numerical predictions can also be tested and verified in the laboratory. It has a relatively simple material behaviour dominated by visco-plastic creep. However, it is commonly perceived to be premature to attempt a larger scale implementation. Reference [7] states that a true *ab-initio* modelling workflow, where the geometries of the mineral phases and their interactions are modelled through their configurational energies to be subsequently homogenised by a renormalisation approach, is beyond the reach of earth sciences at present. The rapid developments over the recent years reported above^(5,6) may be seen as indication that this will change in the near future.

While *ab-initio* modelling is not yet available for a wide range of geomaterials, we recommend to incorporate elements of microstructural information that have been obtained through conventional methods into large scale simulations. The available methods are well established in Material Sciences as they have been conceived more than 20 years ago (see Ref. [8] for a review). Of particular appeal for earth sciences are second-order homogenisation methods where higher-order continuum formulations are used at macro-scale and conventional formulations are used at microscale. The macroscopic velocity gradients and velocities are considered as boundary

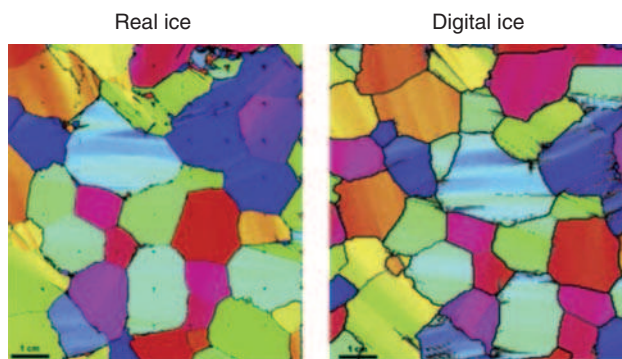


Fig. 1. Deformation experiment of ice at 4 per cent shortening in pure shear, in the laboratory compared to numerical simulation. The colour indicates orientation of the *c*-axis. Reprinted with permission from [7], M. Montagnat, et al., *Journal of Structural Geology* in press (2013). © 2013.

value problem for the microscale simulations. Averaging at microscale leads to the definition of an effective macroscopic stress tensor and its higher order term, which is in turn fed back to the macro-scale calculation. These simulations can for instance be used to assess the effect of visco-plastic anisotropy of individual crystals on the large scale visco-plastic anisotropies observed in the deformation of continents.⁽⁹⁾ The authors used a set of 1000 orthorhombic olivine crystals associated with each macroscopic finite element to show that the developed crystal plastic anisotropy can significantly affect large scale deformation of continents.

A significant drawback of all modelling in earth sciences is that the processes occur on time scales that are not accessible to the laboratory. Geological processes occur at strain rates 10^{-12} s^{-1} – 10^{-16} s^{-1} , several orders of magnitude lower than rates that can be achieved in the laboratory. Hence, there is a gap in multiscale modelling from the above described crystal-scale to macroscopic rock deformation. It is common practice to extrapolate creep laws obtained from rock specimens deformed in the laboratory to geological conditions. The homogenisation step is unfortunately already done through the laboratory experiment, and a suitably large volume is sampled where lattice anisotropies are assumed to cancel each other out. The flow laws are often extracted from end-member mechanical constituents such as quartz, feldspar, pyroxene or olivine and, because of the above described problems in the simple case of ice deformation, no attempt is made to upscale a multiphase material flow law.

The question whether the deformation mechanisms derived from laboratory experiments are at all applicable for upscaling to the large time scales in nature,⁽⁵⁾ or whether other mechanisms or combination of mechanisms prevail, provides ample food for discussion between experimental, field and modelling geoscience disciplines. Deformation mechanisms inferred from natural shear zones evidently include a multitude of processes, which may incorporate a complicated interplay of brittle and ductile material response at different scales.⁽¹⁰⁾ It is therefore not astonishing that Geosciences is not yet ready to fully embrace modern concepts of multiscaling from material sciences because important elements for upscaling appear to be missing.

Geologists report a number of material instabilities that are encountered between the deformation of a crystalline aggregate and the deformation of an entire lithospheric plate. These feature (faults, folds and compaction bands) were introduced briefly in the theory review.⁽²⁾ Although there appears to be a certain similarity in the geometry of instabilities across scales⁽¹¹⁾ it is important to understand whether their manifestations at various scales have mechanical consequence for the overall behaviour of the next larger scale and whether they have a common self-similarity or whether different physics operates at different scales.^(12,13) The main problem that needs to be solved is

hence the question: what controls material bifurcations at multiple scales?

3. DISSIPATIVE STRUCTURES EMERGING FROM MATERIAL BIFURCATIONS ACROSS SCALES

In this section we illustrate the concept of coupling multiphysics processes with their respective length and time scales. In the theory review, we have illustrated the characteristic length scale that results from the error-function solution of the respective 1-D diffusion process of the dissipative mechanism. For a given time scale, we expect to identify the dominant diffusion process that is operating at a given length scale and thereby hope to be able to separate out the governing THMC dissipation mechanism that gives rise to the observed dissipative pattern through its intrinsic energy-feedback process. We test this hypothesis through explicit modelling of the processes, incorporating their various feedback mechanism in the energy equation. First, recall the energy equation of the theory review Eq. (27) in Ref. [2].

$$\frac{D^{(m)}T}{Dt} = D_T \frac{\partial^2 T}{\partial x_k^2} + \frac{\delta_{\text{loc}}}{(\rho C)_m} \pm \frac{r_k}{(\rho C)_m} \pm \rho_m T \frac{\partial^2 \psi}{\partial T \partial \xi_i} \dot{\xi}_i \quad (1)$$

where D_T is thermal diffusivity of the poromechanical mixture, $(\rho C)_m$ is the density of the solid and fluid mixture multiplied by the specific heat capacity $C_m = -T(\partial^2 \psi / \partial T^2)$ of the mixture, ψ is the Helmholtz free energy and ξ_i stands for the state variables. The source/sink term r_k represents volumetric heat production due to chemical reactions or other sources such as electric currents (Joule heating) or radioactive decay. These processes do not generally affect the local mechanical dissipation δ_{loc} , and they can be added as independent heat source/sink terms.

The last term in Eq. (1) opens an alternative, more complete way of calculating the sources and sink terms and has not been discussed in Ref. [2]. In the case where the source/sink terms are tightly coupled to the mechanical deformation, it is impossible to detach local mechanical dissipation from the source/sink terms. In such cases the last term replaces the r_k source/sink terms. If we consider for instance the elastic strain $\xi = \epsilon^e$ as a state variable, then $\rho_m T (\partial^2 \psi / (\partial T \partial \epsilon^e)) \dot{\epsilon}^e$ expresses the thermal-elastic heating effect (negative in contraction and positive in dilation). The stored energy is, however, no longer available for the local dissipation and needs to be subtracted from the local dissipation δ_{loc} . If the state variable stands for a phase $\rho_m T (\partial^2 \psi / (\partial T \partial \xi)) \dot{\xi}$ represents the latent heat release during the phase transition (positive upon heat release and negative while absorbing heat). The latent heat effect of the phase transition is, however, again accompanied by a mechanical contraction/dilation, which needs to be considered in the shear-heating term δ_{loc} (Eq. (21) in

Ref. [2]), which reads:

$$\delta_{\text{loc}} = \sigma_{ij} \dot{\epsilon}_{ij}^p - \frac{\partial \psi}{\partial \xi} \dot{\xi} \geq 0 \quad (2)$$

where the second term describes the above mentioned mechanical feedback, more generally speaking, the power that is stored in the microstructure, which is therefore not available for shear heating.

If we consider shear-heating feedback, it follows from the energy equation Eq. (1) that the length scale for a TM dissipative pattern for thermal-mechanical feedback is defined by the energy equation being in equilibrium with the steady-state shear-heating term δ_{loc} and the thermal diffusion term $D_T(\partial^2 T / \partial x_k^2)$, if all other terms are in steady state. Similarly, but less obviously, if we consider a chemical reaction triggered by the shear-heating term, the energy equation, in conjunction with the fully coupled equation for chemical diffusion, delivers an equilibrium TCM chemical dissipative pattern, with a length scale defined by the chemical diffusivity. Likewise, if we consider a poromechanics problem, where fluid flow triggers a dissipative pattern, the emerging dissipative length scale for a THM pattern can be described by the pore pressure diffusion equation.

This simple logic allows a potential separation of scales of geomaterial instabilities, which is illustrated in the following series of field examples. The coupled feedback between the different THMC mechanisms potentially leads to the emergence of new effective diffusivities altering the dominant mechanism at a given scale. We illustrate in the following only the basic length scales and start with a chemical dissipative pattern. Chemical diffusivities are the smallest of all the THMC processes. They range from $D_c = 10^{-19}$ to $10^{-15} \text{ m}^2 \text{ s}^{-1}$. Considering a characteristic time of $t = 10^{12} \text{ s}$ for a fast geological process in the ductile regime, we would expect from the error function solution of the diffusion equation $L = 2\sqrt{D_c t}$ a length scale of chemical feedback processes of 0.6–60 mm. A chemo-mechanical simulation for contraction of a strong feldspar-rich granitoid layer embedded in a soft quartzite matrix is shown in Figure 2.

The numerical TCM solution produces indeed the style of observed folding pattern at the cm-scale, and it is therefore likely to be caused by the chemical feedback processes that lead to localised deformation in and around the area of the stiff layer. However, we have also highlighted in Ref. [2] that the internal microstructure can cause localisation phenomena such as shear bands at the same scale because, in a granular medium, shear band width is of the order of 15 times the mean grain size⁽¹⁵⁾ as indeed shown in Figure 3.

Brittle deformation mechanism as shown in Figure 3 are fast processes that happen on time scales of less than 10^2 s . Classical modelling of brittle faulting uses Mohr-Coulomb or Drucker-Prager yield envelopes. When implemented in finite-element models, these continuum models

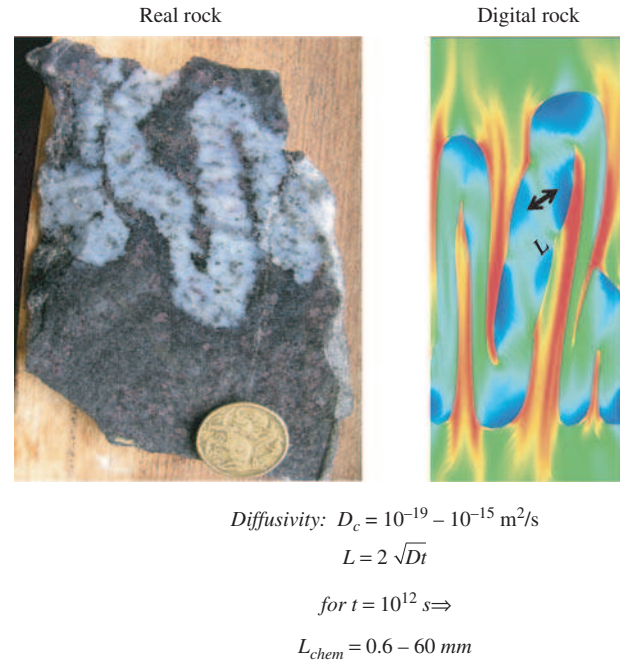


Fig. 2. Compression of a strong layer in a soft matrix with chemical decomposition (mica breakdown) and chemical diffusion (water)⁽¹⁴⁾ reproduces the style of tight folding observed in a gold-bearing vein of a hand specimen. The digital rock image is shown at the same scale as the real rock (two Australian dollar coin for scale). Contours of effective strain illustrate the role of a shear band of width L in the formation of the tight fold. The early stages of folding show a similar pattern as the one shown in Figure 7 which clearly illustrates the role of shear bands in the folding process. Peter Schaub is thanked for supplying the photo of the rock sample.

have notorious mesh dependence because they neglect the intrinsic length scale of faulting, which can be defined either through the enriched Cosserat continuum or through the diffusive length scale that emerges out of the above described energy feedbacks. Let us consider for instance hydro-mechanical coupling. Typical pore pressure diffusion of the coupled THM process has diffusivities of the order of $10^{-5} - 10^{-1} \text{ m}^2 \text{ s}^{-1}$. Therefore, from the diffusion

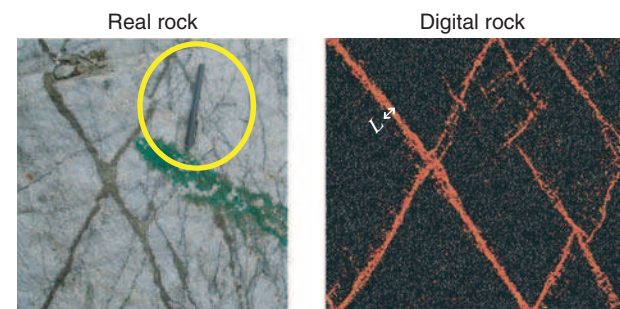
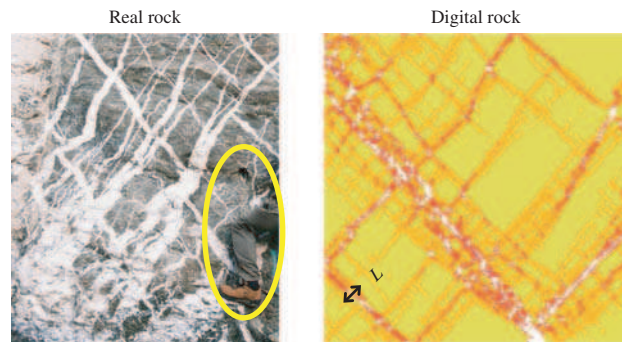


Fig. 3. Compression of a quartzite in the brittle regime calculated with a discrete element code.⁽¹⁶⁾ The natural example shows a copper vein in the Shi-Lu copper deposit Guangdong, China. The width of the shear band is as expected by Ref. [15]. Yanhua Zhang is thanked for providing the example.



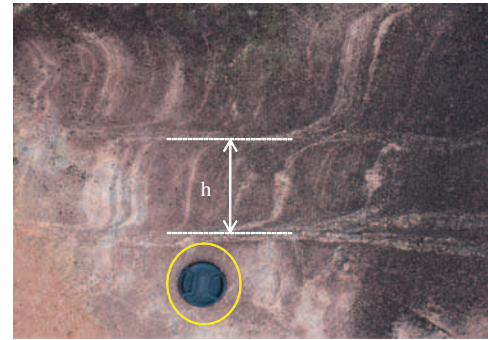
$$\begin{aligned} \text{Diffusivity: } D_H &= 10^{-5} - 10^{-1} \text{ m}^2/\text{s} \\ L &= 2 \sqrt{D_H t} \\ \text{for } t &= 10^2 \text{ s} \Rightarrow \\ L_{\text{Hydro}} &= 0.06-6 \text{ m} \end{aligned}$$

Fig. 4. Brittle fractures with fluid precipitation (calcite) at m-scale (leg for scale). The features can be reproduced by a Drucker-Prager solution enriched by an explicit modelling of fluid pressure. The colour scale illustrates the total strain rate dominated by plastic deformation at 10^{-10} s in the red portion while the yellow part is deforming elastically at strain rates lower than 10^{-16} s.

equation one would expect shear bands with widths of the order of 0.06–6 m if the localisation phenomenon is controlled by the fluid pressure diffusion and the time is 10^2 s. An example solution is shown in Figure 4.

At high temperatures the role of crustal fluids can be replaced by partial melts. The THM feedback is then characterised by viscous creep rather than brittle faulting. Yet again, dissipative patterns emerge the characteristic distance of which can be described by a diffusional length scale. We show in Figure 5 a case of compaction bands in a partially molten rock. The compaction length, $\delta_c = \sqrt{(k_\pi \mu_s)/\mu_f}$ where μ_s is the viscosity of the solid matrix, μ_f is the viscosity of the pore fluid/melt, and k_π is the permeability of the system. The compaction length^(17,18) defines the width of a boundary layer in which compaction of the matrix occurs. In Figure 5 we show an example of a compaction band in a layered partially molten lower crustal rock. A typical solid viscosity is 10^{18} Pa s and that of melt is 10^5 – 10^{10} Pa s. We can invert the compaction length for the unknown lower crustal permeability. From the tens of cm spacing as in Figure 5, we obtain a range of effective permeability of the melt between 0.001–100 Darcy, which would be sufficient to render compaction bands an important ingredient in the segregation of melts. Although this length scale is well known in melt physics,^(17,18) its full implication for geological field observations requires an understanding of the fundamental multi-physics of the instability.⁽¹⁹⁾ This example makes a clear case for the need of combining an understanding of solid mechanical and fluid dynamic instabilities, which will be discussed in depth in the applied case studies.

The next scale up is defined by the thermal-mechanical TM feedback processes, which, because of their simplicity



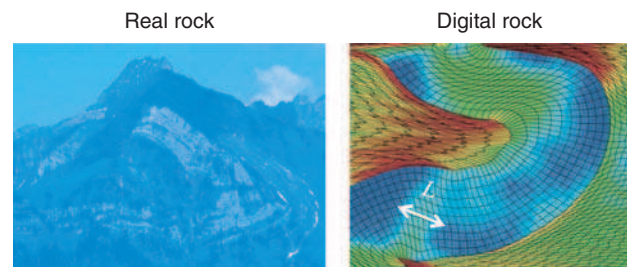
$$h \approx \sqrt{\frac{\mu_s k_\pi}{\mu_f}}$$

Fig. 5. Deep in the crust, rock temperatures reach the point of partial melting where intense deformation forms a rock known as a migmatite. Here, we show a migmatite with layered whitish bands of crystallised melt of different generations. The faintly visible tight vertical banding is part of the gneissic texture of the original rock formation. The two distinct whitish horizontal bands are compaction bands.⁽²⁰⁾ The distance between the bands is proportional to the compaction length h proposed by Refs. [17, 18]. Note that this instability is described by a fluid dynamic approach enriching the classical solid mechanical failure modes. Camera lens cap for scale. We thank Roberto Weinberg for the photo.

and obvious consequence of Eq. (1), was the first fundamental time-dependent length scale identified in geomaterials.⁽²¹⁾ Thermal diffusivities of rocks are of the order of $D_T = 10^{-6} \text{ s}^{-1}$, and time scales of ductile deformation are of the order of 10^{12} s. The feedback is expected to occur on the km-scale. An example is shown in Figure 6.

Thermal-mechanical dissipative patterns in geology often appear as folds or as shear bands or as combinations thereof. Examples, where shear bands contribute to the formation and appearance of folds, are shown in Figure 7.

Chemical TCM dissipative patterns have a very similar appearance to both thermal-mechanical TM patterns and thermo-hydro-mechanical THM patterns. In geological



$$\begin{aligned} \text{Diffusivity: } D_T &= 10^{-6} \text{ m}^2/\text{s} \\ L &= 2 \sqrt{D_T t} \\ \text{for } t &= 10^{12} \text{ s} \Rightarrow \\ L_{\text{thermal}} &= 1 \text{ km} \end{aligned}$$

Fig. 6. Km-scale fault in a carbonate in the Swiss Alps compared to a TM-feedback simulation.⁽²²⁾ The colour scale and the deformed reference mesh illustrate the heterogeneous strain.

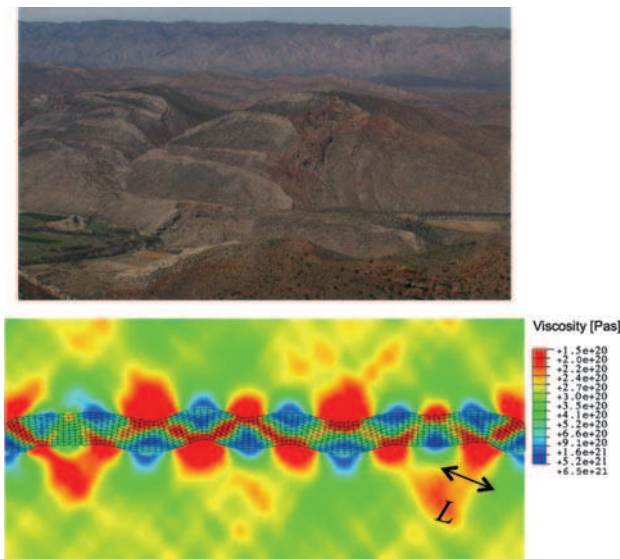


Fig. 7. Folds in the Cape Fold Belt South Africa with prominent hinges in the left part of the figure compared to numerical solutions of folds.⁽²³⁾ The formation of shear bands in and around the stiff bands is illustrated by low viscosity values. These low viscosity channels have a significant influence on the wavelength and formation of the fold.

terms, they all appear as folds, ductile or brittle faults, micro-shear bands, or compaction bands. This is not astonishing because their basic physical behaviour can be described with similar partial differential equations. The main difference is that they are scaled by different diffusivities and therefore occur on different length scales. Other than the vast separation of length scales and the possible cross-scale couplings all of the THMC feedback processes exhibit similar geometric features which makes the identification of the THMC mechanism responsible for their formation difficult in the field. It is therefore not astonishing that in geosciences only the shear heating feedback mechanism, which has first been identified in the 60's⁽²⁴⁾ appears to be more widely accepted with some reservations owing to the lack of direct geological evidence of the heat. It is fair to say that the other mechanism are still hotly debated.⁽²⁵⁾

Geologist therefore often adopt a long-wavelength view and disregard the above described solid-mechanical instabilities. This follows from the widely accepted *lex parsimoniae* (Occam's razor),⁽²⁶⁾ which calls for choosing the simplest theory until simplicity can be traded for greater explanatory power. However, we point out that if data driven perspective is used as the only point of reference to interpret the physics of a model and a physically wrong model fits the data, then Occam's razor is not going to discriminate against it. An example is a data driven or inverse problem where the complexity of the model is regulated by the data. If one can explain all geological observations given noise on them with a one parameter model then there is no need to go to a more complex model. In this sense what determines the amount of explanatory power required

in a model is the data we are trying to fit. It is therefore not astonishing that currently physics-based dissipative pattern that can be derived from the convection of the planetary interior occurring on hundreds of millions of year time scale are widely adopted in Geodynamics. An additional advantage is that the fundamental mode of heat transfer of planets can be assessed by comparing terrestrial planets such as Earth, Venus and Mars which are each in different stages of evolution. The earth has life-sustaining plate tectonics, Venus appears to have a surface that has resurfaced multiple times, and Mars appears to be in a stagnant lid regime where the planetary surface no longer partakes in the convection of the planetary interior.⁽²⁷⁾ The dissipative patterns that arise from these long time-scale phenomena are distinctly different to those described above. An example is shown in Figure 8.

This summary of models of material bifurcations across scales illustrates the two competing concepts currently used in Geosciences, which have both shown promise to extend predictability beyond engineering time scales with constitutive relationships derived from the laboratory. One promising approach is to use the fundamental thermodynamics concept of heat transfer in planets and to understand patterns on a global scale using the theory

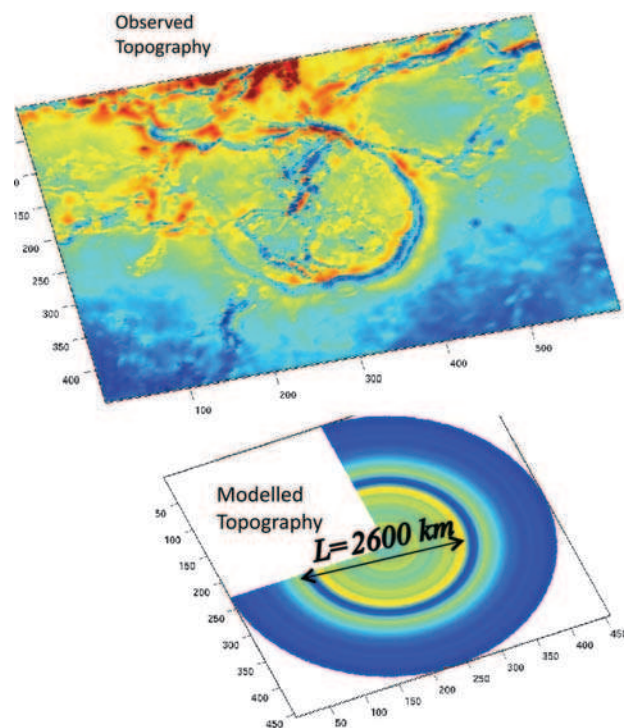


Fig. 8. Artemis Corona on Venus is a 2600 km wide circular feature with a centrally elevated domain surrounded by a deep depression followed by another high. The topography can be modelled by assuming a plume that penetrated the planetary surface and causes circular subduction of the cold rims into the planetary interior. The plume flux explains the centrally elevated platform while the subduction process gives rise to both a flexural bulge on the outside of the Corona as well as the deep trough under the leading edge of subduction.⁽²⁹⁾

of fluid dynamics. The other emerging research hot spot entails explicit calculations of THMC feedbacks implemented in a solid mechanics framework to explain Earth's interesting patterns at multiple scales. A third exciting path is to use microstructural simulations derived from phase-field and force-field models with the purpose of upscaling enriched continuum models that correctly identify the role of microstructure in long time scale behaviour of the planet.

Clearly, the challenge in earth sciences is to explicitly consider thermodynamics in formulating solid-mechanical models of the earth and use these in conjunction with modern upscaling concepts that can encompass the enormous span of multi-physics and multiscale coupling. A robust theoretical framework has been laid out in the theory section.⁽²⁾ The problem of bridging the theories of fluid dynamics and solid mechanics⁽²⁸⁾ is another major challenge. We have only alluded to this problem in the theory section⁽²⁾ through the identification of diffusive and convective length scales that require a formulation for explicit coupling of fluids and solid dynamics. In the sections to come we provide specific examples for the concepts laid out in the theory section and extend the approach to bring solid and fluid modelling concepts together.

4. TOWARDS MODELLING OF GEOMATERIALS ACROSS SCALES

The above described examples have illustrated the main challenges of multiscale modelling of geomaterials. We summarise the main points:

- (a) True *ab-initio* based multiscale modelling is needed in earth sciences but it is often considered beyond the reach of current computational resources. Semi-empirical statistical mechanics based formulations Ref. [4] potentially coupled with a meso-scale thermodynamic variational approach⁽⁶⁾ provide an exciting path for the future.
- (b) A significant drawback of all modelling in earth sciences is that the processes occur on time scales that are not accessible to the laboratory.
- (c) Deformation mechanisms inferred from natural structures are themselves multiscale and multi-physics processes and can incorporate brittle and ductile material response at different scales.
- (d) Geological bifurcations form dissipative structures such as folds, faults, micro-shear bands, and compaction bands, which have similar geometric features but are presumably formed through THMC feedbacks with distinct time-space relations.
- (e) Classical solid-mechanical modelling of bifurcations in Geosciences use quasistatic continuum mechanics models, which have notorious mesh dependence, and being scale-invariant, do not reproduce the observed time-space relations of dissipative structures.
- (f) Classical fluid-dynamic models of the earth have the advantage of being self-consistently driven by heat but can only explain features observed on hundreds to thousands

km in length scale, operating on time scales of hundreds to thousand of million years.

Consequently, earth sciences suffer from a battle on two fronts. The geodynamic-modelling community prefers a fluid-dynamics view, while the structural-geology and the seismology-modelling community prefer the solid-mechanical modelling approach. The obvious next step is to move on from these simple, yet elegant theories, to a more complete geomaterial modelling approach combining both theories and providing the missing link between geomaterial observations at multiple scales. Reference [2] introduced a microstructure homogenisation workflow for upscaling and an irreversible-thermodynamics approach for downscaling. In the following section, we will illustrate this workflow.

5. MICROSTRUCTURE HOMOGENISATION WORKFLOW

At the microscale, geomaterials generally show significant heterogeneity because they are made of components with different material properties and geometries. Microtomography permits the observation of the 3D internal structure of rocks on micro- to nano-scales.⁽³⁰⁾ It opens a new way to quantify the relationship between the microstructure of rocks and their mechanical and transport properties.

Reference [31] first gave computations of linear elastic properties from microtomographic images. Their results show good agreement with experimental data. Reference [32] also computed the linear elastic properties of random porous materials with a wide variety of microstructure. Reference [33] analysed the correlation of properties such as diffusivity, elasticity, permeability and conductivity of three-dimensional digitized images of real cellular solids. Some more examples of studying the elastic response of rocks using microtomography include Refs. [34–39].

There are two implicit assumptions in the above-mentioned studies. The first assumption is that the analysed volume sizes are Representative Volume Elements (RVEs). Since this assumption is often not verified it might cause significant variance in the results of analyses, and different volumes might give different results. The second assumption is that there is no difference in deriving mechanical properties for conservative and non-conservative thermodynamic forces. This is a crude simplification as multiscale, multi-physics geomaterials often have vastly different thermodynamic properties depending on whether they are in equilibrium or far from equilibrium.

Recent work has removed these over-restrictive assumptions, and a comprehensive workflow for the analysis of digital X-ray CT of earth materials^(40,41) see Figure 9 has been developed. The workflow extends the classical X-ray CT workflows through newly developed high-performance computational analysis of (time-lapse) microstructural analysis based on percolation theory. Percolation theory

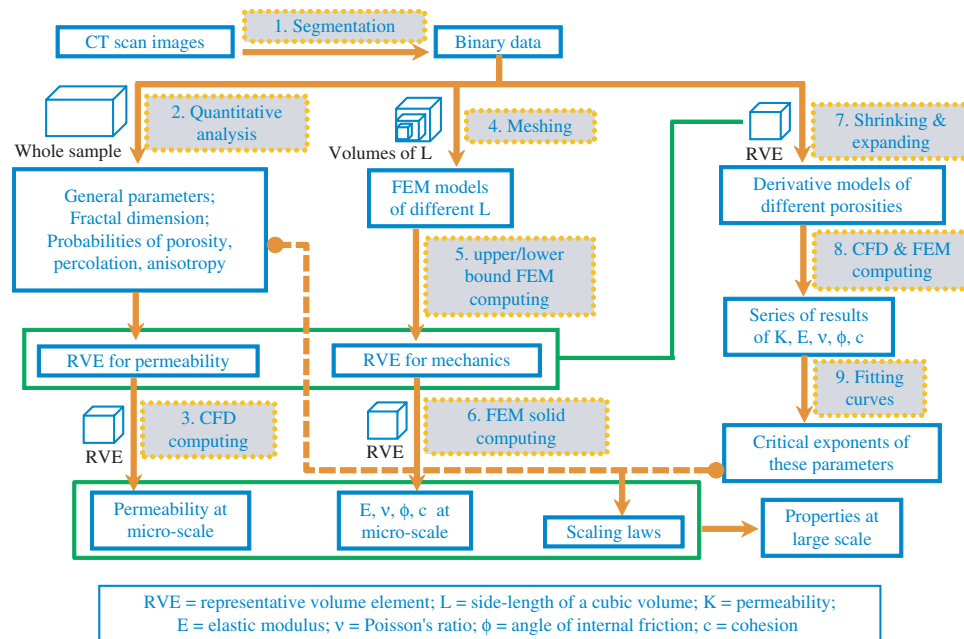


Fig. 9. Workflow for micro-characterization of rock specimens. The first column illustrates standard statistical analyses based on percolation theory, cluster analysis and fluid flow consideration; the second column shows the equivalent solid matrix analysis; the third column shows the scaling analysis for either solid or fluid networks.

allows derivation of (time-lapse) microstructural characteristics for networks of fluid and solid matrix as well a well defined output of statistical microstructural properties including an identification of the minimum size of the analysis volume to extract statistically relevant average material properties.

The workflow is using standard procedures well established over the last twenty years with some new additions. In the following we will focus our review only on the new elements and refer to the literature for in-depth reading. The first step of the workflow is X-ray microtomography, an imaging technique that provides three-dimensional (3D) information on the internal structure of materials (e.g., Ref. [30]). It is increasingly applied to image and quantify the porosity distribution in rocks (e.g., Ref. [42]). A microtomographic dataset of the three-dimensional structure of a sample is reconstructed from rasterised radiographic projections that record the attenuation of X-rays in the sample.

Microtomographic imaging is followed by step 1. “Segmentation” in Figure 9. For segmentation the attenuation pattern is mapped on a Cartesian grid of voxels (volume pixels). Phases with a distinct attenuation, such as pores, can easily be segmented from the patterns (e.g., Ref. [43]) and the resulting binary voxel distributions form an ideal 3D cubic lattice model for percolation theory. The objective of the segmentation process is to identify a target phase. If the target phase is a solid matrix or a multiphase fluid sample the segmentation and statistical analysis has to be repeated for each target phase in order to derive the load bearing framework, the polymineralic network or the multiphase fluid distribution.

Segmentation provides input for all microstructural analyses based on statistical methods such as percolation theory and similar. We focus in this review on a short presentation of the statistical network in terms of percolation theory. This allows time lapse microstructural characterisation of percolation networks for fluid and solid matrix and well defined output of statistical microstructural properties including an identification of the minimum size of the analysis volume to extract statistically relevant average material properties.

Segmentation is followed by step 2. “Quantitative Analysis” in Figure 9 consisting of the following components: (a) Percolation theory and Hoshen-Kopelman algorithm are used to define clusters and percolation of the target phase. A cluster is a group of sites of the same phase that connected each other and separated from other groups. The mathematical definition of percolation refers to the nature of the connectivity in lattice models. A model is defined as percolating when a cluster reaches the opposite boundaries of the model. Based on the concept of cluster of percolation theory, all individual objects of the target phase are identified in the model.

(b) Multi-parameter output, including volume fraction, specific surface area, particle size distribution, percolation, position, size and anisotropy of each cluster. All parameters listed in the above can be calculated in our codes. These parameters give detailed description of the complicate microstructure.

(c) Stochastic analysis is used to derive the probabilities of distribution of main parameters in the model and their scale-dependent features. The size of the representative

volume element (RVE) can be determined if the probabilities are convergent. RVE's are here defined as statistically representative volumes containing a sufficiently large set of microstructure elements such that their influence on the average macroscopic property (porosity, elasticity, permeability, etc.) has converged. Simulations of properties based on the RVE are deemed to be reliable as they encapsulate the intrinsic material heterogeneity.

Presently a normal micro-CT dataset is 2048³. Volumes of 4096³ are now available and it is expected to be 8192³ in the near future. With high performance computation techniques of parallelisation and data decomposition, we can deal with extremely large datasets.

For the statistical characterisation of the percolation network computational techniques are used as inputs for homogenisation of the fluid and solid material properties. If the target phase is a pore space the so characterised percolation network lends itself for a computational fluid dynamic computation (step 3. "CFD computing" in Fig. 9) with the purpose of delivering a permeability value of the sample. A variety of numerical methods are used for this step such as finite difference, finite element, smooth particle hydrodynamics and Lattice-Boltzmann techniques. For finite element analysis standard preprocessing in step 4. "Meshing" of Figure 9 is required.

The steps 5 and 6 in Figure 9 rely on computational homogenisation techniques generalised for geological systems. They are based on well established concepts for multiscale convergence of microstructure in material sciences⁽⁴⁴⁾ with the main difference being that geological processes need to consider additional multiscale and multiphysics couplings on time scales that are inaccessible to the laboratory verification. This adaptation to geological systems is mainly enabled through the application of percolation theory introduced in steps 7–9 of Figure 9.

5.1. Percolation Theory in Real Space

Since its establishment in mathematical physics in the 1950s, percolation theory⁽⁴⁵⁾ has been applied to earth science for investigating various phenomena in heterogeneous media, including various flow phenomena in porous media,⁽⁴⁶⁾ transport, reaction, diffusion and mineral precipitation, distribution of earthquakes, and mechanical properties.⁽⁴⁷⁾ In this section we focus on the application of percolation theory combined with X-ray microtomography.

Fluid flow through pores and/or cracks in solid media is one of the most important topics in earth sciences, with important applications to energy industries and mineral deposits. The prerequisite of fluid flow is the connectivity of voids. Percolation theory describes the global connectivity of models and thus can be used to analyse microtomographic datasets of porous materials. Reference [48] first used the local porosity theory⁽⁴⁹⁾ to analyse 3D microtomographic data and compared the pore space geometry of different samples quantitatively, where

the local percolation distribution is the probability of pore-connectivity of sub-volumes. Reference [50] developed a program for the analysis of pore connectivity and anisotropic tortuosity of porous rocks. Reference [51], studied the evolution of porosity and hydraulic diffusivity during weathering of basalt. Reference [40] extended the local porosity theory to consider anisotropic permeability in percolating sub-volumes.

The key parameter in percolation theory is the percolation threshold, which describes the (minimum) porosity of a connected network of pores.⁽⁴⁵⁾ The percolation threshold can be derived from a percolation cluster analysis in which a group of face-connected cells forms a cluster. Labelling clusters that belong to a target phase in a segmented microtomography dataset is the process whereby all voxels (i.e., cells) in a cluster are given a unique label. After cluster labelling, each individual structure of the target phase is resolved, including the position, size, and orientation. The percolation threshold can then be determined by analysing percolation in a series of datasets with different volumes as laid out in the theory review.⁽²⁾ We review here the first applications of percolation theory. In a first attempt to analyse realistic natural geometries with percolation theory, Ref. [52] studied the geometrical percolation threshold of porous media by assuming grains to be overlapping ellipsoids. The authors found the percolation threshold to be ranging from 0.06% to 28.5% for different aspect ratios and ellipsoid sizes.

Reference [53] created virtual permeable microstructural models of hard-core-soft-shell grains and obtained a percolation threshold of 4% for concrete-like porous media. These mathematical models are limited to ideal or special structures. Reference [51] collected a suite of weathered basalt samples with porosities between 3% and 30% and found that 9% is the percolation threshold of the specific weathered basalt. Reference [54] tested cumulates of sea ice single crystals at different temperatures and determined percolation thresholds of $(4.6 \pm 0.7)\%$, $(9 \pm 2)\%$, and $(14 \pm 4)\%$ in different directions. For most natural samples, a series of models with different porosity and similar structure is not available. Reference [41] proposed a new method for determining the percolation threshold of natural structures by generating a virtual series of digital samples from a single dataset. By volumetrically shrinking or expanding the pore-structure of the static images, a series of models with similar structures but different porosities are created. The percolation threshold can be determined for each individual structure.

5.1.1. Applied Case Study: Critical Percolation Phenomenon in a Granitic Shear Zone in Australia

This analysis was applied to assess the role of micropores in the deformation of a granite at the base of the seismogenic zone in the continental lithosphere.⁽⁵⁵⁾ A deformed

granite from Central Australia that has been subject to 400–500 °C environment was analysed. The granite is now exposed to the surface so that the shear zone can be sampled and analysed using Synchrotron-based X-ray microtomography (Fig. 10). Microtomographic data was used to describe the porosity distribution across the shear zone and interpreted in combination with a classical microstructural study of the evolution of the strained rock.

The analysis with the percolation theoretical approach supported three important discoveries:⁽⁵⁵⁾

- (1) Porosity evolves with progressive deformation, and different mechanisms contribute to the overall porosity at different stages of the microstructural evolution. In the most deformed rock samples with the finest grain sizes, grain boundary pores, formed by creep cavitation, dominate the porosity architecture;
- (2) the maximum porosity in the centre is close to but just below the percolation threshold of 5%;
- (3) Minerals precipitated in the pores evidence synkinematic fluid migration and the redistribution of chemical components on a length scale significantly exceeding the dimensions of pores and minerals (see Fig. 3 in Ref. [56]).

These observations contribute to explaining fluid flow in interseismic periods at the base of the seismic zone. At the highest strains creep cavities self-organise in ductile shear bands, analogues to ductile failure phenomena in metals and ceramics,^(57–59) and promote fluid transfer over distances significantly larger than individual grain diameters. This behaviour can be best modelled using a damage mechanics approach, and a suitable formulation will be reviewed in the constitutive modelling section. In the next section we will show that detection of the percolation

threshold plays an important role in extracting critical exponents and scaling laws for upscaling.

We analyse percolating porosity clusters in the geological hand specimen described in Ref. [55]. Figure 10 shows the specimen together with the locations and labels of subsamples scanned by Synchrotron X-ray microtomography. Note that the hand specimen covers a strain gradient into a highly deformed shear zone. In the centre of the shear zone, the minerals are much more finely grained than at the perimeter, which is an effect of deformation. The fractal dimension d of the pore size distribution in the subsamples can be calculated from the segmented microtomographic data using percolation cluster analysis. The results are listed behind the label in Figure 10. It appears that a higher fractal dimension exists in the centre than at the margin of the shear zone. According to the energy scaling law⁽⁶⁰⁾ the dissipated energy of the fragmentation process of a solid is proportional to its volume to the power of $d/3$. That implies that the higher the fractal dimension the higher is the dissipated energy. This result demonstrates the link between the fractal dimension, energy dissipation, and deformation. Although it is not a surprising result that the centre of the shear zone exhibits the highest energy dissipation this analysis gives the geologist a quantitative tool to analyse and describe the dissipation. The quantification of the dissipated power is an important diagnostic tool to compare model predictions from constitutive modelling with natural samples.

Other quantities that are of interest and can be extracted from microtomographic data analysis are the anisotropy of permeability in a sample, which is related to the anisotropy of the geometry of percolating cluster. The anisotropy of a cluster is described by an orientation tensor.^(2,40,61) Still more information can be revealed from the cluster analysis, such as how much pore space is connected, whether non-percolating clusters are oriented or what their specific shapes are. The fractal dimension can also be calculated by additional methods showing self-consistent and scale-independent characteristics. Additional information derived from the above rock sample is for instance the size distribution of pores and the volume percentage of different cluster-sizes (Fig. 11). This analysis reveals a bimodal distribution of pore sizes which is again a result of the dynamic recrystallisation in the centre of the granitic shear zone.

5.2. Renormalisation Using Percolating Theory

Renormalisation is achieved by using the stochastic analysis of the moving window method, and probabilities of porosity, percolation and anisotropy are derived. Reference [40] illustrates the use of percolation theory for the derivation of RVEs. For better comparison with mathematically constructed reference models a synthetic sandstone sample was selected. The excellent agreement between mathematically predicted and numerically estimated values, derived by applying the X-ray CT workflow

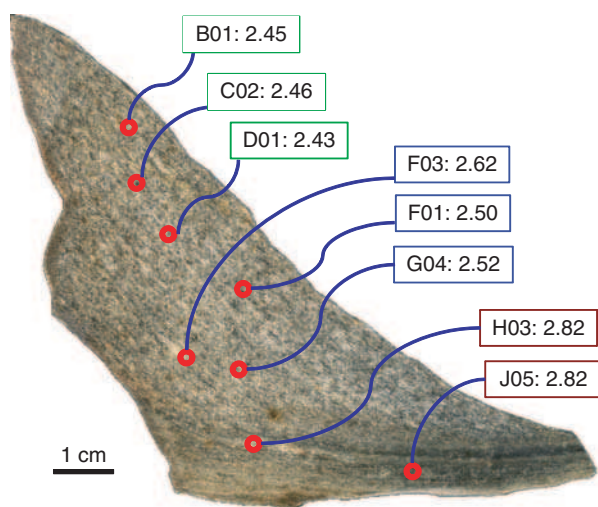


Fig. 10. A hand specimen of a mylonitic shear zone and the fractal dimension obtained through X-ray CT analysis of micro-cores subsamples (B01–J05), locations shown in red. The fractal dimension is increasing towards the centre of the shear zone where the smallest grain size is observed.

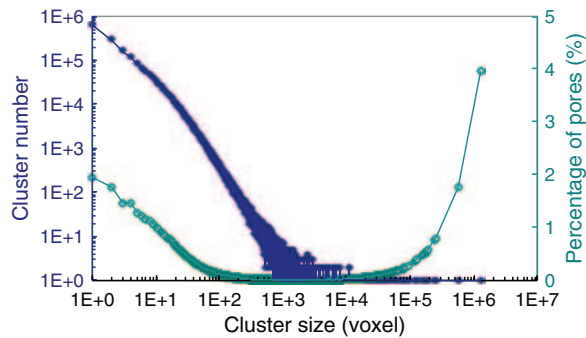


Fig. 11. Histogram of cluster size and the volume percentage of different cluster-sizes.

on the synthetic sample, gave credence to the technique. In this ideal example of renormalisation, probabilities of porosity, percolation, isotropy index and elongation index are all convergent when the sub-volume-size is larger than 400^3 voxels or 1 mm^3 . Thus the RVE size of geometry of pore-structure was determined.

The derivation of the critical exponent of correlation length was investigated in subsequent work using the same synthetic sample.⁽⁴¹⁾ Probabilities of percolation of different sizes and finite-size scaling scheme were used to derive the critical exponent. The critical exponent of correlation length was found to be 0.885, which is very close to the theoretical expected result of 0.88.⁽⁴⁵⁾ Combined with other scaling parameters, such as the percolation threshold, crossover length, and the fractal dimension, the scaling laws of the sample were identified and the permeability at microscale was proven to be usable for large scale directly, without rescaling.

5.3. Asymptotic Homogenisation

The asymptotic computational homogenization method postulates that through a stepwise increase in microstructural cell size the apparent material property can be derived asymptotically.⁽⁴⁴⁾ In such a method the material property such as Young's modulus is found to be consistently

overestimated (stronger) for a displacement boundary condition (constant thermodynamic flux) while the property is consistently underestimated through the choice of a traction boundary condition (constant thermodynamic force).⁽²⁾ The solution of such a boundary value problems delivers a lower bound for a constant force boundary while the constant displacement boundary problem gives an upper bound of the work done. When the volume is sufficiently large the two bounds converge.⁽⁶²⁾ Thus the mechanical RVE size is identified. The limit theorems were initially formulated for specific dissipative elastoplastic systems but they have been shown to be applicable to generalised thermodynamic forces and fluxes.^(2,63) The generic computational asymptotic homogenisation procedure is illustrated in Figure 12. In the following, applications of asymptotic homogenisation for conservative force and non-conservative force are given.

5.3.1. Asymptotic Homogenisation for Conservative Forces

Since the critical percolation threshold corresponds to a scale-dependent phase-change in the physics of the investigated process, the renormalisation procedure of percolation theory allows a robust assessment of fundamental changes in material parameters and their scaling relationships before and after the critical point. Some parameters are changing exponentially when the volume fraction is approaching the percolation threshold. These parameters include permeability, elastic modulus, yield stress, and more.

We first consider the simple case of non-conservative thermodynamic forces, such as the linear elastic response of microstructures for a carbonate sample from an oil field with a heterogeneous microstructure is shown in Figures 13 and 14. The classical asymptotic homogenisation procedure of material sciences should be applicable to this class of problems.⁽⁴⁴⁾ Note that this consideration is not common practice in geosciences. Much simpler methods such as the Gassmann equation are the current state of the art in geophysics.^(64,65) For sand

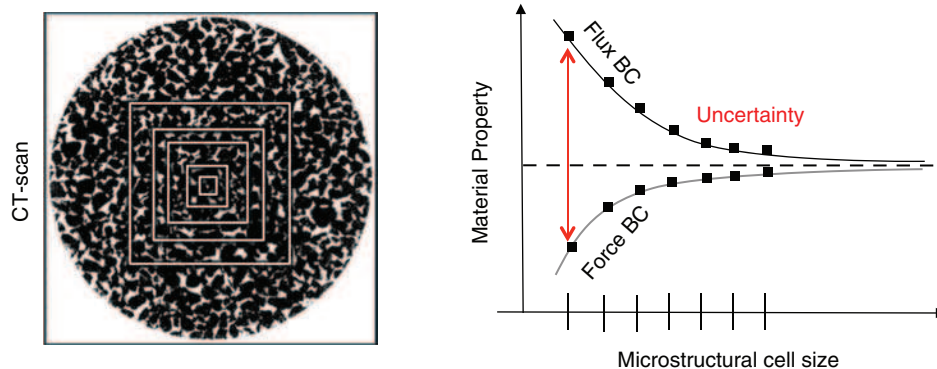


Fig. 12. Thermodynamic homogenisation procedure considering combinations of constant thermodynamic force and constant thermodynamic flux. Modified diagram inspired by a plenary lecture by Robert L. Taylor, University of Berkeley, California, on Computational Mechanics Today 2008.

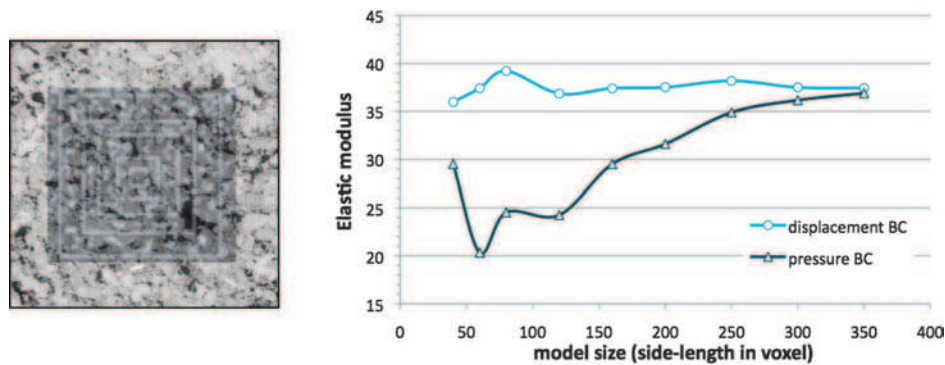


Fig. 13. A carbonate sample and the homogenisation of upper and lower bounds of elastic modulus. The bulk porosity of the sample is 26%. Image resolution is 1.85 micron. Input solid elastic modulus is 50 GPa and Poissons ratio is 0.2.

shale reservoirs Gassmann has proven its value. However, for carbonates, which contain more than 60% of the worlds oil reservoirs,⁽⁶⁴⁾ one has yet to find a similarly successful technique. Currently, elastic properties used for the detection of these reservoirs through seismic methods still use Gassmann formulation⁽⁶⁶⁾ in full acknowledgments that carbonates are notoriously difficult to describe via homogenisation methods. The main problem is the multiscale heterogeneity of carbonates at micro- and macro-level. At micro-level the grain contacts and inclusions cause heterogeneous mm-scale microstructure and at macro-level vugs and solution cavities yield cm-scale heterogeneity. We propose here to use the above described workflow in two different stages. In the first

stage we use the micro-CT data analysis to homogenise the heterogeneity at micro-level and in a second step (not yet performed) we propose to use sonic logs and bore-hole images or similar for an upscaling of the next scale using the effective properties derived from the micro-level as matrix properties. Because of the wavelength of seismic waves this would be the main scale of interest. For this purpose we discuss a carbonate sample of a major oil field found through deep ocean drilling and attempt a mm-scale homogenisation procedure (Fig. 13).

Two criteria are used for cropping volumes in the asymptotic homogenisation procedure: (1) each smaller volume must be a subset of a larger volume; (2) each cropped volume has the porosity close to the bulk porosity,

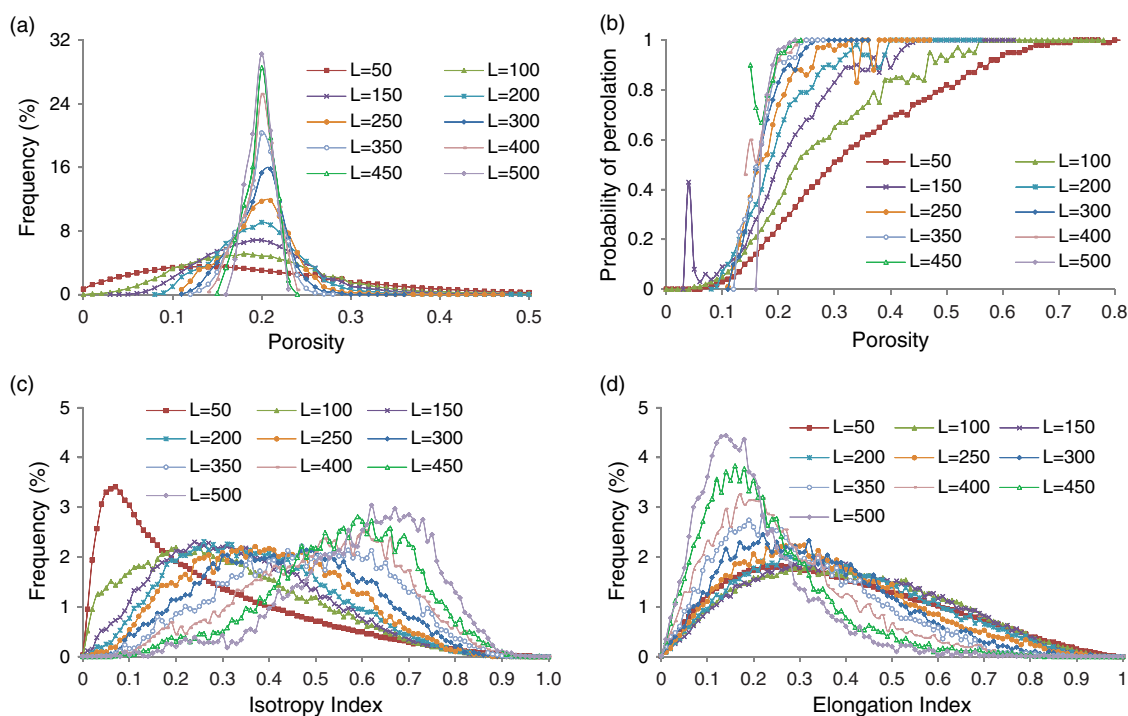


Fig. 14. Probabilities of porosity (a), percolation (b), isotropy index (c) and elongation index (d) of a carbonate sample.

i.e., difference is less than 1.0%. For each cropped model, we use the finite element method to analyse the response of displacement and pressure loads, respectively. The following boundary conditions are given:

- (1) normal displacement constraint on the surfaces of $x = x_{\min}$, $y = y_{\min}$ and $z = z_{\min}$,
- (2) free surfaces on $x = x_{\max}$ and $y = y_{\max}$,
- (3) displacement or pressure load on the surface of $z = z_{\max}$, where min and max denote two end boundaries in x , y and z directions.

Mathematically, this constraint is equivalent to the unconstrained compression of a sample with mirror images of the volume in x , y and z directions. The magnitude of loading is small enough to ensure that only elastic deformation occurs. In our model, only the solid skeleton is considered while pores are void and have no strength. Thus only elastic parameters of the solid are necessary as input.

Two kinds of meshes are used for these volumes for finite element computation. Hexahedral elements are used for small volumes of side-length < 100 voxels, which are easy to create and the computing time is acceptable. Tetrahedral elements are used for large volumes, generally for ≥ 100 voxels. This entails extra procedures and manual work but it can dramatically reduce the computation time. For creating tetrahedral element meshes, we considered the balance of keeping the precision and the fineness of the mesh and reducing the computing time. A typical example is shown in Figure 13. For small volumes the oscillations clearly indicate the limit of validity of the thermodynamic assumption. The lower bound shows stronger oscillation than upper bound, however, the micro-elastic properties appear to converge. For this specific case, the mechanical RVE size can be determined as 200^3 voxels.

5.3.2. Percolation Theory Applied to Non-Conservative Forces

Having identified the micro-mechanical RVE for conservative forces we extend the study to problems with dissipation. We consider Drucker-Prager plasticity of the rock over a mechanical RVE. In order to conduct cohesion and the angle of friction of rock samples, two cases of different pressures are simulated. Using the relationship of $\sigma_y = \sigma_n \tan \varphi + c$ where σ_y and σ_n are the yield stress and normal stress computed from Drucker-Prager plasticity, and c and φ are cohesion and the angle of internal friction of the rock. With the two groups σ_y and σ_n resulting from model calculations of the macroscopic model c and φ can be deduced. Numerical simulations with at least two different pressures are necessary to detect cohesion and the angle of friction of the microstructure.

For each case, the stress-strain relationships are computed (Fig. 15). We label von Mises stress and pressure at the yield point as S_y and S_n in Figure 15. The yield point of Case 1 is identified as shown in Figure 15(a); the yield point of Case 2 can be arbitrarily selected at the strain of

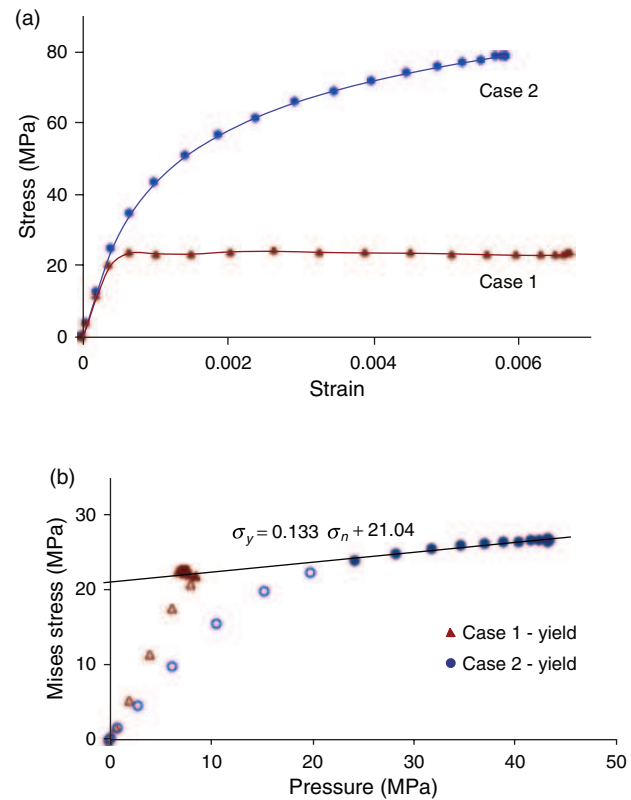


Fig. 15. Plasticity analysis: (a) Stress-strain relationships of two cases; (b) von Mises stress and pressure relationships of two cases and the fitting of cohesion and the angle of friction.

0.1% or 0.2%. For Case 1, the relationship between von Mises stress and pressure is linear before yielding occurs; both von Mises stress and pressure are constant after yielding. For Case 2, von Mises stress and pressure show two linear relationships before and after yielding. The linear trend after yielding is connected to the point after yielding of Case 1. It implies that any two points after yielding of Case 1 and Case 2 will give the same result of cohesion and the angle of friction. A fitted line is shown in Figure 15(b). The slope and the intercept are 0.133 and 21.04, respectively. Thus the cohesion is 22.04 MPa and the angle of friction is 7.8° .

Shrinking/expanding algorithms⁽⁴¹⁾ are used to create a series of derivative models with different volume fractions. Then the derivative models close to the percolation threshold are used to simulate the deformation and yielding. With a series of results of yield stress of models, it is possible to fit the critical exponent of yield stress. Figure 16(a) shows five examples of stress-strain relationships of derivative models close to the percolation threshold. Different derivative model have different responses, from elastic-perfect plasticity to plastic hardening behaviour, and the yield stress is increasing with higher volume fraction of solids. Figure 16(b) shows the example of fitting the critical exponent of yield stress. The

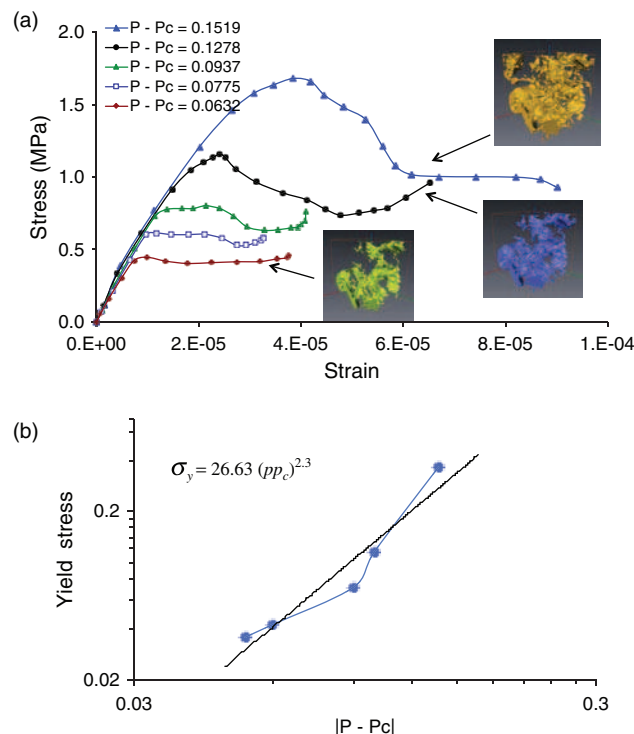


Fig. 16. Plastic response of derivative models close to (a) percolation threshold; (b) the fitting of the critical exponent of yield stress.

best fit of the critical exponent of yield stress is 2.3, which is close to the theoretical result.

5.4. Example Numerical Upscaling for Westerly Granite

This section summarises an example for numerical upscaling using a classical benchmark material of the rock mechanics community: Westerly granite.^(67–70) In this case study, we investigated the magnitude and longevity of thermal-elastic internal stresses in granite induced by slow burial over geological time scales.⁽⁷¹⁾

It is well known that heating and lithostatic loading of rock lead to high internal stresses.^(72,73) The reason lies in the mismatch and anisotropy of the thermal-expansion and elasticity tensors of the constituent minerals. These material properties may differ by as much as one order of magnitude, and consequently, heating of rocks by several hundred degrees and compression induce internal contact stresses of order 100 MPa. Thermal-elastic internal stresses have mainly been examined in the context of brittle rocks, where they generate micro-cracks.⁽⁶⁸⁾ Micro-cracks affect rock properties such as elasticity, strength, thermal conductivity, and permeability and thus attracted research efforts in the fields of mining, drilling, nuclear waste disposal and reservoir stimulation.^(68,69,74) Beyond confining pressures of about 100 MPa, reached at ca. 3 to 4 km depth in the continental crust, micro-cracking is largely suppressed.⁽⁷⁵⁾ At greater depths, the continental

crust of the earth assumes temperatures $> 50\%$ of the melting temperature of its main constituents (quartz and feldspar). At these temperatures (ca. 300 °C for quartz, usually attained at a depth of ca. 10 km), rocks begin to deform by crystal-plastic, ductile deformation processes such as dislocation and diffusion creep (e.g., Ref. [76]). It is often assumed that these creep processes relax thermal-elastic internal stresses quickly, and hence they are usually disregarded in deformation processes in the ductile crust.^(77,78)

Since ductile creep is a slow, time-dependent process, we decided to examine how large thermal-elastic internal stresses can become and how long they are sustained when granite is buried slowly and deeply in the continental crust. Because of the low burial rates and large relaxation times associated with creep processes, these questions cannot be addressed by laboratory experiments. Hence, we resorted to combining physical heating experiments on Westerly granite observed with time-series 3D Synchrotron computed micro-tomography and numerical upscaling. The physical experiments were used to determine the effective material properties of the mineral constituents of our Westerly granite sample. Empirical 2D numerical inversion models mimicking the physical experiments achieved this goal. Finally, the calibrated numerical models were extended to simulating the loading conditions and ductile creep encountered during the slow burial of granite in nature. This workflow is summarised in the following.

5.4.1. Physical Heating Experiment: Time Scale of Hours, ΔT of 200 °C, and Δp of 0 MPa

A mm-scale cylindrical sample of Westerly granite was slowly heated over an interval of 200 °C from room temperature and at ambient pressure (see Fig. 17(a)). The slow expansion of the sample and the formation of micro-cracks were monitored at a resolution of 1.3-micron voxel edge length with a tomograph. Bulk radial expansion and crack statistics were employed as constraints for 2D numerical calibration experiments, which served to determine the material properties of the mineral constituents.

5.4.2. 2D Numerical Calibration Experiment: Time Scale of Hours, ΔT of 200 °C, and Δp of 0 MPa

Knowledge of the thermal-expansion and elastic properties of the individual mineral constituents of granite is paramount for estimating thermal-elastic stresses at the grain scale. These properties change with increasing temperature and were estimated with Gibbs energy minimisation⁽⁷⁹⁾ or taken from laboratory experiments.⁽⁸⁰⁾ These methods presume that the minerals are intact. Our tomograms showed that our sample contained a significant amount of pre-existing intracrystalline pores and cracks (see Fig. 17(a)), resulting in a noticeable reduction of the undamaged elastic moduli and thermal-expansion

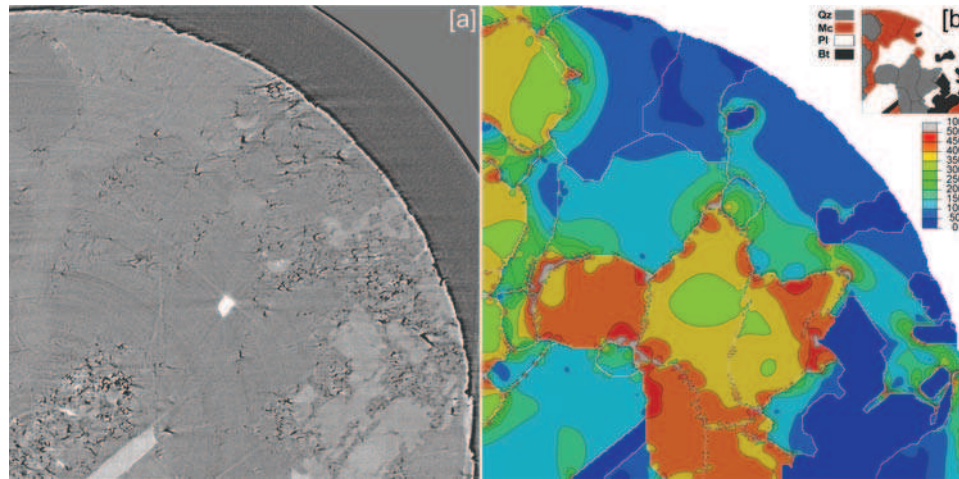


Fig. 17. (a) This image shows the tomogram considered in our numerical experiments. It is a quarter of a horizontal cross-section through the sample cylinder prior to heating. Sample radius is ~ 1.2 mm. The inset in (b) provides a key to the minerals observed in the image (Qz = quartz, Mc = microcline, Pl = plagioclase, Bt = biotite). Note the significant amount of pre-existing damage (regions with black or dark grey colours), both along grain boundaries and within grains. Plagioclase (for example, grain in the lower left corner of the tomogram) exhibits a particularly large amount of intracrystalline pores. (b) 2D contour plot of maximum differential stress (colour scale in MPa) predicted for slow burial with a typical geological velocity of 1 cm/a along a simplified equilibrium geotherm corresponding to a surface heat flow of 70 mW m^{-2} . This stress state is reached at a temperature of 350°C and a confining pressure of ~ 400 MPa.

coefficients. We used an empirical method for inverting effective material properties, as described in the following.

A horizontal 2D cross-section through the centre of the physical sample was digitised, thus reproducing the actual microstructure of the rock and approximating plane-strain conditions (see inset of Fig. 17(b)). For simplicity, only a quarter of the circular cross-section was modelled. The model grains were simulated as isotropic, homogeneous, linear-elastic materials. However, their surfaces were modelled with contact mechanics, allowing for dissipative grain-boundary cracking. An elastoplastic damage law was employed, constrained by experimental data on the tensile yield strength of Westerly granite.⁽⁸¹⁾ and atomic binding models.⁽⁸²⁾ The thermal expansion coefficient of quartz is the most important factor for the expansion behaviour of granite.⁽⁷⁸⁾ Thus, and for computational convenience, we regarded it as tuning parameter. The thermal expansivity of quartz was reduced linearly across the relevant temperature range while the expansion coefficients of the remaining minerals were decreased proportionally to quartz. We used the bulk radial strain observed in the physical experiment as main constraint (Fig. 18).

The quality of the fitted calibration model was tested against independent physical experiments on the same material conducted on much larger (cm-scale) samples.⁽⁷⁴⁾ Our calibrated model predicted the temperature evolution of the upscaled bulk thermal expansivity of Westerly granite very well (Fig. 19). In addition, the modelled grain-boundary crack behaviour matched independent laboratory experiments well.^(69,83)

5.4.3. Upscaled 2D Numerical Burial Experiment: Time Scale of Millions of Years, ΔT of 500°C , and Δp of 670 MPa

In the final step, the calibrated 2D model was used to simulate grain-scale thermal-elastic stresses in granite under loading conditions characteristic of geological burial. Therefore, the time scale of the experiment was upscaled from hours to millions of years. The burial model was augmented in two ways: (a) a time-dependent lithostatic pressure was applied to the model surface, and (b) in addition to elasticity and grain-boundary damage, the rheology of the minerals included rate-dependent plastic

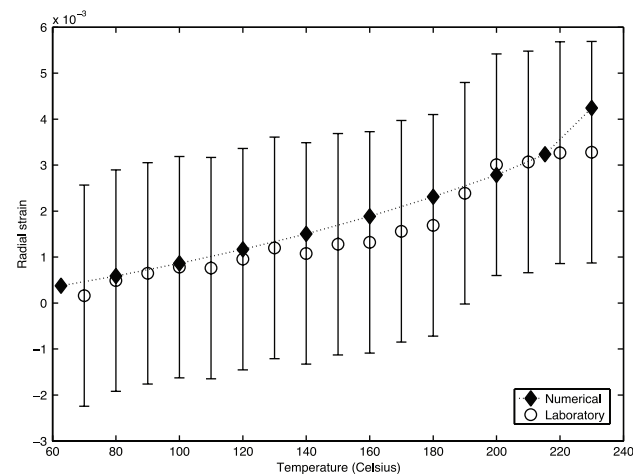


Fig. 18. Comparison of mean radial strain as a function of temperature measured in the physical experiment (error bars denote resolution error of the tomograms) and predicted by the 2D numerical calibration experiment.

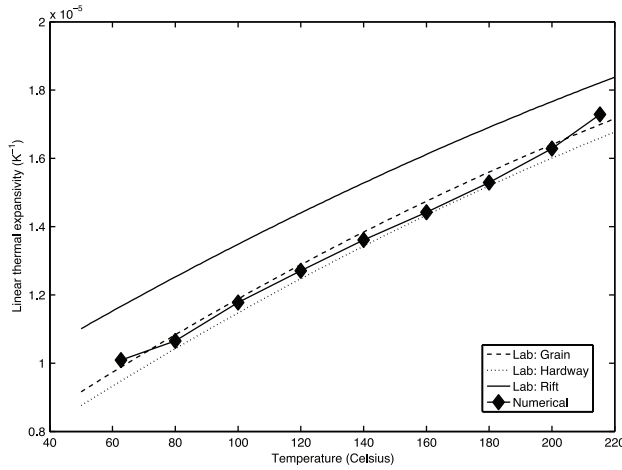


Fig. 19. Linear thermal expansivities as reported by Ref. [74] versus mean linear expansivity predicted by our calibration experiment. The terms *rift*, *grain*, and *hardway* are quarrying terms and denote planes of easiest, intermediate, and hardest splitting of the rock. Experimental linear expansivities were measured normal to these planes.

creep from fairly well established laboratory experiments (e.g., Refs. [84, 85]).

The results of this case study suggest that thermal-elastic stresses can reach magnitudes well in excess of 100 MPa with lifetimes of order millions of years, even at temperatures where ductile relaxation occurs see Figure 17(b). We thus identified an important potential source for upward-cascading of thermal-mechanical instabilities. This particular upscaling problem highlights the importance of the relative diffusive time and length scales involved in rock deformation problems, here those of thermal and stress diffusion.

6. MICROSTRUCTURALLY ENRICHED CONTINUUM FOR GENERALISED RATE DEPENDENT SOLIDS

The above described microstructure homogenization workflow provides the basic framework for upscaling of properties of materials with microstructures based on well established theories. We have shown how the quantification of the scale dependence of uncertainties of these material properties can be derived from thermodynamic extrema of minimum and maximum entropy production. However, the complementary approach of constitutive modelling is less straight forward. This is because classical continuum models do not have a length scale and therefore cannot accommodate microstructure.⁽⁸⁶⁾ This poses a significant problem as the classical constitutive laws cannot be used without modifications, and the question arises what are the fundamental material properties to be homogenised by the workflow and how do they incorporate the microstructure information. As a prerequisite for using thermodynamic extrema for deriving material properties, we need to ideally close the loop using

a constitutive approach that is also primarily based on thermodynamics.

The classical constitutive modelling approach from plasticity theory, which is excellent for deriving constitutive models from controlled laboratory experiments, therefore needs to be extended to being closer to basic thermodynamics and the underlying physics. In classical plasticity, constitutive laws are for instance derived by postulating an elasticity law, a yield function and a plastic flow rule (see Fig. 20). The consistency with the 2nd law of thermodynamics is implied either through the orthogonality principle or verified thereafter to validate the resulting incremental relationships. This approach provides an implicit or indirect consideration of thermodynamic constraints.

Over the recent years, an alternative approach has matured, which uses a direct consideration of the thermodynamic principles.^(87–91) We have already briefly discussed the concepts in the theory review.⁽²⁾ In the following, the key ideas are summarised in further detail. We begin with the local, weak form because the integral of the strong form already encapsulates the Min/Max theorem of entropy production that arises because of the path dependence. Consider the first law of thermodynamics:

$$dU = \delta W^{\text{tot}} + \delta Q \quad (3)$$

where dU is the differential internal energy, δW^{tot} is the increment of work done and δQ the heat supply to the system. In full analogy to the comment on path dependence in the strong form (Ref. [2]), we emphasise here that the increments of work and heat are not complete differentials while the internal energy is a proper differential. The second law of thermodynamics states that:

$$\delta Q \leq T dS_{\text{tot}} \quad (4)$$

Note that the total entropy differential $dS_{\text{tot}} = \delta S_{\text{rev}} + \delta S_{\text{ir}}$ is a complete differential while the irreversible entropy δS_{ir} and the entropy flux on the boundaries δS_{rev} are incomplete differentials. Since $\delta S_{\text{rev}} \equiv \delta Q/T$, the second law can be rewritten as $S_{\text{ir}} \geq 0$. Likewise, by using the definition of the internal energy, we can relegate the path dependence to the irreversible entropy differential δS_{ir} or its time derivative, the irreversible entropy production $\tilde{S}_{\text{ir}}$. This is an important constraint for the uncertainty of the system expressed in the incomplete differential $\tilde{S}_{\text{ir}}$. Another important finding is that, in the non-isothermal case, we have to track the total entropy production in a non-isothermal system.

$$U \equiv \Psi_{(\alpha_i)} + TS_{\text{tot}} \quad (5)$$

$$\dot{U} = \dot{\Psi}_{(\alpha_i)} + \dot{T}S_{\text{tot}} + T\tilde{S}_{\text{ir}}$$

Substituting the internal energy in Eq. (3), we obtain the classical equation for the rate of work done in the system:

$$\tilde{W}^{\text{tot}} = \dot{\Psi}_{(\alpha_i)} + S_{\text{tot}}\dot{T} + T\tilde{S}_{\text{ir}} \quad (6)$$

where the independent state variables α_i may have tensorial values. The total work rate $\tilde{W}^{\text{tot}} = \dot{W}^{\text{cons}} + \dot{W}^{\text{diss}}$

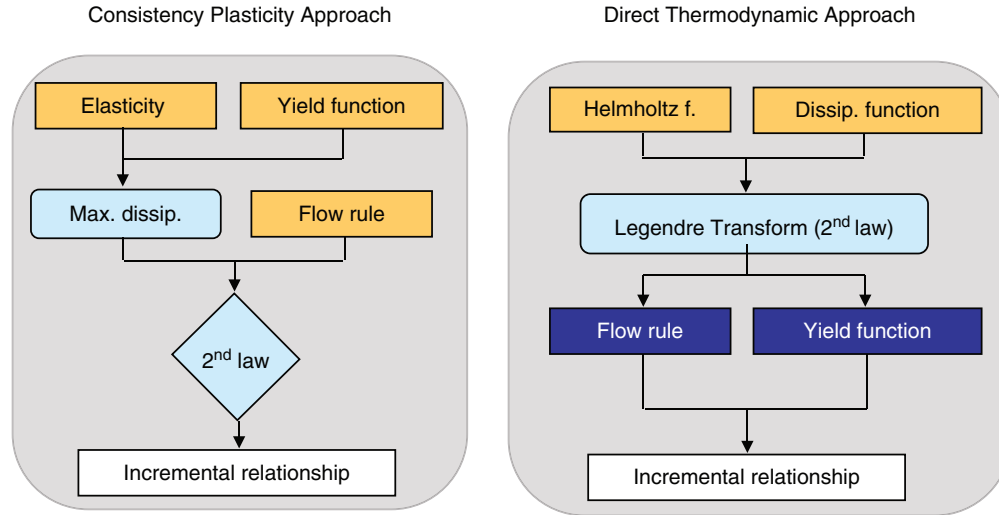


Fig. 20. The consistency plasticity approach postulates elasticity, yield and flow rule independently and verifies the consistency with the second law indirectly either through Ziegler's orthogonality principles (maximum dissipation) or through independent verifications. The direct thermodynamic approach postulates the thermodynamic energy potentials (Helmholtz free energy and dissipation potential) and derives the yield function, elasticity law and flow rule through the Legendre transform.

is composed of a recoverable and a dissipative term described by the rate of Helmholtz free energy plus the work associated to the entropy production. The dissipated work rate $\dot{W}^{\text{diss}}$ is again composed of a work rate related to the net entropy change of the system through fluxes on its boundaries plus the internal irreversible entropy production inside the volume. In a classical mechanical analysis, it is often assumed that the process is isothermal, hence $T = \text{constant}$ and the work term corresponding to the change of the total entropy of the system $S_{\text{tot}} \dot{T} = 0$. In either case, isothermal or non-isothermal, we are left with a complete thermodynamic description where the total work rate $\dot{W}^{\text{tot}}$ can be described by two potential functions, the free energy production plus the dissipation potential.

This opens the way for direct consideration of thermodynamic constraints leading to the proposed development of constitutive models for microstructurally enriched continua. The procedure originally suggested for rate-independent materials^(90,91) is as follows: first, postulate a Helmholtz free energy function $\Psi_{(\alpha_i)}$; second, postulate an internal source of dissipation $\Phi_{(\alpha_i, \dot{\alpha}_i)} = \tilde{\mathcal{S}}_{\text{ir}}$ (note that in addition to the dependence on the state variables α_i the dissipation function can depend on the rate of change of internal variables $\dot{\alpha}_i$); third, deduce the yield function and thereby implicitly the elasticity law, hardening rule and flow law.

We first assume the dissipative potential $\Phi_{(\alpha_i, \dot{\alpha}_i)}$ to be rate-independent, so homogeneous to first order in the internal variables α_i . We can use Euler's Theorem and write the dissipation:

$$\Phi_{(\alpha_i, \dot{\alpha}_i)} = \frac{\partial \Phi_{(\alpha_i, \dot{\alpha}_i)}}{\partial \dot{\alpha}_i} \dot{\alpha}_i \quad (7)$$

where the dissipation is the product of a stress $q_d = \partial \Phi_{(\alpha_i, \dot{\alpha}_i)} / \partial \dot{\alpha}_i$ (sometimes called generalised dissipative stress) and a generalised strain rate $\dot{\alpha}_i$. In the same way we identify the fraction of the rate of work done (released) for the dissipation by the Helmholtz free energy as:

$$\dot{\Psi}_{(\alpha_i)} = \frac{\partial \Psi_{(\alpha_i)}}{\partial \alpha_i} \dot{\alpha}_i \quad (8)$$

where $\partial \Psi_{(\alpha_i)} / \partial \alpha_i$ is identified as the generalised stress and $\dot{\alpha}_i$ is the generalised strain rate. We note that there is no dissipative power without commensurate rate of release of Helmholtz free energy:

$$\frac{\partial \Psi_{(\alpha_i)}}{\partial \alpha_i} \dot{\alpha}_i = \frac{\partial \Phi_{(\alpha_i)}}{\partial \dot{\alpha}_i} \dot{\alpha}_i \quad (9)$$

for every $\dot{\alpha}_i \neq 0$. This is also known as Ziegler's orthogonality or the maximum entropy production principle. As proposed by Ref. [90], the roles of the incremental change of Helmholtz free energy $\partial \Psi_{(\alpha_i)} / \partial \alpha_i$ (generalised stress) and the rate of change of the state variable $\dot{\alpha}_i$ (generalised strain rate) can be interchanged by a Legendre Transform to give the yield function $y_{(\alpha_i)}$:

$$\lambda y_{\alpha_i} = \frac{\partial \Psi_{(\alpha_i)}}{\partial \alpha_i} \dot{\alpha}_i - \frac{\partial \Phi_{(\alpha_i)}}{\partial \dot{\alpha}_i} \dot{\alpha}_i = 0 \quad (10)$$

where λ is a non-negative multiplier. Taking the time derivative of the yield function, we obtain the flow rule:

$$\dot{\alpha}_i = \lambda \frac{\partial y_{(\alpha_i)}}{\partial q_d} \quad (11)$$

In *consistency plasticity* the constant λ is classically evaluated through the consistency and loading condition

$y = \delta y = 0$, ensuring that the stresses do not exceed the yield surface. Later on, we will describe an alternative formulation where stresses are allowed to exceed the initial yield surface. This formulation is called the *overstress plasticity* formulation.

Both formulations enable so-called enriched continuum formulations with the direct consideration of dissipative processes and their energy feedbacks. Such feedbacks may lead to localisation of deformation into narrow shear bands with a small but finite width. Other than describing the homogenisation of the material microstructure, the enriched continuum model embeds a characteristic material length scale that acts like a localisation limiter. The material length scale is, in the above described formulation, directly based on the consideration of the internal energy interactions and dissipation through the microstructure. An example was given in the theory review⁽²⁾ where, for instance, the energy incorporated in rolling and sliding mechanisms of granular media reveals that shear bands are formed due to the rolling component, with their thickness varying with the rolling resistance. Much like in the case of the partial melts (see Fig. 5), where a length scale emerges out of hydro-mechanical diffusion, a new diffusive mechanical length scale emerges. In classical enriched continuum formulations, the shear band width can be shown to depend on the size of the individual grains, which affects the momentum diffusion. We have discussed that the energy considerations also reveal a rate effect not described by the classical Cosserat theories. Because the rates of processes play an important role in earth science, we will focus on the presentation of the various effects of rate sensitivity on localisation phenomena in the following.

7. COMBINING SOLID AND FLUID DYNAMICS FOR GEOMECHANICS

In order to introduce rate sensitivity, we review the different solid-mechanical and fluid-dynamic approaches. We will summarise two theories describing this transition, *consistency plasticity* and *overstress plasticity*. We also highlight their relationship and introduce the fundamental thermodynamically inspired strain-rate decomposition.

7.1. Definition of a Solid versus a Fluid RVE

First, we define the fundamental difference between a solid and a fluid in terms of the overall mechanical behaviour of a RVE. In the microstructure homogenisation workflow, we have discussed the statistical description of distributed networks, percolating or not, of an arbitrary volume of rock. We can use percolation criteria to define solid and fluid behaviour if we consider the material network to be made of force chains^(86,92) that can be pictured as small solid connector rods (hereafter called skeleton) in one subvolume of the RVE. At the level of the RVE, a solid is then defined by the case where percolation of the force chains through the considered volume occurs, i.e.,

the force chains are connected from one end of the volume to the other, and the force chain network can support an applied stress. When the force chains on the level of the RVE are not connected (no percolation), and the material defining the force chains (solid skeleton) is unable to support an applied stress. The material must react to this stress by flow and is, at the scale of the RVE, called a fluid. A solid may turn into a fluid through the application of a load by breakage of the force chains.

In order to describe this difference in mathematical terms, we use the formulation of Love⁽⁹³⁾ of an effective stress tensor of the RVE $\bar{\sigma}'_{ij}$ acting on a RVE of a⁽⁵⁴⁾ having a sub-skeleton of radius R , and sharing the forces across the set N of the contacts:

$$\bar{\sigma}'_{ij} \approx \frac{1}{V_{\text{RVE}}} \sum_{c \in N} l_i^c f_j^c \quad (12)$$

where V_{RVE} is the volume of the RVE, $l_i^c \approx R$ is the vector connecting the centres of each two contact skeleton materials and f_j^c the force acting on the contact.

We use the Terzaghi relation⁽²⁾ of the effective stress $\sigma'_{ij} = \bar{\sigma}'_{ij}$ and the total stress tensor σ_{ij}

$$\sigma_{ij} = \sigma'_{ij} + b p_f \delta_{ij} \quad (13)$$

where p_f is the fluid pressure, δ_{ij} Kronecker's delta and b Biot's coefficient.

For convenience we describe the problem by the strain rate tensor $\dot{\epsilon}_{ij}$ through the velocity field of the solid skeleton $V_k^{(s)}$, which is related to the fluid phase velocity through the consistency relationship.⁽⁹⁴⁾

$$\dot{\epsilon}_{ij} = \frac{1}{2} \left(\frac{\partial V_i^{(s)}}{\partial x_j} + \frac{\partial V_j^{(s)}}{\partial x_i} \right) \quad (14)$$

In Eq. (41) of Ref. [2] we have introduced the additive strain rate decomposition of poromechanics. This follows directly from the generalised decomposition of the total work rate by the rate of Helmholtz free energy and the dissipation.

$$\dot{\epsilon}_{ij} = \dot{\epsilon}_{ij}^e + \dot{\epsilon}_{ij}^p \quad (15)$$

In Eq. (1) of the theory review⁽²⁾ we have also introduced the generalised incremental relationship $\dot{\sigma}'_{ij} = C_{ijkl}^{ep} \dot{\epsilon}_{kl}$ where C_{ijkl}^{ep} is the elasto-plastic compliance modulus.

Assuming a smooth function of the effective stress, temperature T , and additional internal variables ξ_k , we can write the constitutive behaviour of the material as:

$$\epsilon_{ij}^p = f(\sigma'_{ij}, T, \xi_k) \quad (16)$$

Further assuming, without loss of generality, that the deformation is isothermal at a temperature T_0 and at microstructural steady-state, $\xi_k = \xi_k^0$, a Taylor expansion of Eq. (16) around the effective yield stress σ'_Y yields:

$$\delta \epsilon_{ij}^p = f' \cdot \delta \sigma'_{ij} + \sum_{m \geq 2} f^{(m)} \cdot \delta \sigma_{ij}^{'m} \quad (17)$$

where $\delta\epsilon_{ij}^p = \epsilon_{ij}^p - \epsilon_Y^p$, $\delta\sigma_{ij} = \sigma'_{ij} - \sigma_Y$ and $f^{(m)} = (1/m!)(d^m f(\sigma'_{ij}, T_0, \xi_k^0))/d\sigma_{ij}^m|_{\sigma'_{ij}=\sigma_Y}$.

7.2. Solid-Fluid Overstress Plasticity versus Solid Consistency Plasticity

The leading order of the linear expansion of Eq. (17) leads to the classical *consistency plasticity* formulation:

$$\delta\epsilon_{ij}^p = f'(\sigma'_{ij}, T_0, \xi_k^0)\delta\sigma'_{ij} \quad (18)$$

where stresses cannot exceed the yield surface. Consequently, all the evolution parameters such as plastic hardening and softening are tied to an evolution of the yield surface and thus an evolution of the elasto-plastic modulus C_{ijkl}^{ep} . The latter is commonly derived from laboratory experiments and not from considerations of the physical dissipation mechanisms.

By considering the higher order terms of Eq. (17), the stress can exceed the yield surface, and the description of the physics of the dissipation processes can be captured. This is also known as *overstress plasticity*⁽⁹⁵⁾ where $\sigma_{ij} > \sigma_Y$, $\delta\sigma'_{ij}$ is not an increment any more but it represents the overstress $\pi_{ij} = \sigma_{ij} - \sigma_Y$. This approach allows departure from the theory of solid mechanics: areas that exceed the yield stress are deforming in a manner defined by classical fluid dynamics.

Near the yield stress the two approaches are identical. However, the important difference is that after yielding the *consistency plasticity* approach lumps the non-linearities of the material response into the hardening modulus, whereas in *overstress plasticity* the stress-strain increment response is directly implemented in rate form. Classical solid mechanics deals with the fluid-like response after yield as if it were a solid (*consistency plasticity*) while in *overstress plasticity* a dual material behaviour can be considered: solid before yield and fluid after yield.

8. SELECTED APPLICATION TO PROBLEMS IN EARTH SCIENCES

8.1. Compaction Bands: HM Coupling

We start with a discussion of the possible effects of fluid dynamics properties of materials without consideration of an enriched continuum model to illustrate the important effect of rate dependence in Geology. For simplicity, we only consider hydromechanical coupling and investigate whether rate effects of this coupling can lead to localisation phenomena. We also investigate whether a characteristic length scale related to the spacing of these localisation bands emerges. We ignore the energetic processes within the localisation band and do not wish to solve for the width of the localisation band itself. We have presented in Ref. [2] compaction bands in sedimentary rocks, which can be explained by classical constitutive theories. The solution presented here elaborates on a possible alternative model (Fig. 5), where the same observations can be derived from fluid-dynamic rate effects alone.

8.1.1. Bifurcation Criterion

Since compaction bands (CBs) are caused primarily by material non-linearities, current models for mechanical compaction focus on the behaviour of the solid matrix. Common mechanical driving forces of compaction include grain breakage,⁽⁹⁶⁾ bond breakage,⁽⁹⁷⁾ pore collapse and pressure solution^(98–101) under a compressive loading. A frequently used criterion for the formation of CBs as a nonlinear response (bifurcation) of the skeleton to mechanical loading was given by Ref. [102], as an extension of the bifurcation theory used by Ref. [103]. A rate-independent elasto-plastic material, obeying a standard stress-strain incremental response of $\dot{\sigma}'_{ij} = C_{ijkl}^{ep}\dot{\epsilon}_{kl}$, would according to this criterion show material bifurcations in the form of localised deformation if the determinant of the acoustic tensor ($L_{jk} = n_i C_{ijkl}^{ep} n_l$, n_i being the unit vector normal to the localisation band) is zero. This provides non-trivial solutions to the eigenvalue problem $L_{jk} u_k = 0$, u_k being the tensor's eigenvectors.^(102,104)

Through this criterion, areas of shear-dominated localisation (shear bands) and volumetric-dominated localisation (compaction or dilation bands) are identified, as functions of the loading conditions and the material parameters (Fig. 21). Since this criterion provides the conditions for the onset of a single localisation band, recently Ref. [105] extended this continuous bifurcation theory⁽¹⁰³⁾ to discontinuous bifurcation, in order to account for the formation of a discrete, periodic set of compaction bands.

The response of a saturated porous medium to mechanical loading entails expulsion of the fluid from the porous matrix, controlled by its hydraulic diffusivity. Hence, when the aforementioned bifurcation criterion is extended to a saturated elasto-plastic material, it provides the necessary hydromechanical conditions in 2D loading for periodic dilating and contracting instabilities, either in the form of layered strips or checkerboard cells (Vardoulakis⁽¹⁰⁶⁾

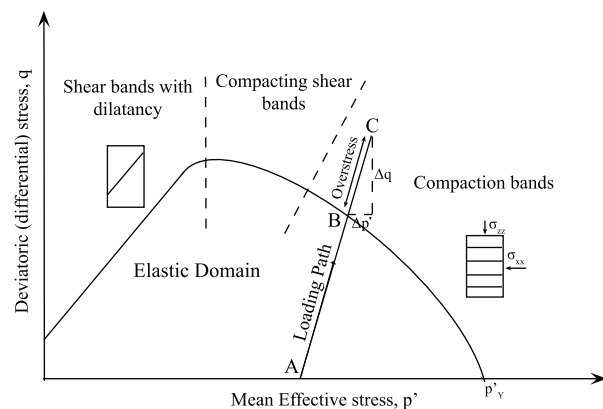


Fig. 21. Yield envelope in generalised stress space p' - q . This study focuses on the area of compactions bands and uses the concept of overstress for loading paths exceeding the yield stress. Insets: failure modes for shear and compaction bands. Reprinted with permission from [109], J. Fortin, et al., *J. Geophys. Res.* 111, B10203 (2006). © 2006.

and Vardoulakis and Sulem⁽¹⁰⁷⁾ Figs. 5.7.1 and 5.7.2). These conditions include, apart from the mechanical properties expressed through the acoustic tensor, the hydraulic parameters of the problem (viscous drag coefficient or hydraulic diffusivity), providing a hydromechanical criterion (see Ref. [107], Eqs. (5.7.47)–(5.7.49)) as an extension of the eigenvalue problem related to the acoustic tensor. The new, hydromechanical problem consists of the eigenvalue problem $A_{ij}u_j = 0$, where A_{ij} is an augmented tensor including the acoustic tensor L_{ij} and the hydromechanical properties, whereas the eigenvector u_j includes the pore pressure.

When applied to simple uniaxial compression of a saturated elasto-plastic material, this method reveals that layered instabilities, emerging in the softening domain of the σ – ϵ -curve of the material, correspond to a regime where pore fluid pressure obeys a backward-in-time diffusion equation (see Ref. [107], Section 5.8.2). Backward diffusion is known to be a mathematically ill-posed problem, which is usually regularised by resorting to viscous and/or gradient considerations.⁽¹⁰⁸⁾

8.1.2. Failure Pattern

It is well known from biaxial and triaxial experiments that shear bands are pairs of inclined localisation bands, appearing in the shear band area of Figure 21 and dipping at Arthur-Vardoulakis angle^(110,111) as defined in Eq. (6) of the theory review.⁽²⁾ When loaded in the compaction band area of Figure 21, CBs will form in an almost horizontal direction (perpendicular to the maximum principal stress, here set to the axial stress σ'_{zz}), as denoted in the inset of Figure 21 (see for experimental evidence Baxevanis et al.⁽¹⁰⁴⁾ Fortin et al.⁽¹⁰⁹⁾ Oka et al.⁽¹¹²⁾). Thus, although deviatoric stresses are required to cause pure compaction, the failure mode involves negligible shear deformation, as measured by Ref. [112].

We may therefore calculate the incremental mechanical dissipation for this failure pattern,

$$\delta W^p = \sigma'_{ij} \delta \epsilon_{ij} \quad (19)$$

to deduce that it is a manifestation of volumetric (and, more specifically, of axial) deformation since $\delta W^p \approx \sigma'_{zz} \delta \epsilon_{zz}$. These considerations indicate that the basic physics of failure in horizontal compaction bands could be approximated by a simple 1D model of uniaxial compression.

8.1.3. Model Formulation

In order to cast these concepts into a simple mathematical model, we consider an infinite slab of height $2H$ consisting of a fluid-saturated, porous geomaterial subjected to uniaxial compression. We define the stress increment above the initial compressive yield stress σ'_Y needed to cause the first emergence of CBs as the overstress $\Delta \sigma'_{zz} = \sigma'_{zz} - \sigma'_Y$, where $|\sigma'_{zz}| > |\sigma'_Y|$ (see the branch B–C of the

overstress definition in the triaxial space in Fig. 21). In this framework, we ascribe a viscoplastic response to the solid skeleton, given in the lines of the overstress plasticity presented by Ref. [95]. If CBs form at zero or negligible overstress, the mode of deformation is interpreted as brittle failure. On the other hand, if the overstress admits a nonzero value, CBs are formed in the ductile (cataclastic flow) regime. For further simplification, compressibility of fluid and solid skeleton and inertia are neglected. Compressive fields (stresses and strain rates) are negative in this study.

8.1.4. Stress Equilibrium

Stress equilibrium in the z -direction, combined with Terzaghi's effective stress $\sigma'_{zz} = \sigma_{zz} + p$ ($p > 0$ whereas $\sigma'_{zz}, \sigma_{zz} < 0$), provides

$$\frac{\partial \sigma'_{zz}}{\partial z} = \frac{\partial p}{\partial z} \quad (20)$$

The volumetric strain rate always has the same sign as σ'_{zz} (to satisfy the second law of thermodynamics).

8.1.5. Mass Balance Considerations

In a porous medium, we define the partial stresses $p_1 = (1-n)p_s$ and $p_2 = np_f$ of the solid and fluid phase, respectively, n is the porosity, and p_s and p_f denote the densities of the solid skeleton and the fluid, respectively.

In the 1D setting considered the mass balance equations for each of the phases (refer to Eqs. (9) and (10) in the theory review⁽²⁾) reduce to the following expressions, when incompressible solid and fluid are considered (i.e., when $p_s = \text{const.}$ and $p_f = \text{const.}$):

$$-\frac{\partial n}{\partial t} - \frac{\partial n v_z^s}{\partial z} + \frac{\partial v_z^s}{\partial z} = 0 \quad (21)$$

$$\frac{\partial n}{\partial t} + \frac{\partial n v_z^f}{\partial z} = 0 \quad (22)$$

where v_z^i , $i = s, f$ the partial velocities of the solid and fluid in the z -direction. By adding them we obtain the mass balance equation of the mixture:

$$\frac{\partial n(v_z^f - v_z^s)}{\partial z} + \frac{\partial v_z^s}{\partial z} = 0 \quad (23)$$

We accept Darcy's law (Eq. (14)) in the theory review⁽²⁾ for the Gersevanov filter velocity $n(v_z^f - v_z^s)$, neglecting higher order terms,

$$n(v_z^f - v_z^s) = -\frac{k_\pi}{\mu_f} \frac{\partial p}{\partial z} \quad (24)$$

and define $\partial v_z^s / \partial z = \dot{\epsilon}_{zz} = \dot{\epsilon}_v$, to get,⁽¹⁰⁷⁾

$$\frac{k_\pi}{\mu_f} \frac{\partial^2 p}{\partial z^2} = \dot{\epsilon}_v \quad (25)$$

where p is pore pressure, $\dot{\epsilon}_v$ the (compressive) volumetric strain rate, μ_f the viscosity of the fluid (in Pa·s), and k_π the permeability, assumed constant in this study as commonly done in consolidation theory.⁽¹⁰⁷⁾

The assumption of constant permeability is adopted here since this work is limited to the study of the failure pattern (onset of failure) rather than the post-failure behaviour of the system. In the general case where permeability, being primarily a function of the porosity, varies in space, Eq. (25) reads:

$$\frac{k_\pi}{\mu_f} \frac{\partial^2 p}{\partial z^2} + \frac{1}{\mu_f} \frac{\partial k_\pi(z)}{\partial z} \frac{\partial p}{\partial z} = \dot{\epsilon}_v \quad (26)$$

Hence, the assumption of constant permeability considered in this study (Eq. (25)) implies that CBs emerge in areas of extremum (minimum or maximum) values of permeability, where $\partial k_\pi(z)/\partial z = 0$ holds.

8.1.6. Fluid-Dynamic Instabilities Through Rate-Dependent Material

For the 1D formulation we assume that the solid mechanical problem is solved, and the orientation of our 1-coordinate system is selected according to the elastoplastic characteristics for a potential solid-mechanical compaction band following the approach of Ref. [102]. Having identified the potential solid-mechanical framework we can, with the overstress formulation, neglect the role of elasto-plasticity and focus on the fluid dynamic solution. This analysis allows us to investigate whether through the consideration of a viscoplastic branch new solutions can emerge, which through the consideration of the factor time, allow fluid dynamic instabilities to occur at lower (over)stresses than predicted by the solid-mechanical solution. Therefore, we accept rate-dependent (viscous) regularization, by correlating the plastic volumetric strain rate to the overstress through a general viscoplastic power law of the form⁽¹¹³⁾

$$\dot{\epsilon}_v = \eta \left[\frac{\Delta \sigma'_{zz}}{\sigma_{ref}} \right]^m \quad (27)$$

where η is the creep parameter (in s⁻¹) and σ_{ref} a reference stress-like quantity, the loading strain rate at the boundary where loading $\Delta \sigma'_n$ is applied, is

$$\dot{\epsilon}_n = \eta \left[\frac{\Delta \sigma'_n}{\sigma_{ref}} \right]^m \quad (28)$$

If we accept σ_{ref} as the applied overstress at the boundary, $\Delta \sigma'_n$, then $\eta = \dot{\epsilon}_n$ and the constitutive model could be written as

$$\dot{\epsilon}_v = \dot{\epsilon}_n \left[\frac{\Delta \sigma'_{zz}}{\Delta \sigma'_n} \right]^m \quad (29)$$

where $\Delta \sigma'_n$ and $\dot{\epsilon}_n$ are reference quantities for the overstress and strain rate at which rate dependent behaviour is

validated, here set equal to their values at the boundary, ensuring clarity in the mathematical treatment.

Note that the effect of poroelasticity was neglected in this study, in order to emphasise the conditions for the appearance of spatially periodic failure patterns that can potentially emerge in the fluid dynamic regime. We have shown in Figure 5 an example where a fluid dynamic length scale known as the compaction length may be interpreted as a criterion for periodic instabilities. This would allow a much richer solution space where upon neglect of the rate dependence solid mechanical compaction bands emerge, while for the consideration of rate dependence both solid and fluid dynamic instabilities are possible.

Reference [113] calibrated the rate-dependency equation Eq. (27) through isotropic compression data, at various strain rates. They deduced that the creep parameter η should incorporate the void ratio e , following $\eta = \eta_0/(1+e)$, and that the rate sensitivity exponent m should be correlated with the slopes of the virgin compression and recompression curves. Based on this definition, we may obtain that $m > 1.5$ for mudstones,⁽¹¹²⁾ $m \sim 10$ for porous sandstones⁽¹¹³⁾ and $m \sim 100$ for kaolinite clay.⁽¹¹⁴⁾ Although phenomenologically introduced, we need to note that the assumed rate-dependency reflects the importance of the textural composition and mineralogy of the earth materials, arising through the presence of even the smallest clay fractions, as well as micro-mechanical processes like grain angularities inducing rotations and wear, grain breakage, bond breakage, pore collapse and pressure solution at soft rocks. When upscaled, these micromechanical properties yield the necessary rate effects to regularise the problem.^(115,116)

8.1.7. Mathematical Considerations

By combining Eqs. (25), (20), (29) and assuming that our material is homogeneous and isotropic (hence σ'_y is the same across the height of the specimen), we obtain the dimensionless effective-stress equation:

$$\frac{d^2 \sigma'}{dz^{*2}} - \zeta \sigma'^m = 0 \quad (30)$$

where $z^* = \frac{z}{H}$, $\sigma' = \Delta \sigma'_{zz}/\Delta \sigma'_n$ and

$$\zeta = \frac{\dot{\epsilon}_n \mu_f}{k_\pi \Delta \sigma'_n} H^2 \quad (31)$$

8.1.8. Hydraulic Diffusion in the Plasticity Regime

The parameter ζ involves the ratio of the characteristic rates of the problem at hand, namely the loading rate (strain rate $\dot{\epsilon}_n$) over the diffusion rate of the pore fluid within the specimen of height H when subjected to loading $\Delta \sigma'_n$. The diffusion rate is given as the inverse of the characteristic diffusion time

$$t_D = \frac{\mu_f H^2}{k_\pi \Delta \sigma'_n} \quad (32)$$

Note that the rate t_D^{-1} is indeed the ratio of the modified, poro-plastic hydraulic diffusivity of the medium $c_V = k_\pi \Delta \sigma'_n / \mu_f$ (in absolute accordance with the classical poro-elastic expression $c_V = k_\pi K / \mu_f$, K being the bulk modulus of the specimen) over the height H over which the load is applied.

This modification of the hydraulic diffusivity means that the hydraulic diffusion time (32) is not a material property (as in classical poro-elasticity), but is mainly controlled by the applied overstress. For example, in a fluid saturated ($\mu_f = 10^{-3}$ Pa·s), laboratory experiment ($H = 0.1$ m, as typical for laboratory specimen), a permeable ($k_\pi = 10^{-12}$ m²) sample would have a very small diffusion time ($t_D \sim 10^{-2}$ sec) for large overstresses, of the order of the bulk modulus ($\Delta \sigma'_n = 1$ GPa), whereas the same sample would have a very large diffusion time ($t_D \sim 10^2$ sec) for small overstresses ($\Delta \sigma'_n = 100$ kPa).

Hence even for high permeability values, such as 1 Darcy, and for realistic laboratory conditions where usually the applied overstress does not exceed 1 MPa, the hydrodynamic control turns the problem considered as elastodynamic instability (expressed through the acoustic tensor's eigenvalue problem) into a creeping flow problem controlled by the dissipative properties and time scales defined above. In the present context, this transition is expressed through the dimensionless parameter ζ (Eq. (31)). When $\zeta \ll 1$ the specimen is loaded in the elastodynamic domain, since the loading rate is lower than the diffusion rate and the specimen has the time to diffuse away any pressure variations induced by the loading conditions. At the other extreme, in the creeping flow regime where $\zeta \gg 1$, the loading rate is faster than the diffusion rate, pore pressure variations cannot be diffused away and overpressure can be produced in the specimen leading to hydro- mechanically induced failure in the form of CBs.

8.1.9. Analytical Solution

The solution of Eq. (30) depends on the value of the rate sensitivity coefficient m . For all $m > 1$ (strong rate-dependency), the solution for σ' is non-trivial, presenting singularities, as common for this class of equations (see Appendix A of Ref. [117] and literature therein for the uniqueness and characterisation criterion of diffusion-reaction/absorption equations). For integer values of $m < 4$ this equation admits closed form solutions. For the sake of mathematical simplicity we focus in this text only on the relevant solutions producing CBs for strong rate-dependence, under drained loading conditions and choose without loss of generality a power law exponent of $m = 3$ and drained boundary conditions ($\sigma'(1) = 1$ and $d\sigma'/dz^*|_{z^*=0} = 0$). The analytic solution of Eq. (30) is

$$\sigma' = \pm C_2 \operatorname{sn} \left[\left(\sqrt{\frac{-\zeta}{2}} z^* + \frac{Icn(0, \iota)}{C_2} \right) C_2, \iota \right] \quad (33)$$

Table I. Analytic solutions for varying rate sensitivity m . Functions $\wp(u, \omega_1, \omega_2)$ and $\operatorname{sn}(u, k)$ are the Weierstrass P and Jacobi SN function, respectively⁽¹¹⁸⁾ and ι is the imaginary number.

Rate sensitivity m	Solution
$m = 1$	$\sigma' = C_1 e^{\sqrt{\zeta} z^*} + C_2 e^{-\sqrt{\zeta} z^*}$
$m = 2$	$\sigma' = \frac{6}{\zeta} \wp(z^* + C_1, 0, C_2)$
$m = 3$	$\sigma' = \pm C_2 \operatorname{sn} \left[\left(\sqrt{\frac{-\zeta}{2}} z^* + C_1 \right) C_2, \iota \right]$
Any other m	Num. solution

where Icn is the inverse Jacobi CN function and C_2 the solution of the transcendental equation $\sigma'(1) = 1$. Refer to Table I for solutions to other exponents m .

8.1.10. Compaction Criterion

For strong rate-dependencies, $m = 3$, the profiles of the normalised effective stress depend evidently on the parameter ζ appearing in the solution. From the plots of Figure 22 we may notice a complex response, providing a multiplicity of singularities for the normalised effective stress as ζ increases. It has to be noted that for small values of ζ (approximately $\zeta < 13$) the effective stress profiles present a smooth solution with a minimum at the origin ($z = 0$), as shown in Figures 22(a) and (b). Past the point $\zeta > 13$ the effective stress presents multiple singularities, the number of which increases with ζ . When the rate of loading is significantly larger than the rate of hydraulic diffusion (i.e., when $\zeta \gg 1$) the pore fluid can be trapped inside the porous matrix and force the medium to periodic patterns of hydraulic compaction acting as fluid flow discontinuities.

The reason for the emergence of CBs is given by the mechanical compaction outpacing the fluid diffusion. In order to describe it mathematically, we retrieve from Ref. [118] (Eq. (16.5.7)) that in Eq. (33) stress tends to infinity when the argument of sn becomes equal to $\iota K'(\iota)$, $K'(\iota)$ being the complementary complete elliptic integral of the first kind. In our case this would mean that CBs appear when $\zeta = \zeta_{cr}$, where

$$\zeta_{cr} = -2 \left[\frac{K(\iota) - K'(\iota)}{z^*} \right]^2 cd \left[\frac{K(\iota) - K'(\iota)}{z^*}, \iota \right]^2 \quad (34)$$

and $K(\iota) - K'(\iota) = 1.31\iota$ whereas cd is Jacobi's CD function. The expression ζ_{cr} of Eq. (34) presents a minimum $\zeta_{cr}^{\min} = 12.7$ at $z^* = 0.52$, in accordance with the stress profile depicted in Figure 22(c). For all $\zeta \geq \zeta_{cr}^{\min}$ stress singularities (CBs) will appear at different z points, verifying the results of Figure 22.

From the definition of ζ , Eq. (31) this gives us directly the criterion for the onset of periodic CBs which is

$$\frac{\dot{\epsilon}_n \mu_f}{k_\pi \Delta \sigma'_n} H^2 \geq \zeta_{cr}^{\min} \quad (35)$$

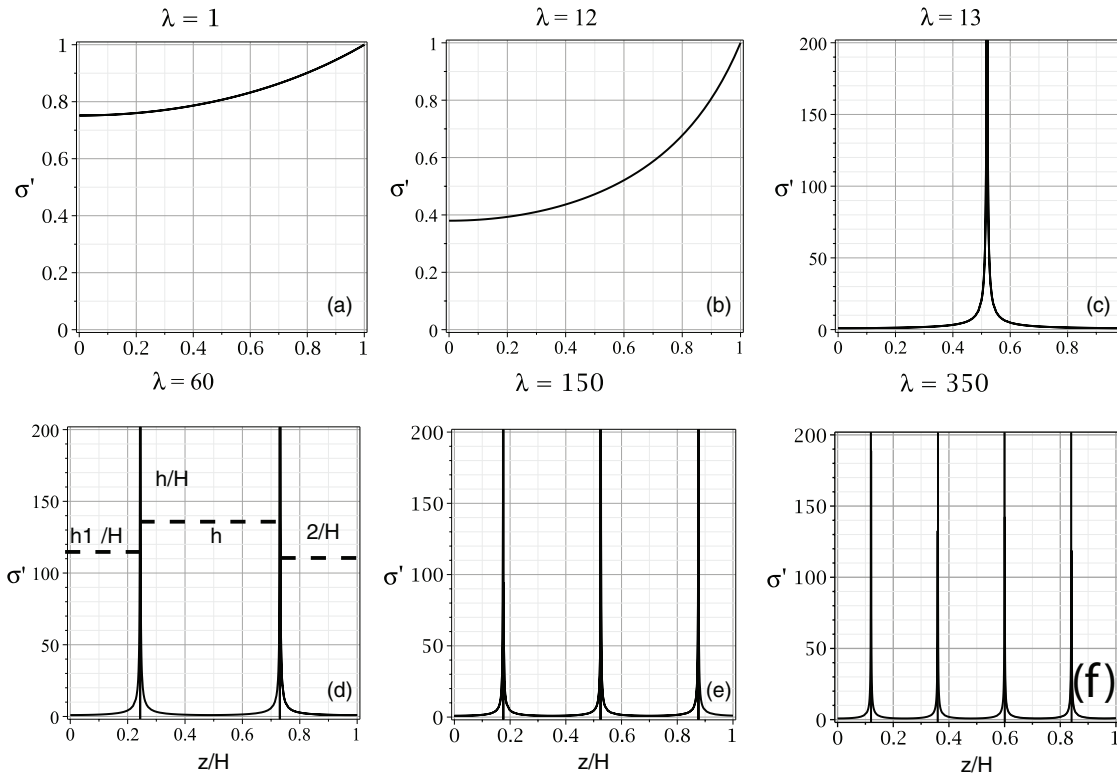


Fig. 22. Distribution of the normalized effective stress σ' (Eq. (33)) inside the specimen for six different values of ζ . In (d) the dimensionless distance h/H between the stress singularities in CBs is highlighted.

A diagnostic element for geological applications in the field is the spacing h between the CBs as annotated in Figure 22(d). Although not periodically placed across the specimen, the CBs divide the space into equal layers of distance h (Figs. 22(b) and 23), since as annotated in Figure 22(d), $h_1 + h_2 \approx h$. As a consequence, h is then defined as the inverse density of bands, $h = H/N_C$ (Fig. 23), or:

$$h = \frac{1}{0.26} \sqrt{\frac{k_\pi \Delta \sigma'_n}{\mu_f \dot{\epsilon}_n}} \quad (36)$$

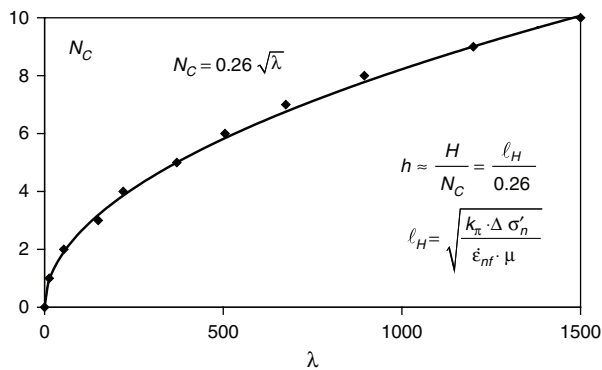


Fig. 23. Number of CBs N_C as a function of ζ . From the definition of ζ , we deduce that the spacing between the stress singularities h scales with the characteristic hydraulic length ℓ_H , providing Eq. (36).

8.1.11. Application to Field and Experimental Observations

When applied to the field and experimental observations, we can draw some conclusions from the observed quantities. Table II lists the average values of permeability, distance of the CBs and fluid viscosity for a series of field examples and laboratory experiments. Notice the absence of further information on the rate sensitivity of the materials and the time of formation of the CBs in the failure patterns reported from the experiments and field data. Due to this fact we use $m = 3$ and accept the monitored CBs patterns as “first generation” CBs thus assuming coeval formation of the patterns. Since the present model doesn’t allow for time evolution of the pattern, we can only model the CB pattern formed initially (first generation of CBs) and exclude later formations from post failure loading. Hence, as the materials could deviate from $m = 3$ and the monitored patterns could include CBs of several generations, the inferred values of the loading conditions could only be qualitatively assessed to deduce whether failure took place in the brittle or the ductile (cataclastic flow) regime.

We start by comparing the results from Eq. (36) versus controlled laboratory experiments on Diemelstadt⁽¹¹⁹⁾ and Bleurswiller^(109,119) sandstones listed as rows A and B respectively. The experimental conditions were at room temperature and the specimens were saturated with ionized water of a viscosity $10^{-3} \text{ Pa} \cdot \text{s}$, the failure mode was

Table II. Laboratory (rows A–C) and Field (D–G) examples of CBs. The first column lists input parameters permeability and viscosity as measured, the second column shows loading conditions, the applied over-stress and loading rate, and the last column shows the distance between CBs. The parameters shown in bold are those inverted from Eq. (36) for each case study. The theory both agrees well with experimental observations and provides reasonable magnitudes of over stresses. The application to the field provides conclusive insights into the source mechanics and rate of deformation. Under a time independent mechanical approach (rows D and F) we expect extreme loading rates that would generate a high-energy end member as a result of the imposed over stress. Our approach clearly shows that if geological time is given for the formation of CBs they can also be the result of smooth loading conditions with extremely low over stress (rows E and G).

Case studied	k_{π} (m ²)	μ_f (Pa s)	$\Delta\sigma'_n$ (Pa)	$\dot{\epsilon}_n$ (s ⁻¹)	h (m)
(A) Diemelstadt Sandstone (central Germany) ⁽¹¹⁹⁾	$3 \cdot 10^{-12}$	10^{-3}	0.0006	10^{-5}	$1.7 \cdot 10^{-3}$
(B) Bleurswiller Sandstone (Vosges Mnt, France) ⁽¹⁰⁹⁾	10^{-12}	10^{-3}	0.2	10^{-5}	0.02
(C) Diatomaceous Mudstone (Noto Peninsula, Japan) ⁽¹¹²⁾	10^{-16}	10^{-3}	400	10^{-6}	Single CB
(D) Navajo Sandstone (Utah, USA) ⁽¹²⁰⁾	$5 \cdot 10^{-12}$ (121)	10^{-3}	$20 \cdot 10^6$	1.5	1
(E) Navajo Sandstone (Utah, USA): This work	$5 \cdot 10^{-12}$ (121)	10^{-3}	1350	10^{-4}	1
(F) Aztec Sandstone (Nevada, USA) ⁽¹²⁰⁾	10^{-12} (121)	10^{-3}	$2.5 \cdot 10^6$	0.74	0.2
(G) Aztec Sandstone (Nevada, USA): This work	10^{-12} (121)	10^{-3}	270	10^{-4}	0.2

brittle (negligible over stress according to our definition) and the spacing between the discrete CBs was measured $h = 1.7$ mm for Diemelstadt and 2 cm for Bleurswiller. To showcase the application of the theory under strong rate dependency ($m = 3$) we use the parameters given in Eq. (36) to calculate the exact numerical value of the over stress, resulting in $\Delta\sigma'_n = 6 \cdot 10^{-4}$ Pa and 0.2 Pa, respectively.

Thus we infer that the failure pattern takes place immediately after the applied stress exceeds the yield limit, in accordance with brittle deformation. In row C we compare the model with a ductile water-saturated diatomaceous mudstone.⁽¹¹²⁾ The sample was loaded at fixed strain rate 10^{-6} s⁻¹. The ductile specimen formed CBs at the coordinates $z_1 \approx 0.35$ H and $z_2 \approx -0.4$ H (see specimen CD5 in Ref. [112]). When the over stress is inverted from the assumed parameters to be 0.4 kPa, we reproduce the observed pattern of CBs obtaining compaction at $z_{1,2} = \pm 0.4$ H. Since the model is one-dimensional, it does not include information on the propagation velocities of the CBs. As a result the failure pattern is appearing to form instantaneously, which is not the case as shown by Ref. [122].

Our simple model therefore provides inverted values of the over stress that are consistent with the laboratory observations. Classical, purely mechanical models, without time considerations, require much higher driving stresses. Table II lists in rows D and F two examples from the field where classical theories⁽¹²⁰⁾ are postulating over stresses of 2–20 MPa to explain the CBs. Under such conditions the compaction energies calculated are of the order of 0.1 MJ/m² (Ref. [120]), comparable to the ones met in major earthquake events (Ref. [123]). With these inferred values and the parameters measured in the field⁽¹¹⁹⁾ we can use Eq. (36) to estimate the required strain rates for the mechanical compaction to outpace the fluid diffusion from the CBs. The required strain rates calculated to accommodate such high stress events, generate the observed pattern of CBs and provide the necessary strain energy release to match the compaction energies calculated are all of seismic nature and would indeed interpret CBs as extreme, short events.

Given that there is no grain crushing observed in the field sites of CBs^(120, 124) the mode of solid deformation is described to be a rearrangement of grains with minor grain cracking. Based on these observations we infer that the formation of CBs may not be necessarily an extreme event. Using the above described model with an upper limit of geological strain rates lower than 10^{-4} s⁻¹ (rows E and G) the required over stress for the observed spacing in the field examples is of the order of 1 kPa. For even lower strain rates maintained over a long geological time scale, over stresses of the order of Pa or less may even be sufficient to generate CBs. This suggests that under geological loading rates CBs maybe a more common feature than previously thought.

The examples in the Table II arguably can be explained both by classical solid mechanical CBs⁽¹⁰²⁾ as well as their fluid dynamic counterparts.⁽¹⁹⁾ The selection of the style of material instability is simply governed by the rates of deformation. If the sample is loaded rapidly as it is done in the laboratory the preferred solution is the solid mechanical CB. However, if the sample is loaded slowly as in nature the fluid dynamic instability may be preferred. The example shown in Figure 5 illustrates an extreme case which strongly supports the dominance of a fluid dynamic instability in a natural deformation case. This is the case of a partially molten rock (migmatite) where a solid mechanical instability seems unlikely as the solid skeleton cannot support a stress owing to the percolation of the melt phase expected from considerations of the wetting angle on grain boundaries. In such cases the overall RVE is better described by the viscosity of the melt^(17, 18) and the fluid dynamic compaction band is the preferred explanation.

8.2. Shear Zones: Formation and Evolution of Faults

Here we show how the consideration of rate effects can lead to strong weakening effects during the formation of

lithosphere-scale shear zones.⁽¹²⁵⁾ For this we consider the role of fluid filled voids inside the solid rocks and their thermo-mechanical couplings. For simplicity the role of the fluid is assumed to be passive, i.e., simply acting as a void filling in stress equilibrium with its surrounding rocks. The voids are considered to be generated by a variety of micro-mechanisms, and their overall effect on the matrix is considered by damage mechanics. We show how the thermodynamic approach can be used to derive the incremental relationships. We also show that the regularisation of the problem is only achieved by the additional coupling to the shear heating instability which controls the width of the shear zone by the thermal diffusion length. By analogy any other of the THMC feedback can control the shear band width if the problem time and length scales are such that the diffusional problem is governed by other length scales such as hydraulic diffusivity or chemical diffusivity.

8.3. Creep Fracture with T(H)M Coupling

In the following we review the construction of a particular lithospheric damage mechanics model based on the new findings reported in the microstructure analysis workflow for the sample shown in Figure 10. In this sample the centre of the shear zone is characterised by creep cavities that are dynamically self-organized in ductile shear bands, analogues to ductile failure phenomena in metals and ceramics.^(57–59)

In earlier models of lithosphere rheology^(126–129) it was shown that shear heating feedbacks have considerable influences on stress and strain localisation. These localisation bands appear as kilometre-wide shear zones dissecting the lithosphere. Thermal-mechanical feedback is reported to weaken the lithosphere by at least a factor of four compared to the classical values derived from consistency plasticity approaches without feedback. The shear heating feedback, therefore can weaken the lithospheric plate sufficiently such that they can deform in response to natural forces available from slab pull or rigid push estimates.⁽¹³⁰⁾

The shear heating weakening mechanism works very well for materials with high activation energy such as mantle rocks where olivine minerals are the main mechanical constituent.⁽⁶³⁾ However, in the granitic crust⁽¹³¹⁾ where quartz and feldspar dominate the rheologies, the shear heating weakening and localisation effects are not enough to explain geological observations. The discovery of grain boundary voids observed in granites deformed under natural conditions⁽⁵⁵⁾ gave first geological support for a ductile dilatant damage mechanical weakening effect. Dilatant damage was first proposed as a solution proposed to solve the problem of fluid transfer through the ductile realm.⁽¹³²⁾ The early model was showing the possible effect of ductile void growth using the classical Gurson criterion for ductile fracture and later extensions thereof.^(133–135)

Similar to the ductile fracture hypothesis of the lithosphere, damage mechanics^(58, 136–138) was also proposed as

a suitable model for instabilities in the brittle crust,⁽¹³⁹⁾ with poroelastic⁽¹⁴⁰⁾ or visco-elastic extensions.^(141, 142) Although suitable for the study seismic events or wave propagation in geological structures under small perturbations, such models do not allow for plastic and visco-plastic analysis where permanent deformations play a crucial role. An alternative model that was entirely based on a fluid dynamical approach was proposed.^(125, 143)

Both brittle/ductile fracture modelling and fluid dynamical approaches include important elements but failed to completely reproduce the field findings from Ref. [55] which clearly indicate that the void growth and coalescence cannot be modelled with an ideal plastic mechanism, but rather needs to explicitly consider the effect of time to accommodate the grain boundary sliding mechanism with a pore fluid derived from precipitation dissolution reactions in the central part of the shear zone. On the other hand, the percolation analysis discussed in Figure 10 also revealed that the shear zone as a whole behaves like a solid. Although being close to the percolation threshold, defining the boundary of solid and fluid dynamics, it does not breach the solid state in the integrated geological record except perhaps for sudden fluidisation events in the very centre of the shear zone during creep fracture.⁽⁵⁶⁾ Hence we cannot model the shear zone simply as a fluid dynamic instability but must consider a dual material behaviour. An elasto-visco-plastic solid mechanical approach needs to be incorporated for a formulation of void nucleation and coalescence dominating the long term response of geological materials. If we are interested in the short time scale phenomenon during fluidisation, the formulation needs to be also able to capture events that may lead to a fluid-like state in the centre of the shear zone. In the following section we will first discuss the long time scale phenomenon and formulate the solid mechanical development of the shear zone. The chemical evolution inside the shear zone will be discussed using a different example.

The following section serves as a specific example for implementation of an additional thermodynamic state variable (here the creep damage parameter D) into the *over-stress plasticity* approach. As the theoretical formulation is suitably generic, we will focus only on the procedure of formulating a thermodynamic force (here the damage force Y) and the associated thermodynamic flux (here the damage flux $\dot{D}$) of the particular state variable chosen. Details on the numerical methods will not be presented and the reader is referred to the cited literature.

8.3.1. Thermo-Mechanical Background

This problem is an ideal showcase for the implementation of the thermodynamic framework⁽⁹¹⁾ to develop the continuum damage mechanics of geological materials subjected to thermo-mechanical loading.^(144, 145) We use the thermodynamically derived additive decomposition of elastic and plastic strain rates (Eq. (15)) and limit ourselves to the

theory of small deformations. The integration procedure presented here works equally well in case of large transformations, as described by Ref. [146]. As stated in Section 6, the specific Helmholtz free energy ψ can be expressed in terms of observable variables (such as strain, ϵ_{ij} , and Temperature, T) and the internal variables (such as damage D , elastic strain, inelastic strain and other dissipation quantities, which we ignore in this study): $\psi(\epsilon_{ij}^e, T, D)$. Hence, the specific internal energy can be expressed as follows:

$$u(\epsilon_{ij}^e, s, D) = \psi(\epsilon_{ij}^e, T, D) + sT \quad (37)$$

where s denotes the specific entropy. In addition, using the local form of conservation of energy shows that:

$$\rho \dot{u} = \sigma_{ij} \dot{\epsilon}_{ij} + r - \nabla q_k^T \quad (38)$$

where ρ is the material density, q_k^T is the heat flux vector, and σ_{ij} is Cauchy's stress tensor. The above equation summarises the first principle of thermodynamics in its local form. It shows that the local internal energy is equal to the internal work augmented with the local heat production and transfer. Combining Eqs. (37) and (38) results in:

$$\rho \dot{\psi} + \rho T \dot{s} + \rho s \dot{T} = \sigma_{ij} \dot{\epsilon}_{ij} + r - \nabla q_k^T \quad (39)$$

The above expression along with the local form of the second principle of thermodynamics results in the following Clausius-Duhem inequality:

$$\sigma_{ij} \dot{\epsilon}_{ij} - \rho \dot{\psi} - \rho s \dot{T} - \frac{q_k^T}{T} \nabla T \geq 0 \quad (40)$$

The above inequality states that the rate of irreversible entropy production is always positive. By using the dependencies of Helmholtz free energy on the state variables of the problem, it can be seen that:

$$\dot{\psi} = \frac{\partial \psi}{\partial \epsilon_{ij}^e} \dot{\epsilon}_{ij}^e + \frac{\partial \psi}{\partial T} \dot{T} + \frac{\partial \psi}{\partial D} \dot{D} \quad (41)$$

Therefore, using the additive decomposition of strain, Eq. (40) can be rewritten as follows:

$$\begin{aligned} & \sigma_{ij} \dot{\epsilon}_{ij}^p + \left(\sigma_{ij} - \rho \frac{\partial \psi}{\partial \epsilon_{ij}^e} \right) \dot{\epsilon}_{ij}^e - \rho \left(s + \frac{\partial \psi}{\partial T} \right) \dot{T} \\ & - \rho \frac{\partial \psi}{\partial D} \dot{D} - \frac{q_k^T}{T} \nabla T \geq 0 \end{aligned} \quad (42)$$

As the tensor products involved in Eq. (42) have to hold for every admissible process, the following relationships can be obtained:

$$\sigma_{ij} = \rho \frac{\partial \psi}{\partial \epsilon_{ij}^e} \quad (a) \quad \text{and} \quad s = - \frac{\partial \psi}{\partial T} \quad (b) \quad (43)$$

These considerations simplify Clausius-Duhem's inequality (40). The remaining terms state that energy dissipation is locally positive:

$$\mathcal{D} = \sigma_{ij} \dot{\epsilon}_{ij}^p - \rho \frac{\partial \psi}{\partial D} \dot{D} - \frac{q_k^T}{T} \nabla T \geq 0 \quad (44)$$

In addition, combining Eqs. (38) and (43) results in the following equation of local entropy variation:

$$\rho T \dot{s} = \sigma_{ij} \dot{\epsilon}_{ij}^p - \rho \frac{\partial \psi}{\partial D} \dot{D} + r - \nabla q_k^T \quad (45)$$

From Eq. (43b) and the dependencies of Helmholtz free energy it can be seen that

$$\dot{s} = - \frac{\partial^2 \psi}{\partial T \partial \epsilon_{ij}^e} \dot{\epsilon}_{ij}^e - \frac{\partial^2 \psi}{\partial T^2} \dot{T} - \frac{\partial^2 \psi}{\partial T \partial D} \dot{D} \quad (46)$$

With the cross derivative in the above equation and considering only the second order derivative terms with respect to time at fixed volume we obtain: $T \dot{s} = -T(\partial^2 \psi / \partial T^2) \dot{T} = C_v \dot{T}$, where C_v is the heat capacity at fixed volume. These simplifications lead to the equation of heat transfer which is often expressed as follows:

$$\rho C_v \dot{T} = \kappa \frac{\partial^2 T}{\partial x_k^2} + \sigma_{ij} \dot{\epsilon}_{ij}^p + Y \dot{D} + r_k \quad (47)$$

where we used the Fourier law $q_k^T = -\kappa \nabla T$ such that κ is the thermal conductivity. We also expressed the thermodynamic force of damage (triaxiality) as: $Y = -\rho(\partial \psi / \partial D)$, which is a scalar in the current study as we consider isotropic materials. Apart from the equation of conservation of energy, understanding the behaviour of geomaterials requires solving the following equation of mechanical equilibrium:

$$\nabla \sigma_{ij} + \rho g = 0 \quad (48)$$

The constitutive behaviour of geomaterials considered herein is damageable elasto-visco-plastic. The elastic component is linear and isotropic which means that Eq. (43-a) reads:

$$\sigma_{ij} = C_{ijkl} \epsilon_{ij}^e \quad (49)$$

where $C_{ijkl} = (K - 2/3G)\delta_{ij}\delta_{kl} + G(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk})$ is the fourth order elasticity tensor of the damaged material, K is the bulk modulus, G is the shear modulus, δ_{ij} is Kronecker's symbol, and (i, j, l, k) are indices representing the Euclidian directions. For simplicity, we use Lemaitre and Chaboche's⁽¹³⁶⁾ description, such that $C_{ijkl} = (1 - D)C_{ijkl}^0$ describes the relationship between the damaged and intact behaviours. Note that this is not the unique option as explained in details by Refs. [139, 142]. Hence the thermodynamic force, Y , associated with damage evolution is given by:

$$Y = -\rho \frac{\partial \psi}{\partial D} = \frac{\sigma_{eq}^2}{2(1-D)^2} \left[\frac{1}{3G} + \frac{1}{K} \left(\frac{p}{\sigma_{eq}} \right)^2 \right] \quad (50)$$

where $p = 1/3 \text{tr}(\sigma_{ij})$ is the hydrostatic pressure, and $\sigma_{eq} = \sqrt{3/2 \sigma'_{ij} \sigma'_{ij}}$ is the equivalent stress.

8.3.2. From Dissipation to Flow Rules

As stated in the last section, we assume that the behaviour of geomaterials at high temperature is elasto-visco-plastic associated with continuum damage mechanics. Implementing this behaviour requires the introduction of a regular potential of intrinsic dissipation $\phi(\sigma_{ij}, Y)$, which depends on the thermodynamic forces which contribute to dissipation. An additive decomposition can be used to account for the contributions of visco-plastic deformation and damage mechanics:

$$\phi(\sigma_{ij}, Y) = g_\sigma(\sigma_{ij}) + g_Y(Y) = 0 \quad (51)$$

The principle of maximum dissipation along with the above constraint shows that:

$$\begin{cases} \dot{\epsilon}_{ij}^p = \dot{\lambda} \frac{\partial g_\sigma}{\partial \sigma_{ij}} \\ \dot{D} = \dot{\lambda} \frac{\partial g_Y}{\partial Y} \end{cases} \quad (52)$$

The behaviour is considered associated in this particular study. This means that the elasticity envelope (or the threshold delimiting the elastic behaviour) coincides with the inelastic deformation up to an additive constant. Therefore, it can be stated that the plastic potential g_σ is defined by:

$$g_\sigma = \tilde{\sigma}_{eq} - \sigma_0(\epsilon_{eq}^{in}) - \tilde{\phi}^{[-1]}(\epsilon_{eq}^{in}) \quad (53)$$

where $\tilde{\phi}$ is an invertible function relating equivalent stresses to equivalent strains, $\tilde{\sigma}_{eq} = \sigma_{eq}/(1-D)$ is the effective stress, and $\epsilon_{eq}^{in} = \sqrt{\epsilon_{ij}^p \epsilon_{ij}^p}$ is the equivalent inelastic deformation. By denoting $\hat{\sigma}_{eq} = \langle \tilde{\sigma}_{eq} - \sigma_0 \rangle$, where $2\langle x \rangle = x + \|x\|$, we can express the invertible function of viscous behaviour as follows:

$$\begin{aligned} \epsilon_{eq}^{in} = \tilde{\phi}(\hat{\sigma}_{eq}) = A_d \frac{\hat{\sigma}_{eq}}{g_d^3} \exp\left(-\frac{Q_d}{RT}\right) \\ + A_p \hat{\sigma}_{eq}^n \exp\left(-\frac{Q_p}{RT}\right) \end{aligned} \quad (54)$$

where A_q ($q = d, p$) are the pre-factors of the diffusion and dislocation mechanisms respectively, Q_q are the activation energies, g_d denotes the grain size, and R the gas constant. As for continuum damage mechanics, we use the potential derived by Ref. [147] based on upscaling considerations^(57, 58, 148)

$$\phi_{De} = Y \left(\frac{1}{(1-D)^{n+1}} - 1 \right) + \Theta(Y) \quad (55)$$

where n is the exponent of the dislocation mechanism and Θ describes the nucleation of voids and micro-defects. The particularity of the above damage potential as compared to expressions postulated by Lemaitre, Chaboche, and Bonora,^(136, 138, 149) is that damage evolution is zero in

the absence of initial damage or voids nucleation. Damage nucleation in this framework can be expressed as: $\Theta(Y) = \mathcal{H}/\kappa + 1(Y/\mathcal{H})^{\kappa+1}$, where $\mathcal{H}$ and κ are material constants. Combining Eqs. (56), (53) and (55) shows that the flow rules read:

$$\begin{cases} \dot{\epsilon}_{ij}^p = \dot{\lambda} \frac{\partial g}{\partial \sigma_{ij}} = \dot{\lambda} \frac{3}{2} \frac{s_{ij}}{\sigma_{eq}} \\ \dot{D} = \dot{\lambda} \left(\frac{1}{(1-D)^{n+1}} - 1 + \left(\frac{Y}{\mathcal{H}} \right)^\kappa \right) \end{cases} \quad (56)$$

where $s_{ij} = \sigma_{ij} - \sigma_{kk} \delta_{ij}$ is the deviatoric stress.

The constitutive model described in this paragraph was implemented numerically using the fortran material subroutine UMAT of Ref. [150]. The purpose of this analysis is to illustrate the effects of creep damage and shear heating on localisation.

8.3.3. The Roles of Shear Heating and Damage in an Idealised Necking Lithosphere

The purpose of this analysis is to illustrate the effects of creep damage and shear heating on localisation. Reference [151] proposed a simple numerical model of a notched lithospheric layer in extension (see Fig. 24). The model is defined by a 100 km deep and 800 km long plane strain cross-section, with a 4 km square notch in the middle of the upper edge. We select free slip boundary conditions on the top and right edges, a free displacement on the bottom side, and an extension velocity of 20 mm/yr at the right hand side of the lithospheric layer are applied while the olivine slab is initially set to 978 K throughout the whole domain.

For simulation, we use the plain strain isoparametric elements which solve for coupled temperature-displacement problems. The initial size of each element is of about 250 m. A convergence study was performed to ensure a proper interpretation of the results. The material properties used for simulation are summarised in Table III.

The case for shear heating only is described by Ref. [151] and in Figure 25 compared to the case of adding additionally the aspect of continuum damage mechanics. In both cases, the faster the loading process the stiffer is the geomaterial behaviour. For comparable loading rates,

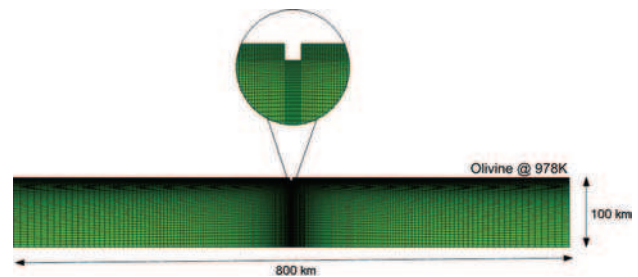


Fig. 24. An idealised lithosphere model for testing the effect of shear heating and creep damage after Ref. [151].

Table III. Simulation parameters: Dissipation constants (*the subscripts $q = p, d$ refer to power law and diffusion respectively. The same order is valid for the following two rows. μm^3 is included only if $q = d$).

Mass	Density ρ (Kg m^{-3})	3300
Elasticity	Young modulus, E (GPa)	10
	Poisson's ratio, ν	0.3
Thermo-mechanics	Thermal conductivity, k ($\text{W m}^{-1} \text{K}^{-1}$)	3.4
	Specific heat, c_p ($\text{J Kg}^{-1} \text{K}^{-1}$)	1240
	Expansion coefficient, α (K^{-1})	1.2×10^{-5}
Thermal feedback	Inelastic heat fraction, ξ	0.9
Creep mechanisms	Universal gas constant, R ($\text{J mol}^{-1} \text{K}^{-1}$)	8.3144
	Mechanism's exponent, n_q^*	$n_p = 3, n_d = 1$
	Prefactor A_q ($\mu\text{m}^3 \text{MPa}^{-n} \text{s}^{-1} \text{K}^{-1}$)	1.5×10^3 4.8×10^4
	Activation energy Q_q (kJ mol^{-1})	470.0 295.0
	Activation energy A_q ($\text{J mol}^{-1} \text{K}^{-1}$)	8.3144
	Grain size g_d (μm)	15
Damage parameters	Critical damage, D_{cr}	0.85
	Normalising term, $\mathcal{H}$ (MPa)	6
	Triaxiality exponent, κ	2
	Critical inelastic deformation, ϵ_0	0.01

geomaterials weaken considerably and produce distinguishable shear zones which appear as shear bands in the numerical simulation. Significant localisation develops in these shear zones owing to the considerable energy dissipation within the shear bands.

As the applied load increases, weakening zones propagate from the notch. Figure 26 shows the contours of temperature and permanent deformation obtained after 1 Myrs and 3 Myrs of loading, both for damageable and non-damageable behaviours. At $t = 1$ Myrs, it can be seen in Figures 26(a), (c) that the temperature and permanent deformation are quite similar. This is because damage did not propagate excessively as can be seen in Figure 27. Yet the slight damage concentration of damage in the vicinity of the notch produced a permanent deformation of about 0.3 in case of damageable materials and 0.25 in case of non-damageable materials. At $t = 3$ Myrs, the difference is more pronounced as shown in Figures 26(b), (d). It can be seen that permanent deformation reached 1.5 in case of damageable behaviour and 0.5 in case of non damageable behaviour. Subsequently, temperature profiles show that shear heating is larger in case of damageable structures.

The results show that damage (see Fig. 27) accelerates weakening of the geomaterial and increases inelastic deformation. Consequently, the amount of shear heating is larger in damageable structures. In addition the overall response is highly affected as for the same rate of loading the resulting forces predicted in damageable structure is much higher.

A surprising result of the simulations is the strong feedback between damage and the modest a shear heating

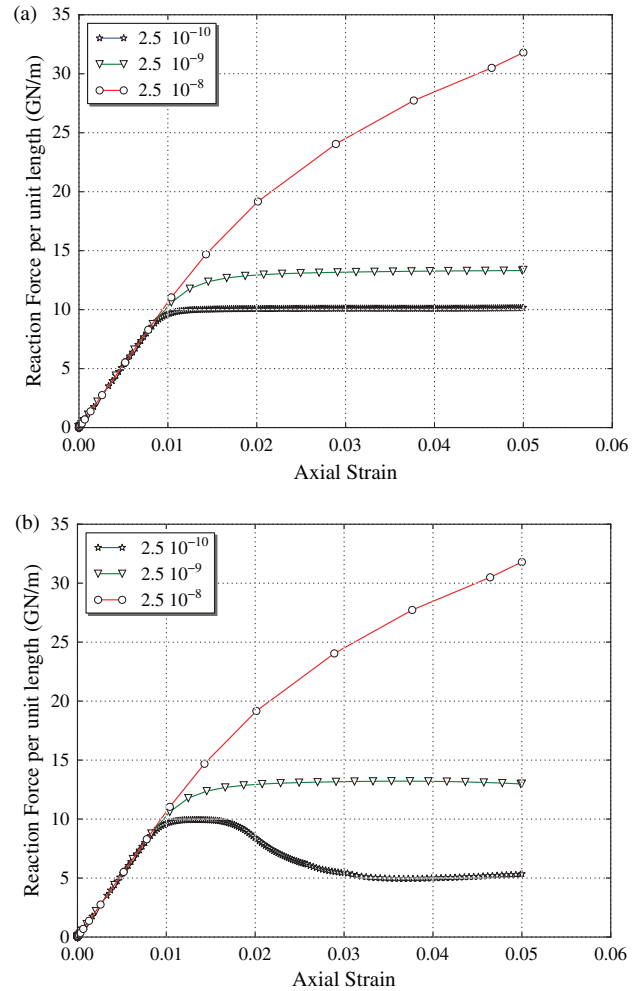


Fig. 25. Rate dependent response of geomaterials in case of: (a) non damageable structures and (b) damageable structures, under different loading strain rates (1/s).

of only a few degrees which may be considered negligible if we are only interested in a thermal solution of the problem. However, the mechanical impact of shear heating is profound. It takes a first order control on the width of the shear band as shown in the following discussion. Figure 28 shows the damage accumulated after 3 million years. The damage varies along a sharp gradient from 0 to 0.85, the maximum critical value allowed in the model. Creep damage considerably weakens the lithosphere and the inelastic deformation is 3 times higher than in the reference case of shear heating only. While the width of the shear zone is still controlled by the thermal diffusion length scale the thermal pattern is similar to the one obtained in the reference shear heating case. However, since a significant portion of the deformational work is converted into damage the overall differential magnitude of shear heating is much lower. It varies from about 7 degrees in the shear heating reference case down to 4 degrees when damage is considered.

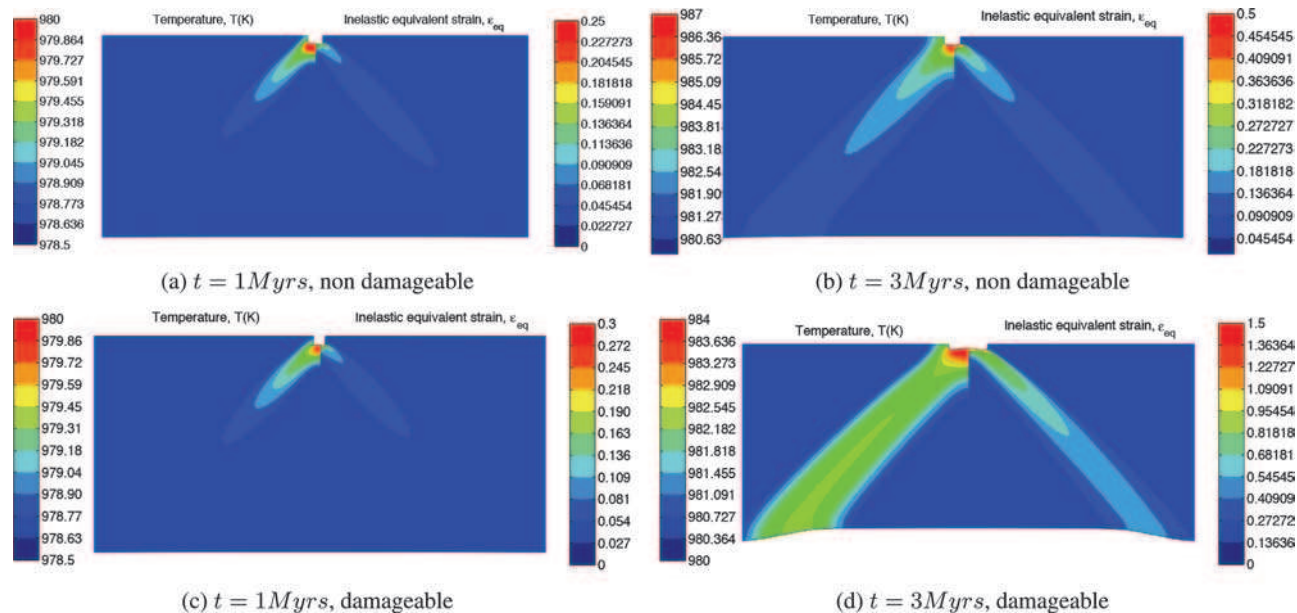


Fig. 26. Response of the structure in terms of temperature and permanent deformation in case of non damageable and damageable structures.

Other than the additional weakening the addition of the damage mechanics effect introduces a strong rate sensitivity of the localisation on loading rate. The creep damage effect is bounded by two extremes. At extremely high rates the creep damage cannot operate since the damage relies

on the activation of the creep processes necessary to grow voids. At extremely low rates the effect of damage is negligible. Creep damage has an optimal time scale of weakening when the deformation process operates at geological rates and lasts for several million years. Creep damage was found to contribute significantly to the material softening and reduces the previously estimated reaction forces in the lithosphere, especially at low plate tectonic loading rates. This effect is entirely overlooked in the laboratory and can only be explored in a numerical analysis.

While this analysis shows the power of the thermodynamic approach it also highlights the transition from a solid mechanical to a fluid dynamic solution at an arbitrary value of 0.85. Ideally this transition needs to be constrained from a percolation analysis such as discussed earlier or from additional theoretical considerations. In the granite case (Fig. 10) the percolation analysis would predict a transition to a fluidized state at significantly lower value than set in the above example. While in classical damage mechanics the material would have negligible strength the advantage of the overstress formulation is that the strength is defined by the effective viscosity in the fluidized state (red portion in Fig. 28). In the next section we turn our attention to the interesting behaviour of the fluidized state which can feature yet another additional localisation mechanism through the post-failure evolution inside the shear zone. We will show in an applied example that this ultra-localised behaviour is common in geological structures.

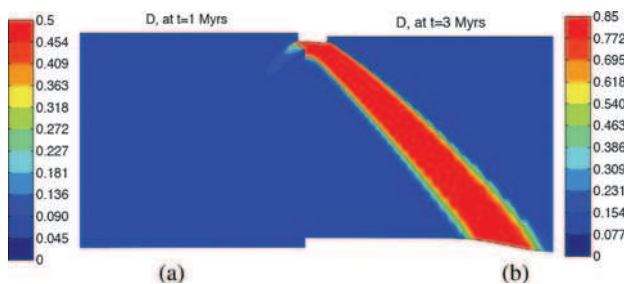


Fig. 27. Contour of damage distribution after (a) 1 Myrs and (b) 3 Myrs of loading.

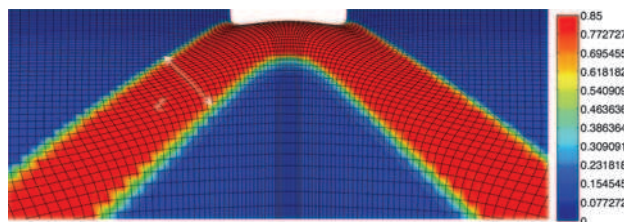


Fig. 28. Damage evolution after 3 Million years extension. The damage variable is capped at 0.85. The notorious mesh sensitivity of damage mechanics (in the initial deformation damage starts on single elements) can be buffered by considering additional feedback processes such as shear heating. The model shows that at steady state the shear band width L is of the same order of magnitude as the thermal diffusion length scale.

9. POST-FAILURE EVOLUTION OF FAULTS

Once the fault has been formed, it continues to deform plastically (irreversible deformation) upon continuation of

loading. Outside the fault however, elastic unloading could take place initially followed by rigid body motion on the fault's boundary. This fact restricts our modelling efforts to the behaviour of the material inside the formed fault, where stress equilibrium requires that the stresses across the fault are uniform, $\sigma'_{ij} = \sigma'_{ij}(t)$.^(117, 152–154)

This post-failure area requires additional information from the material in order to be modelled. Indeed, at stress states above the initial yield stress, $\sigma_{ij} > \sigma_Y$ (or $\pi_{ij} > 0$), the two plasticity approaches diverge. In this regime *consistency plasticity* needs to be provided with an evolution law for the hardening modulus, that expresses the response of the yield surface to post-yield loading (hardening/softening law). This evolution law is introducing weak rate effects and is commonly derived through experiments in laboratory scales.

Overstress plasticity on the other hand does not require this evolution law. Having an intrinsic rate of loading implemented through Eq. (17), the rate of plastic strain is expressed as

$$\dot{\epsilon}^p_{ij} = \sum_m f^{(m)} \pi'^m_{ij} \quad \text{when } \pi_{ij} > 0 \quad (57)$$

where $f^{(m)}$ has now dimensions of $(\text{Pa} \cdot \text{s})^{-1}$, being thus the inverse of the material's viscosity. In analogy to experimental validation required in *consistency plasticity*, Eq. (57) needs to be calibrated through experiments or field data, to obtain the values of the viscosity and the rate sensitivity exponent m .

We remind that f is a function of temperature T and the internal state variables ξ_k apart from the stress, as defined in Eq. (16). Hence, the material's viscosity $\eta = 1 / \sum_m f^{(m)}$ is in principle a function of temperature and ξ_k .

$$\eta = \eta(\sigma_Y, T, \xi_k) \quad (58)$$

This modification requires the evolution laws of temperature and ξ_k to be prescribed. The evolution of the internal variables, as well as their influence on the yield surface, can be provided either through experiments or from first principles. This would provide equations of the form

$$\dot{\xi}_k = \Xi(\sigma_Y, T, \xi_k) \quad (59)$$

at microstructural steady state, where $\Xi(\sigma_Y, T, \xi_k) = 0$ a hypersurface is defined for the evolution of the yield surface with temperature and ξ_k . Such hypersurfaces are frequently used in constitutive modelling of geomaterials.^(155–158)

Unlike the evolution of internal variables, the temperature evolution is not obtained through experiments. Rather than that, it obeys the differential equation derived through the energy balance and the second law of thermodynamics. We generalise Eq. (47) to consider the presence of a fluid phase and other internal variables and write:^(117, 152)

$$(\rho C) \dot{T} = \kappa \frac{\partial^2 T}{\partial x_k^2} + \sigma'_{ij} \dot{\epsilon}^p_{ij} + Y \dot{\xi}_k + r_k \quad (60)$$

where ρ is the density of the material, C and κ its specific heat and thermal conductivity at the x_k coordinate, respectively, Y the energy conjugate of the internal variable ξ_k and r_k heat source/sink terms due to chemical reactions.

In the present study we choose to work in the overstress plasticity framework. In order to complete the system of equations we need to prescribe the loading path for the stresses, the expression of the viscosity (Eq. (58)), the evolution law of the internal variables (Eq. (59)) and the properties of the chemical reaction.

Veveakis et al.⁽¹¹⁷⁾ showed that the laws frequently met in the literature provide similar physical response, with the Arrhenius expression being the most general one. Hence we choose

$$\eta = \eta_0 e^{E_d/RT} \quad (61)$$

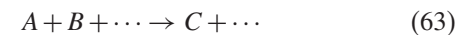
where $\eta_0 = \eta_0(\sigma_Y, \xi_k)$ and E_d/R the thermal sensitivity of the viscosity. Following these considerations, and in the absence of chemical reactions, Eq. (60) now writes

$$(\rho C) \dot{T} = \kappa \frac{\partial^2 T}{\partial x_k^2} + \frac{\pi'^{(m+1)}_{ij}}{\eta_0} e^{-E_d/RT} + Y \dot{\xi}_k \quad (62)$$

9.1. Chemical Reactions

In the above described formalism where post failure conditions are described by the energetics of shear heating and damage the chemical reaction term $\Delta H \cdot r$ is of crucial importance, since the system could predict unrealistic temperature values through the extreme localisation leading to excessive shear heating.^(117, 152) Any endothermic chemical reaction that the material could admit during metamorphism will be triggered at high loading conditions, restricting significantly the maximum temperature reached. The ultimate endothermic reaction is melting buffering the temperatures to finite amplitudes.

As far as the chemical properties are concerned, we briefly summarize the principles of modelling chemical reactions in this section. Chemical reactions typically involve the absorption or release of heat. In geomechanical applications and especially in fault mechanics endothermic reactions like chemical thermal pressurisation, phase transition (decomposition) or melting of the solid phase are processes that require considerations from reaction kinetics and thermodynamics in order to be modeled. A simple arbitrary reaction of the form



has a reaction rate r that in general depends on the temperature and the reactants' concentration. A common supposition for the reaction rate is the Arrhenius term

$$r \sim \exp[-E/RT] \quad (64)$$

where E is the reaction's activation energy. After the above considerations, in the presence of any chemical reaction,

the system of equations that govern the problem are:

$$\begin{aligned} r^* &= c^* \exp \left[\frac{Arw\theta}{1+w\theta} \right] \\ \frac{\partial \theta}{\partial t^*} &= \frac{\partial^2 \theta}{\partial x_k^{*2}} + \left[\frac{Gr}{c^*} \exp \left(\frac{aAr}{1+w\theta} \right) \pm 1 \right] r^* \\ \frac{\partial c^*}{\partial t^*} &= \frac{1}{Le} \frac{\partial^2 c^*}{\partial x_k^{*2}} - \mu_r r^* \end{aligned} \quad (65)$$

where $\pm$ signifies exothermic (+) or endothermic (−) reaction, c is the concentration, and

$$\begin{aligned} z^* &= \frac{x_2}{(D/2)}, \quad t^* = \frac{\kappa_m}{(D/2)^2} t, \\ \theta &= m(T - T_0), \quad c^* = \frac{c^{(a)}}{c_0} \\ m &= \left[\frac{r_0 |\Delta H|}{jk_m} \left(\frac{D}{2} \right)^2 \right]^{-1}, \quad r_0 = k_0 c_0 e^{-E/RT_c} \\ Gr &= m \frac{\tau_n \dot{\gamma}_0}{jk_m} \left(\frac{D}{2} \right)^2 \frac{r_0}{k_0 c_0} = \frac{\tau_n \dot{\gamma}_0}{k_0 |\Delta H| c_0} \\ w &= \frac{1}{mT_c} = \frac{r_0 |\Delta H|}{jk_m T_0} \left(\frac{D}{2} \right)^2 \\ Ar &= \frac{E}{RT_0} a = \left(1 - \frac{x}{N} \right) \\ Gr &= \kappa_m / D_C, \quad \mu_r = \frac{r_0}{\kappa_m c_0} \left(\frac{D}{2} \right)^2 \end{aligned} \quad (67)$$

The dimensionless parameters appearing in (9.1) are discussed in depth in Ref. [117]. Here we emphasize the role of the Gruntfest number Gr , expressing the ratio of the characteristic time scale of heat production over the characteristic time scale of energy transfer due to the heat absorbed/released by the chemical reactions.

In a quasi-static evolution of the system, the Gruntfest number remains constant, while boundary temperature (strain rate) may vary with time. In the case of monotonic increase of the temperature, the response diagrams for various boundary temperatures are depicted in Figure 29. The steady state problem has 1, 2, or 3 solutions depending on the value of Gr . It is to be noted that the middle branch of Figure 29 is unstable corresponding to a localisation instability (see the strain rate profiles in Figs. 29(b)–(d)).⁽¹¹⁷⁾ Hence, as temperature increases, deformation localizes inside the initial fault core (shear zone) of thickness D , which was obtained during initial failure (Section 3). This localisation stops when the chemical reaction is triggered, forming an ultrathin, chemically altered PSZ. This fact verifies the results of the linear stability analysis performed by Veveakis et al.⁽¹⁵⁹⁾ revealing the localizing nature of temperature and the role of the reaction parameters on the PSZ thickness.

In Figure 29 we observe that the lower turning point significantly decreases in Gruntfest number with increasing

θ_{bound} , while at the same time the upper point presents a smaller but apparent reduction. At a specific dimensionless core temperature θ_s they degenerate to a single (inflection) point with Gruntfest number Gr_s . Given an initial boundary temperature, we may recognize three stability areas with respect to Gr (see Fig. 29).

9.1.1. The Area of Low Loading Conditions, $0 < Gr < Gr_s$

In this area the system has a single stable solution, the lower branch of the response curve. It lays between the frozen state of $Gr = 0$ and the stretched threshold, Gr_s . It is expected that in this area a fault creeps stably and aseptically under any initial temperature.

9.1.2. The Area of Intermediate Loading Conditions, $Gr < Gr < Gr_c$

For intermediate Gruntfest numbers the system has three solutions, two stable and an unstable one. For a fault that creeps under monotonically increasing core temperature and admits all these branches, a transition from aseismic creep to seismic slip is possible, since beginning from an initial temperature (e.g., from the point 1 of Fig. 29) there is a critical temperature at which this process becomes unstable (the point 2 of Fig. 29). Past this point, temperature and strain-rate localizes in an ultrathin core shear band with a simultaneous abrupt increase of the temperature, establishing essentially adiabatic conditions (Fig. 29(c)). Since in the present formulation temperature and velocity are coupled through the plasticity law, the unstable thermal evolution corresponds to a rapid velocity increase (acceleration) as well.⁽¹⁵³⁾ This velocity and temperature increase, representing what is known to material science as tertiary creep, ends by triggering inside the fault the chemical reaction (point 4 of Fig. 29) that has the lower activation energy among the possible reactions that the material under study may exhibit. Past this point, the present model loses its validity as chemical reaction sets in and dominates the response of the fault.

9.1.3. The Area of High Loading Conditions, $Gr > Gr_c$

When the initial loading condition is high, the system has one stable solution, the upper branch that corresponds to the initiation of the chemical reaction; in this area of high Gr , after a rapid evolution, the system enters directly the reaction regime and obviously its behavior depends on the nature of the reaction (as described previously). In real problems, a fault would accelerate rapidly due to high loading conditions, and trigger a reaction without creeping. The reaction would take place under adiabatic conditions in a shear band with a width that is defined from the initial and boundary conditions, as well as the characteristics of the reaction.⁽¹⁶⁰⁾

In the following example we will discuss an example and compare it to a chemically induced localisation

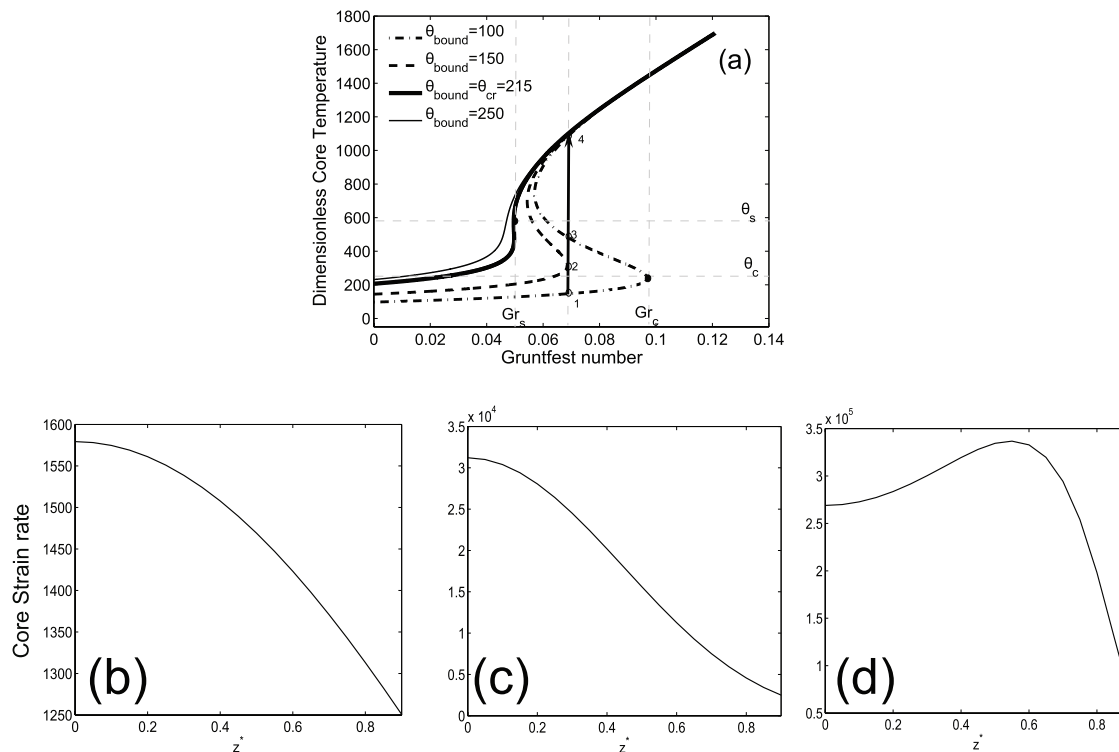


Fig. 29. (a) Response diagram of Eq. (65) for various values of the boundary temperature. The diagrams below present the core strain rate for a specific Gruntfest number, at: (b) the lower stable branch (point 1 of (a)); (c) the unstable middle branch (point 3); and (d) the upper stable branch (point 4). Figure from [117], E. Veveakis, et al., *J. Mech. Phys. Solids* 58, 1175 (2010). © 2010.

mechanism that emerges through the fluid dynamic feedback at the critical Gruntfest number Gr_c triggering a THMC oscillator. The example is at the Glarus UNESCO world heritage site in carbonate rocks of the Swiss Alps where chemical reactions are known to be associated with a zone of intense deformation.

9.2. Shear Zones: Glarus Case

Discussing THMC Coupling

With increasing depth and temperature, carbonates tend to deform by ductile creep under geodynamic loading rather than failing in a brittle manner. Nevertheless, we sometimes encounter highly localised deformation, such as that seen in mountain ranges formed by thrusts, where carbonates that deformed ductilely under temperature conditions in excess of 200 °C are now exposed at the surface. Well-known examples include the Glarus thrust⁽¹⁶¹⁾ in the Alps (Fig. 30), the McConnell thrust in the Rocky Mountains⁽¹⁶²⁾ and the Naukluft thrust in Namibia.⁽¹⁶³⁾ Those highly localised mm-cm thick shear zones (“Slip Planes”) are interpreted to have accommodated a total slip of over 30 km over periods of a few million years. The shear zones are embedded in a pervasively deformed zone, of the order of a metre thickness (termed the “Process Zone”), exhibiting significant chemical alteration and mechanical deformation (Fig. 1(b)). Outside of the Process Zone, chemo-mechanical deformation is much

less intense and in some places the rock even appears intact.

The Process Zone displays evidence for two dramatically different deformation rates.^(161,163,164) The creeping flow regime surrounding the Slip Plane is characterised by slow deformation processes, displacing at millimetres per year. The Slip Plane, however, shows brittle deformation with fluid injection structures (hydraulic fracturing), which imply seismic velocities of millimetres per second or more. It appears therefore that there is geological evidence for earthquakes in the ductile regime, cascading over multiple temporal and spatial scales. An additional observation from the aforementioned sites is that these seismic events appear to occur in a repetitive manner (Herwegh pers.comm.).

We propose a model that reconciles these seemingly paradoxical spatial and temporal cascades. We hypothesise that a single mechanism exists that causes the slowly creeping Process Zone to suddenly localise in its centre, triggering fast slippage on the Slip Plane. By accounting for the reported chemo-mechanical feedback inside the Process Zone, all the apparently uncorrelated events become part of the same temporal sequence.

Laboratory experiments,⁽¹⁶⁵⁾ theoretical studies⁽¹⁵⁴⁾ and geological observations of mineralogy and chemical alteration patterns^(164,166) independently suggest that carbonate decomposition is a key reaction during seismic events.



Fig. 30. The Glarus Thrust in the Swiss Alps (visible as a thin line with a strong colour contrast in the upper part of the photo) has experienced in excess of 30 km thrusting on meter wide creeping zone (here called Process Zone) that shows an internal knife sharp ultra localised deformation zone here called Slip Plane shown in Figure 32. Most of the displacement is presumed to be accommodated on the slip plane. We argue that the meter wide creeping zone is a solid mechanical instability and focus on the modelling of the ultra localised zone in the centre using a fluid dynamic approach. We thank Marco Herwegh for the field photo.

We therefore suggest carbonate decomposition as the reaction controlling the primary alteration in the Process Zone: $XCO_{3(s)} \rightleftharpoons XO_{(s)} + CO_{2(f)}$, with X representing any Group 2 cation, for example calcium or magnesium. Carbonate decomposition is an endothermic reaction and so requires the addition of heat to be activated. The generated CO_2 behaves as a supercritical fluid at these depths, whereas the solid oxide, being highly unstable, could react further. For the purpose of highlighting the principal processes only, we neglect these secondary events and idealise the rock composition inside the Process Zone to be pure carbonate XCO_3 , like calcite ($CaCO_3$).

We approach the problem with a 1D model shown in Figure 31. The undeformed hanging wall and footwall are taken as mechanical boundary conditions, where constant vertical and horizontal stresses are applied. We assume creeping flow inside the shear zone and focus on model the fluid dynamic behaviour inside the creeping zone. Note that the mechanical boundary conditions could be derived from a solid mechanical solution (as discussed above) using explicit THM coupling. The observed meter wide shear zone could be calculated by a fluid diffusional length scale as shown in Figure 4 but we emphasise here on calculating the possible source of the fluid and the mechanisms that may lead to fluid release.

The key ingredients of the physics considered in this simple model are mechanical dissipation appearing as shear heating^(21,24) that produces the heat required to trigger carbonate decomposition, which in turn will release supercritical CO_2 , causing chemical pressurisation. We investigate whether the resulting coupled system

of equations flows in a stable or unstable manner. The problem can be abstracted into two dimensionless parameters^(117,167,168): the Gruntfest number (Gr),

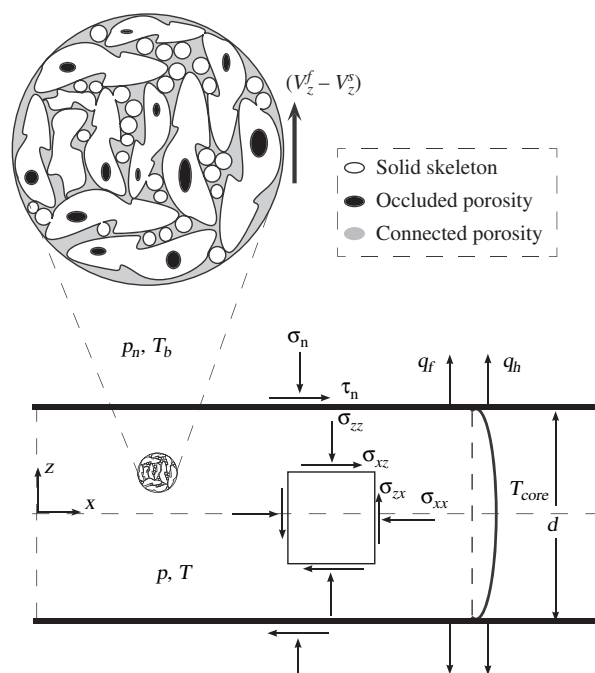


Fig. 31. Formulation of the 1D model after Ref. [117]. In the absence of experimental data at the grain scale, we assume that the chemical reaction is taking place at the grain-pore interface. Hence all the produced fluid contributes to the interconnected pore volume and is concentrated on the grain boundaries.

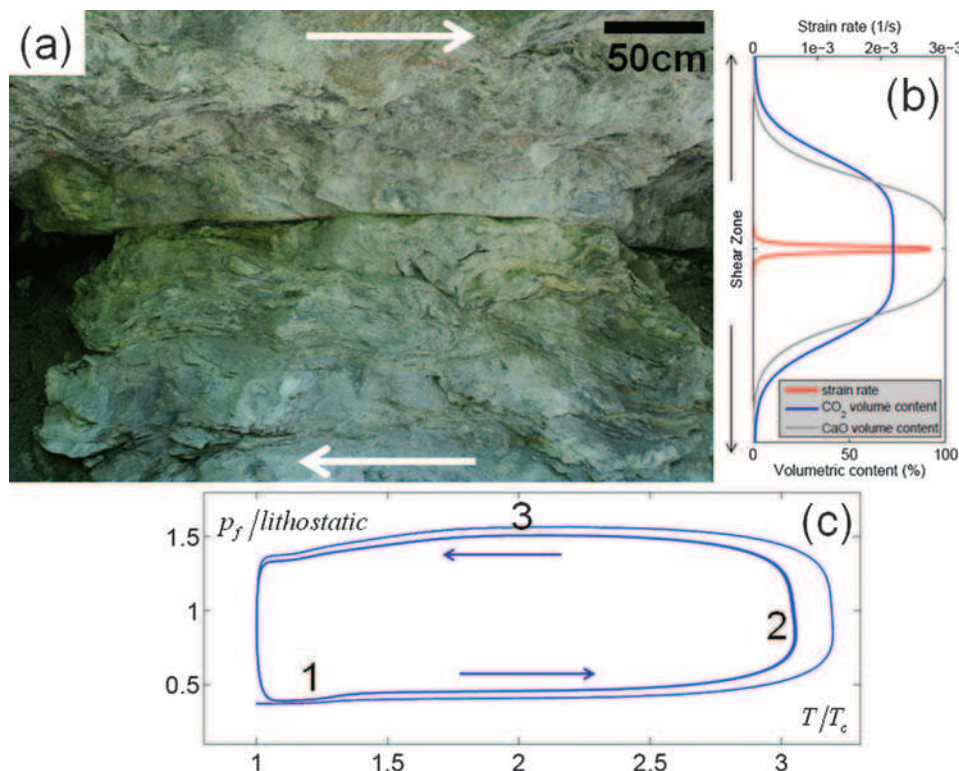


Fig. 32. (a) Photo from the Glarus thrust (Alps) illustrating the metre-wide shear zone (see Figure 30), at the centre of which an ultra-localised centimetre-thick Slip Plane is clearly visible. The chemical gradient (green coloured minerals) indicates a chemically altered zone inside the interface zone and around the Slip Plane. (b) Results of the model for carbonate decomposition. Spatial profiles of the strain-rate, porosity and volumetric content of CaO across the Process Zone at the maximum strain rate (point 2 in c). The co-location of mechanical deformations and chemical alterations is as observed in (a), highlighting the causal relationship between those two phenomena. (c) Calcite oscillator at the core of the Slip Plane. The cycle from shear heating to calcite decomposition (branch 1–2) causing ultra-high pressures (point 3) is sufficient to explain the major spatial patterns and the reasons for hydraulic fractures around the Slip Plane.

characterising the ratio of energy input as shear heating over the energy consumed by chemical decomposition, and the Lewis number (Le), which is the ratio of thermal over hydraulic diffusivity (see Eq. (9.1)), expressing how fast the temperature and pressure can diffuse away from the shear zone. For $Le > 1$ and $Gr > Gr_c$, the analysis reveals intriguing cycles of events.

This regime is succinctly described in Figure 32(c) by a thermal-chemical oscillator in the fluid pressure (P_f)—temperature (T) phase diagram. Starting at any P_f and T , the evolution of the P_f , T conditions will converge on a stable limit cycle. From point 1, close to lithostatic pressure and slightly elevated temperature, the model stays in the creeping flow regime, where temperature slowly increases (from point A to point B) due to shear heating outpacing thermal diffusion. When temperature reaches the threshold for carbonate decomposition (point 2), the cycle evolves rapidly Figure 32. The reaction releases super-critical CO₂, increasing fluid pressure and porosity until a maximum value of porosity and strain rate (point 2). Past this point, temperature decreases while fluid pressure becomes sufficiently high to cause hydraulic fractures (point 3). The fluid now lubricates the depleted central part

of the shear zone and diffusion is therefore the only active mechanism at play, slowly bringing the system back from point D to point A. Since full reversibility of the chemical reaction was assumed, this cycle will then repeat itself indefinitely. In nature, the presence of solid oxides (like CaO, MgO etc.) may trigger secondary reactions, which have not been considered here (e.g., hydration of CaO by the presence of H₂O). We postulate however that the chemical-thermal oscillator model provides the primary mechanism driving the reported shown in Figure 32(a) by the green halo identifying the chemo-mechanical deformation inside the Process Zone.

The puzzling spatial observations of three length-scales (Nappe, Process Zone, Slip Plane) are captured by our simple model. The undeformed kilometre-thick host rock (Nappe) is taken as a boundary condition for the Process Zone and we verify whether the Slip Plane emerges out of the chemical-mechanical oscillator. As input we assume a carbonate XCO₃ (calcite for Glarus) with an activation energy larger than the activation energy for creep.^(117,154) We assume typical pressure and temperature conditions sufficient to cause ductile creep of calcite. With this approach we obtain the profiles of strain rate, oxide

content and porosity displayed in Figure 32(b). Starting with an initial value of 1%, the porosity in the Slip Plane reaches 70% in a short period during the fast timescale instability in which the calcite is depleted. The instability is marked by a rapid increase in strain rate that is tightly localised into a central Slip Plane Figure 32(a) where the carbonate has fully decomposed. The strain rate on this central Slip Plane increases by three orders of magnitude during the instability. This chemo-mechanical localisation explains the spatial observations on the Glarus thrust as well as at McConnell and Naukluft.

10. DISCUSSION

In this review we have reported the rapid development in Geosciences that occurred largely over the past ten years in the area of multi-physics based, multiscale approaches (compare Ref. [169]). Although atmospheric sciences were the pioneers of multiscale formulations in earth sciences, the solid earth was up to the last decade deemed too difficult to describe by a fundamental theoretical approach. We have presented two different approaches that may drastically change this situation in the near future. One trend is a statistical mechanics based upscaling approach that considers discrete energy interactions in a hierarchical nesting of calculations that can be seen as the equivalent of quantum mechanical simulations extended to larger scales (see Ref. [4] for a short review). The other trend is to approach the problem from a thermodynamically based continuum framework. Both approaches are indeed complementary (see Fig. 7 in the theory review⁽²⁾) and can be used in conjunction for multiscale data compaction and data assimilation. Such two way data-based analysis of Earth system processes clearly is a challenge for the future. We have discussed here an example to highlight the potential of overcoming the challenges faced by geomaterial modelling on millions of years time scale. In this example a percolation theory based data compaction approach has been used to identify a creep fracture mechanism in a deformed granite.⁽⁵⁵⁾ This information gleaned from the percolation analysis was then fed into the formulation of a damage mechanics approach based on an irreversible thermodynamics framework.⁽¹⁴⁵⁾

Early attempts of formulating earth material behaviour on the basis of irreversible thermodynamics were inspired by thermographic experiments of Chrysochoos^(170,171) and introduced to the earth science modelling community in Ref. [172]. This approach was widely disseminated in the geoscience community by Ref. [125] and subsequent publications. The seminal work of Collins and Housby⁽⁹⁰⁾ and Rosakis⁽¹⁷³⁾ were completely overlooked. Vardoulakis became a true pioneer in the geomechanics community through his interest in modelling catastrophic landslides.^(174,175) Because of this oversight, and the impossible restrictions to an isothermal assumption⁽⁹⁰⁾ the developments were happening in parallel in two

different communities: Geosciences and Mechanical Engineering.

We have shown how the far from equilibrium thermodynamic view has provided a much improved understanding on how critical point phenomena cascade through vast length and time scales of the Earth. We have described how fundamental THMC feedback processes drive earth processes at multiple scales and how the vastly different THMC length scales allow nested computational forward simulations of the fundamental earth instabilities at different scales. We have also listed first applications of a comprehensive multiscale and multi-physics upscaling and constitutive modelling framework formulated in the theory review.⁽²⁾ The approach although in its infancy promises much improved formulations of computational multiscale data assimilation with consideration of uncertainty for the future.

We have emphasised in particular on the important consideration of the factor time for the earth science problem being far outside the realm of classical mechanics. It was shown to be impossible to incorporate the necessary time dependence in constitutive laws directly from empirical laboratory experiment as processes in nature occur on time scales that are inaccessible in the laboratory. This constraint necessitates particular care in the formulation of the multiscale formalism as each constitutive law at larger scale must be derived through a homogenisation technique that is based on an understanding of the elementary physics of the next smaller scale process.

Although it may not be beyond computational reach, currently, there is a gap in incorporating direct information from modelling at the quantum mechanical scale to make this a truly *ab-initio* based multi-physics framework. This is largely due to the multitude of microphysical processes at microscale and the lack of clarity which microphysical process needs to be incorporated through the nano-chemistry framework to derive a realistic description of earth materials at grain aggregate scale.⁽⁷⁾ We have therefore described a method where grain aggregates behaviour is evaluated through hybrid numerical time-lapse micro-CT scan experiments⁽⁷¹⁾ calibrated by classical laboratory experiments. We have also illustrated how the behaviour of the grain-aggregates may then be assessed on the million year time scale in a virtual experiment. Another extremely powerful method to incorporate micro-CT data directly was described as a computation homogenisation workflow that assesses thermodynamic extrema of conservative and dissipative material properties.

The scales beyond the behaviour of grain aggregates is the classical playing field for geologist. Material instabilities appear as shear zones, fault zones or fault groups, folds and compaction bands which from an observational point of view do not show significant differences from meter to plate tectonic scale.⁽¹¹⁾ We have discussed computational modelling techniques based on the fundamental thermodynamic length scales of thermal, chemical, mechanical and

hydraulic diffusion. A particular emphasis was given on explicitly incorporating the length scales identified in the theory review⁽²⁾ in order to derive the fundamental mode of geomaterial failure at the given scale.

The overwhelming role of time dependence of these processes makes it necessary to employ a formulation that is capable of describing solid and fluid material behaviour. Earth materials are therefore regarded as solids until the point of first yield. After yield their behaviour is described by the flow theory of plasticity.⁽⁹⁵⁾ This dual formalism incorporates all classical material failure modes if the earth material is loaded fast enough but it enriches the solution space through a second set of instabilities that is entirely described by the theory of fluid dynamics and requires long time scales to operate. This allows a surprising richness of instabilities which cover the full plethora of geological observables from melt extraction through ductile compaction bands constrained by orientations of classical compaction bands, shear zones with multiple zones of localisation (central slip zones and chemical process zones). Geologists are, through their basic training, ill equipped to understand the plethora of mechanical and fluid dynamic solutions. As a consequence new research teams are emerging where these challenges are crossed in interdisciplinary collaboration and communication. The main task still relies on breaking the communication barriers. If this is done geologists, can conversely enrich the material scientists and physicist with a new perspective: thinking deep time.

NOTATION

Symbol	Meaning (SI units)
<i>Roman symbols</i>	
t	Time (s)
C_{ijkl}^{ep}	Elastoplastic compliance modulus (Pa)
v_k^a	Mass-average velocity for the a -th constituent and k direction (m s^{-1})
V	Volume (m^3)
V_v	Pore volume (m^3)
n	Bulk porosity (–)
T	Absolute temperature (K)
p	Pressure (Pa)
R	Universal gas constant ($8.3144621 \text{ J mol}^{-1} \text{ K}^{-1}$)
A_0	Pre-exponential factor (s^{-1})
k_π	Permeability (m^2)
k_m	Thermal conductivity ($\text{W m}^{-1} \text{ K}^{-1}$)
q_k^T	Heat flux vector ($\text{J m}^{-2} \text{ s}^{-1}$)
C	Specific heat ($\text{J kg}^{-1} \text{ K}^{-1}$)
D_a	Diffusivity of the a -th physics (THMC) ($\text{m}^2 \text{ s}^{-1}$)
g	Gravitational acceleration ($\text{m}^2 \text{ s}^{-1}$)
K	Elastic bulk modulus (Pa)
G	Elastic shear modulus (Pa)
D	Damage parameter (–)
c	Cohesion (–)
RVE	Representative volume element (–)
$\tilde{S}_{ir}$	Rate of irreversible entropy production inside RVE (W)

Continued

Symbol	Meaning (SI units)
<i>Roman symbols</i>	
$\tilde{S}_{ir}$	Rate of irreversible entropy production inside RVE (W)
$\tilde{S}_{rev}$	Rate of entropy production through fluxes on the RVE boundary (W)
$\dot{S}_{tot}$	Rate of total entropy production (W)
$\dot{W}^{diss}$	Rate of non-conservative work (W)
$\dot{W}^{cons}$	Rate of conservative work (W)
$\dot{W}_{max}^{ext}$	Available work at optimum path (minimum entropy production) (W)
r_k	Heat source/sink term inside RVE (W)
Gr	Gruntfest number (–)
Le	Lewis number (–)
d	Fractal dimension (–)
L	Shear band width (m)
h	Compaction length scale (m)
<i>Greek symbols</i>	
σ_{ij}	Effective stress (Pa)
σ_y	Yield stress (Pa)
$\dot{\epsilon}_{kl}$	Strain rate (s^{-1})
ρ	Density (kg m^{-3})
β	Compressibility (Pa^{-1})
α	Thermal expansion coefficient (m K^{-1})
σ	Piola-Kirchhoff stress tensor (Pa)
ψ	Helmholtz free energy (J)
δ_{loc}	Local dissipation (W)
ϕ	Dissipation potential (W)
λ	Lagrangian multiplier (–)
ξ	State variable (–)
β_T	Taylor Quinney coefficient (–)
μ_f	Fluid viscosity (Pa s)
μ_s	Solid viscosity (Pa s)
ζ	Ratio of strain rate over diffusion rate (–)
η	Creep parameter (s^{-1})
φ	Angle of internal friction (s^{-1})
<i>Superscripts</i>	
p	Plastic
a	a -th mechanism/constituent
f	Fluid species
s	Solid species
<i>Subscripts</i>	
i, j, k, l	Indices
m	Fluid-solid mixture
T	Thermal
H	Hydrous
M	Mechanical
C	Chemical
<i>Mathematical symbolism</i>	
$\dot{x}$	Complete time derivative of x
$\tilde{x}$	Incomplete (path-dependent) time derivative of x
$\partial f / \partial x$	Partial derivative of f with respect to x
D^m / Dt	(Substantive) material time derivative
∇	Nabla operator
β	Imaginary number

References and Notes

1. G. Bell, T. Hey, and A. Szalay; *Science* 323, 1297 (2009).
2. K. Regenauer-Lieb, M. Veveakis, T. Poulet, F. Wellmann, A. Karrech, J. Liu, J. Hauser, C. Schrank, O. Gaede, and M. Trefry; *Journal of Coupled Systems and Multiscale Dynamics* 1, 1 (2013).
3. J. Gale and K. Wright; *Review in Mineralogy and Geochemistry* 71, 391 (2010).

4. J. Amodeo, P. Carrez, B. Devincere, and P. Cordier; *Acta Mater.* 59, 2291 (2011).
5. P. Cordier, J. Amodeo, and P. Carrez; *Nature* 481, 177 (2012).
6. K. Hackl and J. Renner; *Journal of Geophysical Research: Solid Earth* 118, 943 (2013).
7. M. Montagnat, O. Castelnau, P. Bons, S. Faria, O. Gagliardini, F. Gillet-Chaulet, F. Grennerat, A. Grier, R. Lebensohn, H. Moulinec, et al.; *Journal of Structural Geology* in press (2013).
8. M. Geers, V. Kouznetsova, and W. Brekelmans; *Jnl. Comp. Appl. Math.* 234, 2175 (2010).
9. A. Tommasi, M. Knoll, J. Signorelli, C. Thoraval, and R. Loge; *Nature Geosciences* 2, 423 (2009).
10. N. Mancktelow; *Geology* 34, 345 (2006), cited by (since 1996) 25.
11. J. Tchalenko; *Geological Society of America Bulletin* 81, 1625 (1970).
12. J. Hippert; *Journal of Structural Geology* 21, 975 (1999).
13. C. E. Schrank, M. R. Handy, and F. Fusseis; *Journal of Geophysical Research* 113, B01407 (2008).
14. K. Regenauer-Lieb, B. Hobbs, A. Ord, O. Gaede, and R. Vernon; *Physics of the Earth and Planetary Interiors* 172, 43 (2009).
15. H. B. Muhlbach and I. Vardoulakis; *Geotechnique* 37, 271 (1987).
16. A. Ord, B. Hobbs, and K. Regenauer-Lieb; *International Journal for Numerical and Analytical Methods in Geomechanics* 31, 373 (2007).
17. D. McKenzie; *Journal of Petrology* 25, 713 (1984).
18. D. McKenzie; *Journal of the Geological Society London* 144, 299 (1987).
19. M. Veveakis, K. Regenauer-Lieb, and R. Weinberg; *Geophysical Research Letters* submitted (2013).
20. R. Weinberg, M. Veveakis, and K. Regenauer-Lieb; *Geology* submitted (2013).
21. K. Regenauer-Lieb and D.A. Yuen; *Physics of Earth and Planetary Interiors* 142, 113 (2004).
22. B. Hobbs, K. Regenauer-Lieb, and A. Ord; *Journal of Structural Geology* 30, 1572 (2008).
23. B. Hobbs, K. Regenauer-Lieb, and A. Ord; *Geology* 35, 175 (2007).
24. I. Grunfest; *Trans. Soc. Rheol.* 7, 95 (1963).
25. B. Hobbs, A. Ord, and K. Regenauer-Lieb; *Journal of Structural Geology* 33, 758 (2011).
26. S. C. Constable, R. L. Parker, and C. G. Constable; *Geophysics* 52, 289 (1987).
27. V. S. L. Moresi; *Geophysical Journal International* 669, 669 (1998).
28. M. Veveakis and K. Regenauer-Lieb; *Pure and Applied Geophysics* submitted (2013).
29. R. Mettler, Master's Thesis, ETH Zurich (2001).
30. S. Stock; *International Materials Reviews* 53, 129 (2008).
31. C. Arns, M. Knackstedt, W. Pinczewski, and W. Lindquist; *Geophysical Research Letters* 28, 3361 (2001).
32. A. Roberts and E. Garboczi; *Proc. R. Soc. Lond. A* 458, 1033 (2002).
33. M. Knackstedt, H. Arns, M. Saadatfar, et al.; *Proc. R. Soc. A* 462, 2833 (2006).
34. M. Knackstedt, C. Arns, and W. Pinczewski; *Geophysical Prospecting* 53, 349 (2005).
35. P. Oren, S. Bakke, and R. Held; *Water Resources Research* 42, W12S04 (2007).
36. D. Makarynska, B. Gurevich, R. Ciz, C. Arns, and M. Knackstedt; *Computers and Geosciences* 34, 647 (2008).
37. M. Madadi, A. Jones, C. Arns, and M. Knackstedt; *Computing in Science and Engineering* 11, 65 (2009).
38. B. Almqvist, D. Mainprice, C. Madonna, L. Burlini, and A. Hirt; *J. Geophys. Res.* 116, B01204 (2011).
39. V. Shulakova, M. Pervukhina, T. Muller, M. Lebedev, S. Mayo, S. Schmid, P. Golodoniuc, O. D. Paula, B. Clennell, and B. Gurevich; *Physical Prospecting* 61, 278 (2012).
40. J. Liu, K. Regenauer-Lieb, C. Hines, K. Liu, O. Gaede, and A. Squeglch, *G3* 10, Q05010 (2009).
41. J. Liu and K. Regenauer-Lieb; *Phys. Rev. E* 83, 016106 (2011).
42. D. Wildenschild and A. Sheppard; *Advances Water Resources* 51, 217 (2013).
43. A. Kaestner, E. Lehmann, and M. Stampanoni; *Advances Water Resources* 31, 1174 (2008).
44. K. Terada, M. Hori, T. Kyoya, and N. Kikuchi; *Int. J. Solids and Structures* 37, 2285 (2000).
45. D. Stauffer and A. Aharony, *Introduction to Percolation Theory*, 2nd edn. (1994).
46. Y. Bernabé, M. Zamora, M. Li, A. Mainault, and Y. Tang; *Journal of Geophysical Research-Solid Earth* 116 DOI: 10.1029/2011JB008543 (2011).
47. M. Sahimi; *Physics Reports* 306, 213 (1998).
48. B. Biswal, C. Manwart, and R. Hilfer; *Physica A* 255, 221 (1998).
49. R. Hilfer; *Phys. Rev. B* 45, 7115 (1992).
50. Y. Nakashima and Y. Kamiya; *J. Nucl. Sci. Technol.* 44 1233 (2007).
51. A. Navarre-Sitchler, C. I. Steefel, L. Yang, L. Tomutsa, and S. L. Brantley; *Journal of Geophysical Research: Earth Surface* 114, F02016 (2009).
52. E. J. Garboczi, K. A. Snyder, J. F. Douglas, and M. F. Thorpe; *Phys. Rev. E* 52, 819 (1995).
53. D. P. Bentz; *ACI Materials Journal* 105, 297 (2008).
54. D. J. Pringle, J. E. Miner, H. Eicken, and K. M. Golden; *Journal of Geophysical Research: Oceans* 114, C12017 (2009).
55. F. Fusseis, K. Regenauer-Lieb, J. Liu, R. Hough, and F. deCarlo; *Nature* 459, 974 (2009).
56. K. Regenauer-Lieb, D. Yuen, and F. Fusseis; *Pure and Applied Geophysics* 166, 1 (2009).
57. A. Cocks and M. F. Ashby; *Metal Science* 14, 395 (1980).
58. A. C. F. Cocks and M. F. Ashby; *Prog. Mater. Sci.* 27, 189 (1982).
59. R. Raj and M. F. Ashby; *Acta Metall.* 23, 653 (1975).
60. A. Carpinteri and N. Pugno; *Nat. Mater.* 4, 421 (2005).
61. R. A. Ketcham; *J. Struct. Geol.* 27, 1217 (2005).
62. R. Hill; *J. Mech. Phys. Solids* 11, 357 (1963).
63. K. Regenauer-Lieb, A. Karrech, H. T. Chua, F. G. Horowitz, and D. Yuen; *Philosophical Transactions of the Royal Society London, A* 368, 285 (2010).
64. B. L. M. Adam and I. Brevik; *Geophysics* 71, F173 (2006).
65. E. Mavko, T. Mukerji, and J. Dvorkin, *The Rock Physics Handbook: Tools for Seismic Analysis of Porous Media*, Cambridge University Press (2010).
66. R. Salter, D. Sheldner, M. Beller, B. Flack, D. Gillespie, N. Moldoveanu, F. G. Pineda, and J. C. P. Alfaro; *World Oil* 57 (2006).
67. K. Hadley; *Journal of Geophysical Research* 81, 3484 (1976).
68. H. C. Heard and L. Page; *Journal of Geophysical Research* 87, 9340 (1982).
69. M. H. B. Nasser, A. Schubnel, and R. P. Young; *International Journal of Rock Mechanics and Mining Sciences* 44, 601 (2007).
70. F. T. Wu and L. Thomsen; *International Journal of Rock Mechanics and Mining Sciences and Geomechanics Abstracts* 12, 167 (1975).
71. C. Schrank, F. Fusseis, A. Karrech, and K. Regenauer-Lieb; *Geochem. Geophys. Geosyst.* 13, Q09005 (2012).
72. J. T. Fredrich and T.-F. Wong; *Journal of Geophysical Research* 91, 12743 (1986).
73. D. Richter and G. Simmons; *International Journal of Rock Mechanics and Mining Sciences and Geomechanics Abstracts* 11, 403 (1974).
74. H. Cooper and G. Simmons; *Earth and Planetary Science Letters* 36, 404 (1977).
75. T.-F. Wong and W. Brace; *Tectonophysics* 57, 95 (1979).
76. B. Evans and D. Kohlstedt, *Rheology of Rocks*, American Geophysical Union, Washington (1995), pp. 148–165.

77. S. Bratt, E. Bergman, and S. Solomon; *Journal of Geophysical Research* 90, 10249 (1985).
78. W. M. Bruner; *Journal of Geophysical Research* 89, 4167 (1984).
79. J. A. D. Connolly; *Earth and Planetary Science Letters* 236, 524 (2005).
80. B. J. Skinner, Thermal Expansion, Geological Society of America, Boulder, Colorado (1966), pp. 75–96.
81. D. E. Moore and D. Lockner; *Journal of Structural Geology* 17, 95 (1995).
82. Tromans and J. Meech; *Minerals Engineering* 17, 1 (2004).
83. M. Darot, Y. Gueguen, and M. Baratin; *Geophysical Research Letters* 19, 869 (1992).
84. G. Hirth, C. Teyssier, and J. Dunlap; *International Journal of Earth Sciences* 90, 77 (2001).
85. E. Rybacki and G. Dresen; *Journal of Geophysical Research B: Solid Earth* 105, 26017 (2000).
86. A. Tordesillas, D. Walsh, and S. Gardnier (eds.), *Bridging the Length Scales: Micromechanics of Granular Media*, Kluwer Academic Publishers (2004), pp. 539–556.
87. L. Onsager; *Physical Review* 37, 405 (1931).
88. B. Coleman and W. Noll; *Archive for Rational Mechanics and Analysis* 13, 167 (1963).
89. C. Truesdell (ed.), *Rational Thermodynamics*, McGraw-Hill Series in Modern Applied Mathematics (1969).
90. I. Collins and G. Houlsby; *Royal Society of London, A* 453, 1975 (1997), cited by (since 1996); 46 Export Date: 26 October 2006 Source: Scopus Language of Original Document: English Correspondence Address: I. F. Collins, Department Eng. Sci., University of Auckland; School of Engineering Private Bag 92019, Auckland, New Zealand.
91. G. Houlsby and A. Puzrin, *Principles of Hyperplasticity: An Approach to Plasticity Theory Based on Thermodynamic Principles*, Springer Verlag, London (2006).
92. A. Tordesillas, M. Walker, and Q. Lin; *Phys. Rev. E* 81, 011302 (2010).
93. A. Love (ed.), *Treatise on the Mathematical Theory of Elasticity*, Cambridge University Press (1927).
94. O. Coussy (ed.), *Poromechanics*, Wiley (2004).
95. P. Perzyna; *Adv. Appl. Mech.* 9, 243 (1966).
96. A. Das, G. Nguyen, and I. Einav; *J. Geoph. Res.* 116 (2011).
97. M. Arroyo, R. Castellanza, and R. Nova; *Soils and Foundations* 45, 181 (2005).
98. E. Rutter; *Phil. Trans. R. Soc. Lond. A* 283, 203 (1973).
99. S. D. Meer, C. Spiers, and C. Peach; *J. Geol. Soc. Lond.* 157, 269 (2000).
100. R. Katsman, E. Aharonov, and H. Scher; *J. Geoph. Res.* 111 B03204 (2006).
101. L. Angheluta, J. Mathiesen, and E. Aharonov; *J. Geophys. Res.* 117, B08203 (2012).
102. K. Issen and J. Rudnicki; *J. Geoph. Res.* 105, 21529 (2000).
103. J. Rudnicki and J. Rice; *J. Mech. Phys. Solids* 23, 371 (1975).
104. T. Baxevanis, E. Papamichos, O. Flornes, and I. Larsen; *Acta Geotech.* 1, 123 (2006).
105. A. Chemenda; *J. Mech. Phys. Solids* 57, 851 (2009).
106. I. Vardoulakis; *Geotechnique* 46, 457 (1996).
107. I. Vardoulakis and J. Sulem (eds.), *Bifurcation Analysis in Geomechanics*, Blankie Acc. and Professional (1995), pp. 150–180.
108. I. Vardoulakis; *Geotechnique* 52, 157 (2002).
109. J. Fortin, S. Stanchits, G. Dresen, and Y. Gueguen; *J. Geophys. Res.* 111, B10203 (2006).
110. J. Arthur, T. Dustan, Q. Al-Ani, and A. Assadi; *Geotechnique* 42, 395 (1977).
111. I. Vardoulakis; *Int. J. Num. Anal. Meth. Geomech.* 4, 119 (1980).
112. F. Oka, S. Kimoto, Y. Higo, H. Ohta, T. Sanagawa, and T. Kodaka; *Int. J. Numer. Anal. Meth. Geomech.* 35, 244 (2011).
113. R. Hickman and M. Gutierrez; *Int. J. Numer. Anal. Meth. Geomech.* 31, 583 (2007).
114. H. Leinenkugel Doctoral Dissertation, University of Karlsruhe, Karlsruhe, Germany (1976).
115. J. Sulem, I. Stefanou, and E. Veveakis; *Granular Matter* 13, 261 (2011).
116. E. Veveakis, J. Sulem, and I. Stefanou; *J. Struct. Geology* 38, 254 (2012).
117. E. Veveakis, S. Alevizos, and I. Vardoulakis; *J. Mech. Phys. Solids* 58, 1175 (2010).
118. M. Abramowitz and I. Stegun (eds.), *Handbook of Mathematical Functions*, U.S. National Bureau of Standards (1964).
119. S. Tembe, P. Baud, and T. Wong; *J. Geophys. Res.* 113, B09409 (2008).
120. R. Schultz; *J. Geophys. Res.* 114, B03407 (2009).
121. J. Solum, J. Brandenburg, S. Naruk, O. Kostenko, S. Wilkins, and R. Schultz; *AAPG Bulletin* 94, 1453 (2010).
122. E. Townend, B. Thompson, P. Benson, P. Meredith, P. Baud, and R. Young; *Geoph. Res. Lett.* 35, L15301 (2008).
123. E. Tinti, P. Spudich, and M. Cocco; *J. Geophys. Res.* 110, B12303 (2005).
124. P. Mollema and M. Antonellini; *Tectonophysics* 267, 209 (1996).
125. D. Bercovici, Y. Ricard, and G. Schubert; *Journal of Geophysical Research: Solid Earth* 106, 8887 (2001).
126. K. Regenauer-Lieb, D. A. Yuen, and J. Branlund; *Science* 294, 578 (2001).
127. K. Regenauer-Lieb, R. F. Weinberg, and G. Rosenbaum; *Nature* 442, 67 (2006).
128. B. E. Hobbs, H. Mhlhaus, and A. Ord; *Geological Society, London, Special Publications* 54, 143 (1990).
129. A. Sengupta; *Current Science* 98, 1195 (2010).
130. K. Regenauer-Lieb, G. Rosenbaum, and R. F. Weinberg; *Tectonophysics* 458, 96 (2008).
131. K. Regenauer-Lieb and D. Yuen; *Pure and Applied Geophysics* 163, 1 (2006).
132. K. Regenauer-Lieb; *Journal of Geodynamics* 27, 1 (1999).
133. A. L. Gurson; *ASME Transaction, Journal of Engineering Materials* 99, 2 (1977).
134. V. Tvergaard; *International Journal of Solides and Structures* 18, 659 (1981).
135. A. Needleman and V. Tvergaard; *Journal of the Mechanics and Physics of Solids* 32, 461 (1984).
136. J. Lemaitre; *Computer Methods in Applied Mechanics and Engineering* 51, 31 (1985).
137. J. Lemaitre and J. Dufailly; *Engineering Fracture Mechanics* 28, 643 (1987).
138. J. L. Chaboche; *Nuclear Engineering and Design* 105, 19 (1987).
139. V. Lyakhovsky, Z. Reches, R. Weinberger, and T. Scott; *Geophysics Journal International* 130, 157 (1997).
140. Y. Hamiel, V. Lyakhovsky, and A. Agnon; *Geophysics Journal International* 156, 710 (2004).
141. Y. Hamiel, V. Lyakhovsky, Y. Ben-Zion, and D. Lockner; *Geophysics Journal International* 159, 1155 (2004).
142. V. Lyakhovsky, Y. Ben-Zion, and A. Agnon; *Geophysics Journal International* 161, 179 (2005).
143. D. Bercovici and Y. Ricard; *Geophysical Journal International* 152, 581 (2003).
144. A. Karrech, K. Regenauer-Lieb, and T. Poulet; *Journal of Engineering Science* 49, 1141 (2011).
145. A. Karrech, K. Regenauer-Lieb, and T. Poulet; *Journal of Geophysical Research* 116, Art. no. B04205 (2011).
146. A. Karrech, K. Regenauer-Lieb, and T. Poulet; *International Journal of Solids and Structures* 48, 407 (2011).
147. A. Karrech, T. Poulet, and K. Regenauer-Lieb; *Philosophical Magazine* 92, 3439 (2012).
148. T. C. Chang, C. H. Popelar, and G. Staab; *International Journal of Fracture* 32, 157 (1987).
149. N. Bonora; *Engineering Fracture Mechanics* 58, 11 (1997).

150. Abaqus, User Manual version 6.4, Hibbit and Karlson and Sorensen Inc. (2007).
151. K. Regenauer-Lieb and D. Yuen; *Geophysical Research Letters* 25, 2737 (1998).
152. J. R. Rice; *J. Geophys. Res.* 111, B05311 (2006).
153. E. Veveakis, I. Vardoulakis, and G. DiToro; *J. Geophys. Res.* 112, F03026 (2007).
154. J. Sulem and V. Famin; *J. Geophys. Res.* 114, B03309 (2009).
155. T. Hueckel and G. Baldi; *J. Geotech. Eng.* 116, 1778 (1990).
156. T. Hueckel and R. Pellegrini; *Eng. Geology* 64, 195 (2002).
157. C. Cekerevac and L. Laloui; *Int. J. Num. Anal. Meth. Geomech.* 28, 209 (2004).
158. F. Cecinato, A. Zervos, and E. Veveakis; *International Journal for Numerical and Analytical Methods in Geomechanics* 35, 1507 (2011).
159. E. Veveakis, I. Stefanou, and J. Sulem; *Geotechnique Let.* 3, 31, (2013).
160. C. Law (ed.), *Combustion Physics*, Cambridge University Press (2006).
161. A. Ebert, M. Herwegh, and A. Pfiffner; *Journal of Structural Geology* 29, 1164 (2007).
162. L. A. Kennedy and J. M. Logan; *Journal of Structural Geology* 19, 785 (1997).
163. C. D. Rowe, Å. Fagereng, J. A. Miller, and B. Mapani; *Earth and Planetary Science Letters* 333–334, 200 (2012).
164. S. Schmid; *Eclogae Geologicae Helveticae* 68, 247 (1975).
165. R. Han, T. Shimamoto, T. Hirose, J. Ree, and J. Ando; *Science* 316, 878 (2007).
166. V. Famin, S. Nakashima, A.-M. Boullier, K. Fujimoto, and T. Hirono; *Earth. Plan. Sci. Let.* 265, 487 (2008).
167. E. Veveakis, T. Poulet, and S. Alevizos; *Journal of Geophysical Research* submitted (2013).
168. S. Alevizos, M. Veveakis, and T. Poulet; *Journal of Geophysical Research* submitted (2013).
169. K. Regenauer-Lieb and D. Yuen; *Earth Science Reviews* 63, 295 (2003).
170. A. Chrysochoos, O. Maisonneuve, G. Martin, H. Caumon, and J. Chezeaux; *Nuclear Engineering and Design* 114, 323 (1989).
171. A. Chrysochoos and R. Peyroux; *Revue Générale de Thermique* 37, 582 (1998).
172. K. Regenauer-Lieb, J. Petit, and D. Yuen; *Electronic Geosciences* 4 (1999).
173. P. Rosakis, A. Rosakis, G. Ravichandran, and J. Hodowany; *J. Mech. Phys. Solids* 48, 581 (2000).
174. I. Vardoulakis; *International Journal of Solids and Structures* 39, 3831 (2002).
175. I. Vardoulakis; *Mech. Cohesive Frict. Mater.* 5, 443 (2000).